BioSpecInfo

Dossier completo

Samuele Pio Provenzano

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Raccolta integrale dei documenti del progetto, nell'ordine di numerazione. Ogni documento è disponibile anche come PDF separato.

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BioSpecInfo — Technical Dossier

Open-access web platform for computational chemistry and the interactive consultation of chemical data on any device.

Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Profile	Cheminformatics Engineer / Scientific Software Developer
Software type	Progressive Web App (PWA), client-side, local-first, offline-first
Live demo	https://samupropio1-ship-it.github.io/BioSpecInfo-v11/
Repository	github.com/samupropio1-ship-it/BioSpecInfo-v11

> This dossier summarises the application and is the cover of the full technical documentation (SAD, Validation & Benchmark, Security & Compliance, Licensing) attached below.

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1. Executive summary

BioSpecInfo is an **interactive cheminformatics platform** that runs all of its scientific logic **in the client browser**, with no server: molecular analysis, spectral prediction, 2D/3D modelling and educational tools — all **free, installable as an app and working offline**.

Its distinctive value is the **rare combination of advanced chemistry and modern software engineering** (WebAssembly, WebGL, PWA), with spectral predictors **quantitatively validated** against the official SDBS and NIST databases.

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2. Product factsheet

Value proposition — A **≈4.5 MB** web platform (main file) that runs molecular analysis, spectral prediction and 3D visualisation at **zero server cost** and **100% offline-ready**.

Technical architecture	Content & capabilities
Client-side / Local-First (no backend)	**RDKit** cheminformatics engine (WebAssembly)
Single-file HTML5 / ES6, installable as **PWA**	**Spectra Centre**: IR, ^{1}H/^{13}C-NMR, MS + isotope pattern, UV-Vis, Raman
2D (SVG/Canvas) and **3D** (three.js, 3Dmol.js) rendering	**Organic Chemistry**: 25 interactive modules
Service Worker offline-first, versioned	**Retrosynthesis**: 46 strategies + animated 2D/3D mechanisms
Zero copyleft dependencies (MIT/BSD-3 only)	**3D Drug Atlas**: 63 diseases (23 tumours), 3D organs, PubMed
Local storage (localStorage, SQLite via sql.js)	3D astrochemistry, Quantum Lab, Academy (SM-2 quizzes)

Quantitative validation (thesis benchmark)

- **^{1}H-NMR** vs SDBS → **MAE 0.31 ppm** (89% of signals within ±0.5 ppm)

- **IR (C=O band)** vs NIST WebBook → **MAE 11 cm⁻¹** (90% within ±30 cm⁻¹)
- Interactive performance **30–60 fps** from iPhone/iPad to Raspberry Pi 4

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3. Verified capabilities (current code state)

Spectra Centre (molecular editor & analysis) — draw from scratch (full periodic table, templates, drag-to-bond) or load via SMILES/library →

- All chemical notations: formula, canonical SMILES, CXSMILES, SMARTS, InChI,

InChIKey, mol block, common/IUPAC name, CAS (one-click copy).

- Predicted spectra with bibliographic sources; **isotope pattern** (IUPAC

abundances) and MS **adduct masses**.

- Functional groups and properties (H donors/acceptors, aromatic, rotatable,

stereocentres) **highlightable on the 2D and 3D structure**.

- Drug-likeness: **Lipinski, Veber, Egan, Ghose, Muegge**, ESOL solubility,

QED, **Brenk/PAINS structural alerts**, physicochemical radar.

- **Tanimoto** similarity (Morgan ECFP4 fingerprint), elemental analysis, SMARTS

substructure search, SVG/PNG/MOL export.

Synthesis & Retrosynthesis — 46 strategies with an **animated mechanism player** in **2D and 3D**: *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

Organic Chemistry — 25 modules (fundamentals → heterocycles, pericyclic & Woodward–Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups).

Data Science — 3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), ML labs, SQL playground.

Other — 3D astrochemistry, Quantum Lab, Academy with spaced repetition (SM-2), interactive simulations, databases of elements/drugs/biomolecules.

> Overall metrics declared in the thesis (historical data assets): 324 organic reactions in vector SVG graphics, 26 biosynthetic pathways, 140+ drugs, 80+ biomolecules, 118 elements, 10 quantum simulators.

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4. Technology stack

`HTML5` · `ES6/ESNext` · `WebAssembly` (RDKit MinimalLib, sql.js/SQLite) · `WebGL` (three.js, 3Dmol.js) · `SVG` / `Canvas 2D` · `Service Worker` / `PWA` · public API integration (PubChem PUG-REST, NIST, SDBS, MassBank, nmrshiftdb2, NASA/ESA).

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5. Author's declaration (self-attestation)

The undersigned **Samuele Pio Provenzano** declares that:

1. The software **BioSpecInfo** is the author's **original work**, except for the

third-party open-source libraries (RDKit, 3Dmol.js, three.js, SmilesDrawer, sql.js) used under their respective permissive licenses and documented in the *Open-Source Licensing & IP Matrix*.

1. The reported validation data (MAE 0.31 ppm for ^{1}H-NMR vs SDBS; MAE 11 cm^{-1} for

the IR C=O band vs NIST) were measured by the author as part of the thesis work; the predictors are **heuristic group-additivity** models, suitable for educational use and qualitative functional-group identification, with the limitations stated in the *Validation Report*.

1. The architecture is **local-first**: no user personal data is transmitted to

author-controlled servers (alignment with GDPR principles).

> This is an **author's declaration**, not a certification issued by a third party.

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6. Attached documentation

1. **Software Architecture Document (SAD)** — architecture, components, technologies.
2. **Verification & Validation & Benchmark Report** — test method, benchmarks, limitations.
3. **Security, Privacy & Compliance** — local-first, GDPR, OWASP, GAMP 5 (alignment).
4. **Open-Source Licensing & IP Matrix** — dependencies, licenses, intellectual property.

Software license: proprietary — "All rights reserved" (see `LICENSE`).

Software Architecture Document (SAD) — BioSpecInfo

Field	Value
Project	BioSpecInfo
Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Type	Progressive Web App (PWA), client-side, local-first, offline-first
Platform	Modern browsers (Chromium, Firefox, Safari/WebKit) · Desktop and Mobile
Repository	`samupropio1-ship-it/BioSpecInfo-v11`

1. System overview

BioSpecInfo is a **monolithic yet modular** web platform that runs all of its application logic inside the client browser, with no application server. It is distributed as an installable **Progressive Web App** (`display: standalone`) and works **offline** thanks to a Service Worker that precaches the heavy assets.

Architectural pattern: Single-Page/Single-File Application + satellite sub-apps in iframes, with a WebAssembly-based cheminformatics *core* decoupled from the presentation logic.

1.1 Guiding principles

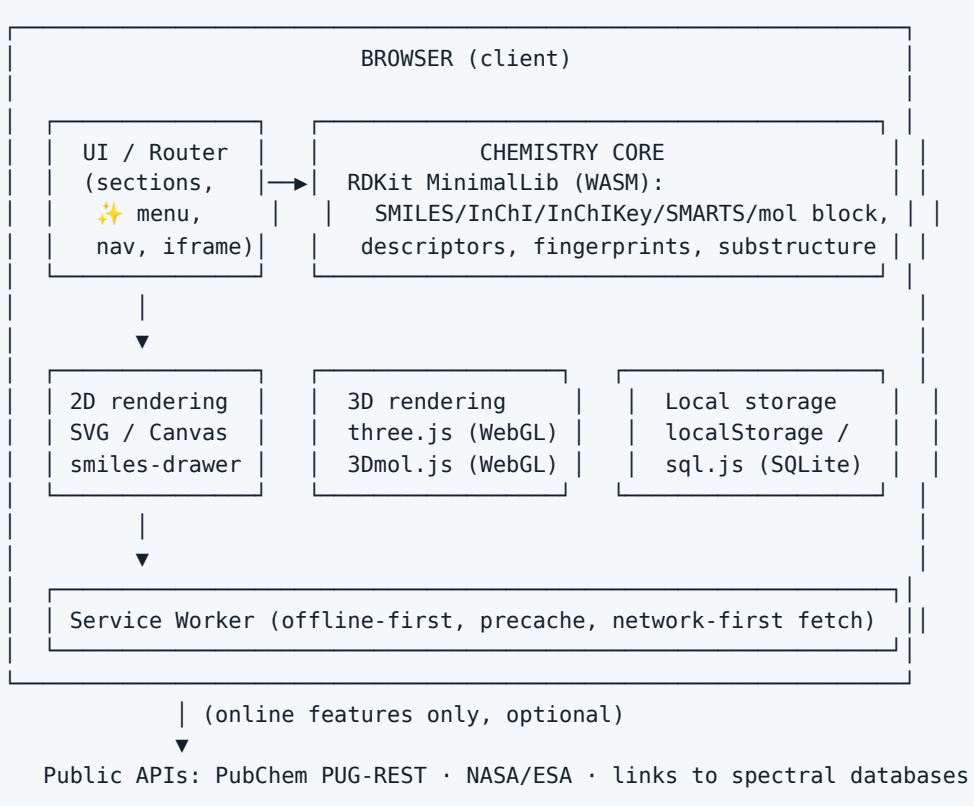
- **Local-First:** data and computations live on the device; no mandatory transmission to third parties.
- **Zero-backend:** no server database, no dynamic hosting cost (deployable on static hosting, e.g. GitHub Pages).
- **Progressive enhancement:** online features (PubChem search, real 3D models, astronomical imagery) degrade gracefully when offline.

2. File structure

File	Role
`index.html`	Main application (chemistry core, Spectra Centre, Data Science, ✨ menu). ~4.5 MB, over 43,000 lines.
`chimorga.html`	Organic Chemistry course (25 modules).
`astro.html`	3D Astrochemistry (three.js).

File	Role
`pro.html`, `rdkit_lab.html`, `accademia.html`, `simulazioni.html`, `guidaret.html`, `file_manager.html`, `Biochimica_Guida_Definitiva.html`, `sr_completo.html`, `sr_essenziale.html`, `download.html`, `changelog_tesi.html`	Thematic sub-apps loaded in iframes with `?v=timestamp` cache-busting.
`manifest.json`	PWA configuration (icons, theme, standalone).
`sw.js`	Service Worker: *network-first* strategy with cache fallback; precache of heavy libraries/assets; cache version bumped on every release.
`RDKit_minimal.js` / `RDKit_minimal.wasm`	Cheminformatics engine (RDKit MinimalLib).
`3Dmol-min.js`	3D viewer for molecules/macromolecules.
`three.min.js`, `three_bloom.js`, `gltf_loader.js`	WebGL 3D rendering (anatomy, astrochemistry, mechanisms).
`smiles-drawer.min.js`	2D depiction of SMILES.
`lib/sql-wasm.js` / `lib/sql-wasm.wasm`	SQLite in WebAssembly (SQL playground).
`models/`, `textures/`	3D assets (organs, écorché, planets) and textures.

3. Component architecture



3.1 Module Router / UI

Navigation between sections via CSS classes (`.section.on`) and a ✨ menu (FAB) that opens the special tools (Spectra Centre, Retrosynthesis, Quantum Lab, Data Science, etc.). Sub-apps are isolated in versioned `iframe` s.

3.2 WebAssembly Engine (RDKit MinimalLib)

Loaded on demand (`window.bsiLoadRDKit`) with `locateFile` pointing to the local `.wasm` files and a CDN as a last-resort fallback. It exposes:

- **Identifiers:** canonical/isomeric SMILES, CXSMILES, InChI, InChIKey, SMARTS,

MDL V2000 mol block.

- **Descriptors:** molecular weight, monoisotopic exact mass, LogP

(Wildman–Crippen), TPSA, HBD/HBA, rotatable bonds, aromatic/aliphatic rings, Fsp^3, molar refractivity, stereocentres.

- **Fingerprints & similarity:** Morgan ECFP4 (1024 bit) and the Tanimoto index.
- **Substructure matching** via SMARTS (`get_qmol` + `get_substruct_matches`).
- **2D depiction** (`get_svg` , `get_svg_with_highlights`) and 2D coordinate

generation (mol block) re-importable into the editor.

3.3 Rendering

- **2D:** proprietary SVG (reaction mechanisms with Bézier arrows), Canvas 2D

(molecule editor, CIP stereochemistry viewer), smiles-drawer.

- **3D:** `three.js` (WebGL) for anatomy (Drug Atlas), astrochemistry and 3D

mechanisms; `3Dmol.js` for molecular models from PubChem SDF.

3.4 Storage Layer

- `localStorage` (with a **safety guard** and in-memory fallback for incognito /

blocked-storage scenarios).

- `sql.js` (SQLite compiled to WASM) for the Data Science SQL playground.

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4. Implemented features (current state)

4.1 Spectra Centre (molecular editor & analysis)

Draw from scratch (full periodic table, ring/group templates, drag-to-bond with snapping) or load via SMILES / library (63 molecules) → full analysis:

- **All chemical notations** (formula, canonical SMILES, CXSMILES, SMARTS, InChI,

InChIKey, mol block, common/IUPAC name, CAS) with one-click copy.

- **Predicted spectra** as sub-panels: IR, ^{1}H-NMR, ^{13}C-NMR, MS with **isotope

pattern **(IUPAC abundances) and** adduct masses** ([M+H]⁺, [M+Na]⁺…), UV-Vis, Raman — each with a bibliographic source.

- **Functional groups** (~30, via SMARTS) and **properties** (H donors/acceptors,

aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.

- **Drug-likeness / ADMET:** Lipinski, Veber, Egan, Ghose, Muegge rules;

estimated ESOL solubility; QED (when exposed by the engine); **Brenk/PAINS structural alerts**; physicochemical radar profile.

- **Elemental analysis** (weight %) and **Tanimoto similarity** against the library.
- **3D models** (ball & stick, spacefill, wireframe) from PubChem; SVG/PNG/MOL export.

4.2 Synthesis & Retrosynthesis

46 disconnection strategies with an **animated mechanism player** in **2D and 3D**: curved *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

4.3 Organic Chemistry (`chimorga.html`)

25 modules: fundamentals, stereochemistry, alkenes/alkynes, aromatics & EAS, SN/E, carbonyls, amines, carbohydrates, amino acids, lipids, nucleotides, animated syntheses, named reactions, heterocycles, pericyclic & Woodward–Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups.

4.4 Data Science

3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), interactive ML labs, SQL playground (sql.js), course roadmap.

4.5 Other sections

3D astrochemistry, Quantum Lab, Academy (gamified quizzes with SM-2 spaced repetition), interactive simulations, guides, databases of elements/drugs/ biomolecules.

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5. Architectural roadmap (NOT yet implemented)

The following items belong to the *enterprise/HPC* vision and require external binaries, models or build pipelines that cannot be hosted inside a single-file PWA:

- HPC parallelism via Web Workers + `SharedArrayBuffer` / `Atomics`.
- WebGPU acceleration for dense/sparse algebra and GNN inference.
- ML predictors in ONNX Runtime WebGPU (GNN for NMR such as ShiftML2/DeepNMR;

hERG, BBB, CYP450, LogS).

- In-browser molecular docking (AutoDock Vina/Webina, PDBQT format).
- Semi-empirical quantum chemistry (xTB/GFN2-xTB in WASM).
- Rule-based computational retrosynthesis engine (SMIRKS).

- SQLite-WASM database at the 1,000,000-structure scale and SDF batch processing via Web Workers.
- Immersive WebXR (VR/AR) visualisation.
- Native packaging (Android TWA, iOS, Electron/Tauri) and CI/CD pipelines.

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6. Requirements and constraints

- **Browser:** WebAssembly and WebGL support; JavaScript enabled.
- **Network:** not required for the core; recommended for PubChem search and real 3D models.
- **Performance:** target 30–60 fps for interactive rendering; heavy computations are delegated to RDKit WASM and kept responsive.

Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Purpose	Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations.

1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

2. Software verification (QA)

Technique	Description	Status
Syntax checking	`node --check` on the critical JavaScript blocks on every change.	✅ In use
End-to-end (E2E) testing	Automation with **Playwright** on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: **0 non-network errors**).	✅ In use
HTTP-server verification	RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.	✅ In use
Manual cross-device testing	Testing on real devices (see §4).	✅ In use
Automated unit-test suite (Vitest/Jest)	—	🔜 Roadmap
Coverage ≥ 90%	—	🔜 Roadmap

> Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

3.1 ¹H-NMR benchmark vs SDBS

Metric	Reported value
Reference database	**SDBS** (Spectral Database for Organic Compounds, AIST)
Mean Absolute Error (MAE)	**0.31 ppm**
Signals within ±0.5 ppm	**89%**

3.2 IR benchmark (C=O band) vs NIST

Metric	Reported value
Reference database	**NIST Chemistry WebBook**
Mean Absolute Error (MAE)	**11 cm^{-1}**
Bands within ±30 cm^{-1}	**90%**

3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol → $C_{27}H_{46}O$, 4 rings, 8 stereocentres*.

> **Benchmark data source:** measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

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4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

Device	Result
iPhone / iPad	30–60 fps
Windows / macOS laptop	60 fps
Raspberry Pi 4	30–60 fps

Size of the main `index.html` file: **≈ 4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

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5. Declared model limitations

In line with the principle of scientific transparency:

1. **Additivity model.** The NMR/IR predictors sum group contributions and do

not fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.

1. **Qualitative/educational spectra.** The simulated spectra serve to recognise

functional groups and understand structure–spectrum relationships, not to replace certified experimental data.

1. **Network dependency for some functions.** IUPAC/common name, CAS number, GHS

pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.

1. **Two-way accuracy.** For rigorous use, direct links to the official databases

(NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.

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6. Release acceptance criteria

- ✅ `node --check` passes without errors on the modified modules.
- ✅ Playwright E2E: 0 non-network JS errors.
- ✅ No visible regression in the touched sections.
- ✅ Service Worker cache version bumped.
- ✅ `tools/verifica-farmaci.js`: no defects; every deviation recorded with its reason.
- ✅ `verifica_guida`: whatever the documentation promises exists in the code.

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7. Verification battery — state at version `bsi-v219`

Verification is no longer only functional end-to-end testing: it includes benches dedicated to the properties a functional test does not observe (stability over time, network degradation, data accuracy).

7.1 Composition

Family	Benches	Subject
Scientific data	`verifica-farmaci`, `test_spettri`, `test_assi`, `test_assi_canvas`, `test_costanti`, `audit_dati`	structure vs molecular weight, functional-group recognition, axis conventions, physical constants
AI agent	`test_ko`, `test_404`, `test_503`, `test_firma`, `test_attesa`, `test_attesalunga`, `test_tetto`, `test_nucleo`, `browser_ko`, `browser_prova`	retired model, CORS, overload, reasoning signatures, token ceiling, long waits, memory of unreachable providers

Family	Benches	Subject
Stability	`audit_stabilita`, `audit_promesse`, `audit_quota`, `test_filemanager`, `test_doppioinvio`, `test_sw`, `test_visore3d`	long sessions, rejected promises, exhausted storage, unavailable archive, overlapping submissions, degraded network, WebGL contexts
Interface	`browser_reset`, `browser_proxy`, `browser_proxyui`, `browser_rdkit`, `browser_lab`, `browser_frontiera`, `test_aggiorna`, `test_guidaproxy`	panels, proxy, RDKit laboratory, updates
Consistency	`verifica_guida`	documentation ⟷ code correspondence, conflict markers across 58 files

7.2 Properties verified that a functional test does not observe

Property	Method	Measured result
Absence of memory leaks	96 sections opened for 5 rounds, DOM node count per round	construction +26,876 nodes on the first round, **+0 on the four that follow**
Survival of exhausted storage	`localStorage.setItem` forced to throw	10 pages out of 10 stay operational
Network degradation (≠ no network)	requests held for 20 s, interception at context level	answer served from cache in **3,507 ms** (threshold 3,500 ms)
Spectrum reproducibility	drawn twice, compared byte for byte	identical
WebGL contexts	viewer constructions counted across 10 molecules	from 7 down to **1**

7.3 Notes on method

A bench that measures nothing passes, and that is the most insidious way to obtain a false guarantee. While the battery was being built, several cases of vacuous measurement were found and corrected, among them:

- Playwright's `page.route` does **not** intercept requests originated by the

Service Worker: `context.route` is required. With the former, the "degraded network" test was measuring a perfectly healthy network and concluding in 16 ms.

- Probing `localStorage` exhaustion with a one-byte write proves nothing: after

saturation, one byte always finds room.

- The ticks of an axis drawn on `<canvas>` are not readable as text: they must

be collected by intercepting `fillText`, and the bottom-most line is the axis *title*, not a tick.

Hence the rule adopted: **every bench that simulates a condition must count how many times the simulation actually fired**, and fail if that count is zero.

Security, Privacy & Compliance — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Model	Client-side / Local-First PWA

1. Privacy & data protection (GDPR)

1.1 Local-First by design

- **No account, no login, no advertising tracking.**
- All user data (drawn molecules, preferences, quiz history, study progress) is

stored **exclusively locally** on the device (`localStorage` / IndexedDB / `sql.js`), with no transmission to author-controlled servers.

- **There is no application backend** collecting or processing personal data.

1.2 Data handling and GDPR alignment

- **Data minimisation:** the app requires no personal data to function.
- **No transfer to the author's third parties:** data never leaves the device for

infrastructure controlled by the author.

- **Optional, transparent external calls:** only upon a user action (searching a

molecule, opening a 3D model) does the app query public third-party APIs (PubChem, NASA/ESA, spectral databases). Such requests contain **only the chemical identifier** (e.g. SMILES/name) and **no personal data**. Use of these services is subject to their respective privacy policies.

- **Local right to erasure:** the user can delete data by clearing the site

storage in the browser.

> **Note.** This alignment describes the local-first design properties that favour GDPR compliance. It does not constitute a legal certification; for regulated contexts a dedicated DPIA assessment is recommended.

2. Application security (OWASP)

2.1 Attack surface

Being a **static application with no backend**, entire classes of server-side vulnerabilities are absent by construction (SQL injection on a server DB, server-side RCE, server authentication/session, IDOR on proprietary APIs).

2.2 Input sanitisation

- Parsing of **SMILES / SMARTS** strings and molecular files is delegated to

RDKit MinimalLib (WASM), which validates the structures and rejects malformed input by returning invalid molecules rather than executing them.

- Strings coming from user input or external APIs, when inserted into the DOM,

are treated as untrusted. Their textual value (names, notations) is shown as text content/`textContent` or with escaping of the `& < >` characters.

2.3 Runtime robustness

- `localStorage` **guard:** every storage access is protected by `try/catch` with

an in-memory fallback, so that incognito mode or blocked storage cannot halt execution (a historical regression that has been fixed).

- **Controlled degradation:** panels never silently stay empty; errors show a

message and, where applicable, a "Retry" button.

- **Sub-app isolation** in `iframe`s.

2.4 Content Security Policy (deployment recommendation)

Since the app can be published on static hosting, it is recommended to serve it with security headers at the hosting/CDN level:

- Restrictive `Content-Security-Policy` (self script/style + WASM),

`frame-ancestors`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: no-referrer`.

- `Cross-Origin-Opener-Policy` / `Cross-Origin-Embedder-Policy` if

`SharedArrayBuffer` and parallel Web Workers are enabled in the future (HPC roadmap).

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3. Integrity and reproducibility (GAMP 5 alignment)

For use in pharmaceutical/regulated contexts, the architecture is **compatible** with GAMP 5 principles:

- **Determinism:** the cheminformatics computations (RDKit) are deterministic and

reproducible for the same input.

- **Version traceability:** code versioning on Git, versioned Service Worker

cache, changelog.

- **Method transparency:** the heuristic predictors are documented with their

sources and limitations (see the *Validation Report*).

> Again: this is **alignment with the principles**, not a formal qualification (IQ/OQ/PQ), which would require a dedicated validation process at the using organisation.

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4. Summary

Aspect	Status
Personal data transmitted to author's servers	**None**
Backend / server database	**Absent**
Molecular input validation	RDKit WASM
DOM output escaping	Yes (untrusted content)
Storage guard / error degradation	Yes
CSP / security headers	Recommended at hosting level
Offline operation	Yes (Service Worker)

Open-Source Licensing & IP Compliance Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author / owner of proprietary code	Samuele Pio Provenzano

1. Third-party dependency matrix

All bundled third-party libraries are **open-source under permissive licenses** (MIT / BSD-3-Clause). **No viral copyleft dependency (GPL / AGPL / LGPL)** is used, so there are no obligations to redistribute the proprietary code.

Component	Repo file	Use	License
RDKit (MinimalLib / rdkit-js)	`RDKit_minimal.js`, `RDKit_minimal.wasm`	Cheminformatics engine (SMILES/InChI/InChIKey/SMARTS, descriptors, fingerprints, 2D depiction)	**BSD-3-Clause**
3Dmol.js	`3Dmol-min.js`	3D viewer for molecules/macromolecules (WebGL)	**BSD-3-Clause**
three.js	`three.min.js`, `three_bloom.js`, `gltf_loader.js`	WebGL 3D rendering (anatomy, astrochemistry, mechanisms)	**MIT**
SmilesDrawer	`smiles-drawer.min.js`	2D depiction of SMILES strings	**MIT**
sql.js (SQLite compiled to WASM)	`lib/sql-wasm.js`, `lib/sql-wasm.wasm`	Client-side SQL playground	**MIT**
PDF.js	`vendor/pdfjs/`	Opening PDF documents in the spectrum reader: every page drawn and its text layer	**Apache-2.0**

> The exact versions are those of the files included in the repository. Each permissive license requires preserving the original copyright notice, which must be kept in the library files or in a `THIRD_PARTY_NOTICES` file.

1.1 Third-party 3D models (ATOMIC PROTOCOL, first person)

File	Work	Author	Licence	Obligations
`models/mani-atomic.glb`	hands animation	hugo colauto (Sketchfab)	Sketchfab Standard	use within the work; do not redistribute the model on its own — to be checked with the author before publication

File	Work	Author	Licence	Obligations
`models/manica-atomic.glb`	Modern Scientist in a Lab Coat (crop)	restore50 (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`
`models/braccia-atomic.glb`	Fists (2023) | First Person Animations	1Matzh (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`
`models/mano-robotica-atomic.glb`	Robotic Hand	SeanNicolas (Sketchfab)	CC-BY 4.0	attribution: in the game's Codex and in `models/CREDITS.md`
`models/mappe/small.glb`	Small Lab	Biohazard Rooms (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`
`models/mappe/scifi1.glb`	Sci-fi lab	Agata Wilczek (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`
`models/mappe/flynn.glb`	Flynn's Secret Lab [Tron Legacy]	jvouillon (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`
`models/mappe/underground.glb`	Location underground laboratory	bazz4r (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`
`models/mappe/overgrown.glb`	Overgrown Laboratory	MichalTroye (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`
`models/mappe/reattore.glb`	Quantum_Laboratory_X_Alpha	-X-ScornGames (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`
`models/mappe/capsula.glb`	Sci Fi lab	IrisProcess (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`

File	Work	Author	Licence	Obligations
`models/pistola-atomic.glb`	Capsule Gun	sc8di (Sketchfab)	CC-BY 4.0	attribution and notice of changes: in the game's Codex and in `models/CREDITS.md`

The derivatives are produced by `tools/prepara-modelli.js`; the entries with their SHA-256 digests are in the SBOM (`docs/07`).

2. External services/APIs (not redistributed)

These services are queried over the network at runtime (their data is not part of the package); their use is subject to the respective terms.

Service	Use
PubChem PUG-REST (NIH/NCBI)	IUPAC/common name, CAS, formula, 3D SDF models, GHS/CLP pictograms
NIST WebBook · SDBS · MassBank · nmrshiftdb2 · SpectraBase	Links to reference experimental spectra
NASA / ESA	Astronomical images and data (Astrochemistry section)

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3. Intellectual property of the proprietary code

- BioSpecInfo's application code (UI logic, the proprietary SVG mechanism engine,

the CIP stereochemistry viewer in Canvas, the heuristic spectral predictors, the educational content, the internal databases) is the **author's original work**, except for the third-party libraries listed above.

- **No copyleft-licensed code** is embedded in the proprietary source: the author

retains full licensing freedom over their own code.

3.1 Project license

The repository includes a **proprietary — "All rights reserved"** `LICENSE` file: the code is viewable and assessable (personal study and professional/academic evaluation), but **copying, modification, redistribution and commercial use require the Author's written permission**. This choice protects the project from unauthorised reuse while keeping it presentable to recruiters and companies. Third-party components remain governed by their permissive licenses.

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4. Compliance summary

Item	Result
Viral copyleft dependencies (GPL/AGPL)	**None**

Item	Result
Dependency licenses	Permissive (MIT / BSD-3-Clause)
Obligation to keep copyright notices	Yes (in library files / NOTICES)
Licensing freedom over proprietary code	Full
Project `LICENSE` file	✅ Present — proprietary "All rights reserved"

AI Agent Architecture "Spectra" — BioSpecInfo

Field	Value
Project	BioSpecInfo — Spectra component (agentic AI copilot)
Author	Samuele Pio Provenzano
Thesis supervisor	Prof. Savino Longo — University of Bari Aldo Moro
Component	`bsi-ai-hub.js` — 6,254 lines, zero runtime dependencies
Type	Multi-provider conversational agent with client-side tool execution
Repository	`samupropio1-ship-it/BioSpecInfo-v11`
Documented version	Service Worker `bsi-v219`

1. Executive summary

Spectra is an **agent**, not a thin wrapper around a language model. The difference is measurable: the model does not answer quantitative questions from memory — it **invokes deterministic tools** that compute the result, and the control loop lets it **chain up to 10 calls**, verifying each step before answering.

The architecture addresses three problems that separate a usable agent from a demo:

1. **Grounding** — a fabricated chemical value can be dangerous. 35 tools cover

computation, public databases and the application's own datasets; the system prompt explicitly forbids quoting numeric values from memory.

1. **Transparency** — the model's reasoning is streamed live and stored

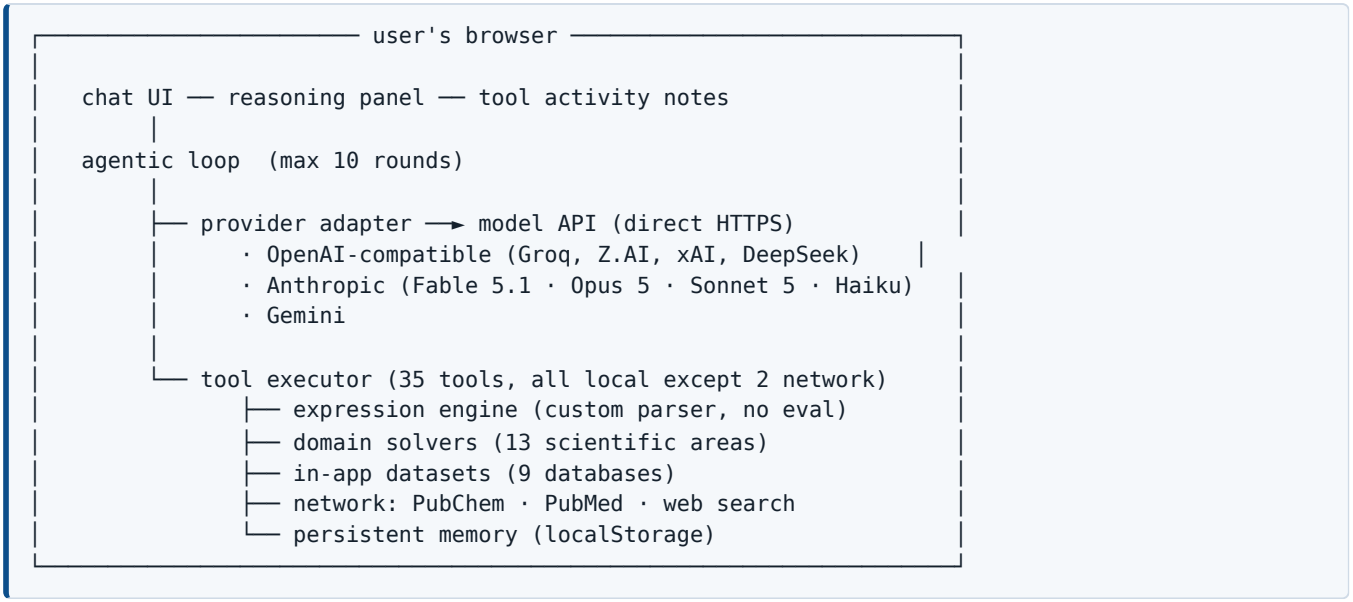
alongside the answer, together with the list of tools actually invoked. The agent's work is auditable after the fact.

1. **Provider portability** — a single tool registry is translated into the

three function-calling formats in use today (OpenAI-compatible, Anthropic, Gemini), so the agent behaves identically across ten model configurations, four of them free.

2. Architecture

2.1 Overview



There is no backend. API keys stay in the browser's `localStorage` and travel only to the selected provider: no data passes through BioSpecInfo servers, which do not exist.

2.2 Agentic loop

Each round: send the conversation history (last 40 turns) plus the system prompt and the schema of all 35 tools → receive the streamed response → if it contains tool calls, execute **all** of them → rebuild the assistant turn in the provider's native format → repeat.

Three exit conditions: no tool calls (final answer), `MAX_ROUNDS = 10` exhausted, or user interruption.

2.3 Provider-neutral tool registry

Tools are declared once in JSON Schema and translated at runtime:

Family	Required shape
Anthropic	`{name, description, input_schema}`
Gemini	`[{functionDeclarations: [...]}]`
OpenAI-compatible	`{type:"function", function:{...}}`

The same applies to messages: turns containing tool calls are serialised into each provider's native form and kept in a `_native` field, so a conversation stays coherent even when switching models.

2.4 No hard-coded model names

A model name written into the code is a time bomb: it works until the provider retires that model, and then the application stops with a 404 the user cannot fix. It has happened twice in this project:

Provider	Error	When
Google	`models/gemini-1.5-flash is not found for API version v1beta`	removed from `v1beta`
Groq	`The model llama-3.3-70b-versatile does not exist or you do not have access to it`	deprecated 17/06/2026

On some services the problem is structural: free model ids change constantly, by design. (That is why OpenRouter was later dropped from the list, see 2.4.)

The fix is not to write the new name — that would be the same bomb with a longer fuse — but to **remove the name** and ask the API, which knows what exists *for that key*. This also resolves the ambiguity in Groq's message: "does not exist" **or** "you do not have access" are different cases, and a per-key listing tells them apart.

Every configuration with `modelliCandidati` starts at `model: null` and is resolved at runtime. The two families use different strategies because their names are different:

Gemini — version scoring. Names follow a regular scheme (`gemini-<major>.<minor>-<family>`), so they can be ordered: newest version first, `flash` family preferred, with penalties for experimental, `preview` and `-lite` variants; models without `streamGenerateContent` are discarded, as are non-conversational ones (embedding, image, native audio, Live API). When Gemini 3.0 ships it will be picked automatically.

OpenAI family — candidates in preference order. Here names are not comparable to each other (`openai/gpt-oss-120b` versus `qwen/qwen3.6-27b`) and automatic scoring would choose badly. `GET /models` is queried and the **first candidate that still exists** wins: the order of the list *is* the preference, and the listing serves to skip the ones that vanished. Only if no candidate survives does a generic score run over the available models — discarding transcription, speech, embedding and guard models, and preferring the larger model all else being equal.

Three fallback levels in both cases: model listing → candidate probing → first candidate, letting the real error from the generation call speak instead of inventing one.

The choice is cached for seven days per provider, keyed to a 32-bit FNV-1a fingerprint of the API key (different keys see different catalogues; the key itself is never duplicated). If the model is retired *while* cached, the `404` — or a `400` whose text mentions the model, as some providers return — invalidates the cache, re-resolves and retries **once**: a flag prevents an infinite loop when the new model fails too.

Anthropic is the deliberate exception: its models are paid, explicitly chosen by the user, and deprecated with long notice.

The three free services, chosen one by one

Free quality is not all alike, and the difference matters: an 8-billion-parameter model does not hold up on a multi-step physical-chemistry problem. The list has been **shortened**, not extended — two services were removed because they made answers worse, not better.

Service	Models	Its role
Groq	GPT-OSS 120B, Qwen3	The workhorse: 131K context, ~30 requests/minute
Google Gemini	Gemini Flash	Up to 1M context: whole documents and PDFs
Z.AI GLM	GLM-4.7-Flash	Free **with no expiry**: the reserve still there when credits are gone

Removed. *GitHub Models*: it was here, and was removed because GitHub **retired it entirely on 30 July 2026** — playground, catalogue and inference API switched off for everyone. It had been added two days earlier on the strength of pages describing the still-live service: the lesson is that verifying *a thing exists* is not verifying *it is still alive*, and the search to run is "<name> retired / shutdown / deprecated <year>". The failure did not stop at the dropdown entry: the rank assigned to it also made it Core Mode's automatic choice.

Mistral: mid-tier quality and a free plan requiring consent to data training — not worth paying for unpublished thesis material when five alternatives carry no such clause. *OpenRouter*: small free models with ids that change constantly, unsuited to an agent chaining ten tool calls. *Cerebras*: since August 2026 there is no card-free plan, and free context is capped at 8K — Spectra sends 32 tool definitions on top of history, which does not fit.

None of the five matches a paid frontier model on the hardest problems; GitHub Models come closest, at the cost of low request ceilings. That is why the choice stays the user's rather than imposing one service.

The paid frontier

Free tiers are enough for studying; for a hard problem they are not. The selection criterion was **GPQA Diamond** — graduate-level physics, chemistry and biology questions — because it is the only benchmark measuring what this app actually asks.

Service	Why it is here
GPT-5.6	94.6% on GPQA Diamond: the highest score available today. ~$4/M input
Gemini 3 Pro	Over 1M context, and **the same key** as free Gemini. ~$2/M
DeepSeek V4	High-tier reasoning at ~$0.66/M: the best quality/price ratio
Grok 4.6	First on agentic tool calling and lowest hallucination rate — exactly the profile an agent chaining tools needs. ~$2/M
Claude (Fable 5.1, Opus 5, Sonnet 5)	One key for all three; the only ones with the Python sandbox in Core Mode

Two configurations can use **the same key**: the four Claude models are one Anthropic account, and Gemini Flash and Gemini 3 Pro one AI Studio key. `chiaveCondivisaCon` points them at the same entry, so the user pastes it once. Clearing it clears it for all twins — leaving one behind would make the key reappear on the other.

The two Gemini entries forced the scorer to be generalised: they share family, endpoint and key but want **opposite** models. The preference is now declared in configuration (`/flash/` versus `/pro/`) and outweighs the rest of the score. Without it the free configuration would have picked `gemini-3-pro` — a newer generation, therefore a higher score, but **paid**: a model the free key cannot use.

The same pass surfaced a second issue: the minor version in the name is **optional**. From generation 3 Google dropped it (`gemini-3-pro`, not `gemini-3.0-pro`), and the regex requiring the dot scored the *newest* models as generic aliases — so they lost to older ones, and the automatic upgrade described in 2.4 would never have triggered.

The provider audit, and why it must be repeated

The system repairs itself on **model names** — it queries the catalogue and picks among what exists. It does not repair itself against a **dead service**: that must be checked by hand, and GitHub Models' retirement showed what not doing so costs.

September 2026 check, provider by provider:

Provider	Result
Groq	Active. `llama-3.3-70b-versatile` **is no longer served** as of August 2026 and `llama-3.1-8b-instant` is deprecated: removed from candidates, leaving Groq's own replacements
Google Gemini	Active
NVIDIA NIM	Removed 09/2026: alive, but without CORS headers it cannot be called from a web page
Z.AI	Active, GLM-4.7-Flash and 4.5-Flash still free
OpenAI, DeepSeek, Anthropic	Active
xAI	Active, but candidates were **two generations** behind: the top is `grok-4.6` since 12/08/2026

Two traps recorded so they are not repeated. First: never put a paid model among a free configuration's candidates — true for `glm-5.x` on Z.AI as it was for `gemini-3-pro`, or the configuration picks something the free key cannot use. Second: a dead provider with a high rank does not merely break a dropdown entry, it **becomes Core Mode's automatic choice**.

When a provider does not answer

A `fetch` that fails **before** receiving a response means one of three things: no network, the service does not accept direct browser calls (CORS), or the service no longer exists. The browser, for security reasons, does not let you tell which — the error is deliberately generic.

This used to stop the turn. But if the three causes cannot be told apart, the right response is the same for all three: **move to another provider** the user holds a key for, exactly as for exhausted quota. It is the behaviour that would have masked GitHub Models' retirement entirely, instead of leaving every request stuck on a dead service.

Errors that would repeat identically everywhere stay out of the fallback — a wrong key, a malformed request: trying them on every provider and then reporting a random last error would hide the real cause.

Removing a service from the list is a breaking operation, and deserves a note: whoever had it selected has that name saved in `localStorage`, and without validation would hit `PROVIDERS[undefined]` — Spectra would no longer open. `getSavedProvider()` always validates against the registry and falls back to an existing service.

2.5 The optional proxy — Spectra without a key

The API key lives in the user's browser: with no backend there is no alternative, and it is the limitation stated openly at the end of this document. Every visitor has to obtain their own free key and paste it in.

`proxy/spectra-proxy.js` is the way out, and it sits **outside** the page: a Cloudflare Worker (free plan) holding the keys as server-side secrets. When it is configured the browser carries no key at all, and anyone opening BioSpecInfo uses Spectra without entering anything.

It is not a plain forwarder:

- **The path passes through untouched.** The route is `/<provider>/<path>` and

the rest is relayed as received: the proxy knows nothing about API shapes and keeps working when Spectra changes model or parameters.

- **Several keys per provider, with automatic takeover.** Each secret is a

comma-separated list; on `429` (quota) or `401/403` (revoked key) it moves to the next one within the same request. Not on `400` — that is a bad request and would fail identically.

- **Streaming is never buffered**: the response body is passed straight

through, otherwise answers would appear all at once at the end.

- **It is not an open relay.** Credentials arriving from the client

(`Authorization`, `x-api-key`, `?key=`) are discarded and replaced: nobody can use the proxy to push a key of their own.

- **Three brakes** against abuse: an allowed-origins list, a per-IP limit and a

daily cap.

On the Spectra side the graft is in one place. `GET /stato` reports which providers the proxy actually covers, so a model whose secret is missing is never offered as "no key needed"; `buildRequest` routes to the proxy **and omits authentication entirely**; `chiaveDaUsare()` returns a placeholder so the "missing key" guards do not block sending — a placeholder that never leaves the browser.

Both paths coexist: without `PROXY_URL` everything works as before, and for providers the proxy does not cover Spectra keeps using the local key.

2.6 Starting over

Everything Spectra remembers lives in `localStorage`: chats, history, API keys, persistent memory, review cards. A "clear history" button that removed only the conversations would not start over — reopening the app would restore the same state.

The delicate part is that keys exist in **two formats**: the current map (`bsi_api_keys`, one key per provider) and the single-key format of earlier versions (`bsi_api_key`). `getKeysMap()` migrates the latter into the former, so clearing only the map **brings the old key back** on the next access. Both must go — along with the resolved-model caches, which mean nothing without a key.

That is why the list of everything the application writes lives in one place, `DATI_CANCELLABILI`, split into five groups with a label and a description. The confirmation panel is built from it: it shows how many entries actually exist per group, disables the empty ones and requires a second explicit click. Only chats and keys are pre-selected; memory, review scheduling and the rest of the app's progress must be chosen deliberately, because they erase work unrelated to starting the conversation over.

Deliberately excluded: the PRO licence, the trial period and the device identity — data a user does not expect to lose by emptying a chat, and a licence must never be deleted by accident. The panel says so.

After deletion the in-process state is reset too — the list of providers the proxy covers, the resolved model per provider — otherwise it would stay valid until the page reloads, and Spectra would keep using a model chosen with a key that no longer exists.

2.7 Making free keys carry real load

Free tiers impose two different limits, and each needs a different answer.

The per-request ceiling. GitHub Models accepts 8,000 input tokens. Spectra's fixed cost is ~8,150 — 2,009 for the system prompt plus **6,128 for tool definitions alone** — so that service would not even start: it would fail before the user typed a word. The measurement came before the fix, and changed what the fix had to be.

The answer is not to trim history — it is to trim **the fixed cost**. Tools are paid on every turn; the conversation is the content. Of the two, the fixed cost must shrink first. `adattaAlBudget()` selects the tools **relevant** to the question instead of all 32, then shortens history with what remains.

Relevance is scored against a keyword map kept **outside** the tool registry: the words people use to *ask* for something are not the words a tool is *documented* with. The description of `astrofisica` talks about Wien and Stefan-Boltzmann; the user writes "what temperature is this nebula". Matching the question against descriptions alone picked `farmacocinetica` for an astrochemistry question — measured, not assumed. Matching is by stem (first five characters) because Italian inflects: "nebulosa" and "nebulose" must count as the same word.

Rate limits. A `429` on a free tier is almost always the *per-minute* limit, not an exhausted quota: waiting a few seconds clears it. The provider says how long in `Retry-After`, or in the error text ("*Please try again in 7.5s*"); when it says nothing, exponential backoff with a little jitter, so every open tab does not retry at the same instant. If the requested wait is absurd, it does not wait at all: better to say "quota exhausted" than to leave the user watching an hourglass for an hour.

Provider fallback. When a quota really is exhausted, the turn is redone on another service the user holds a key for. This is why several free keys together carry load that none of them carries alone.

Two non-negotiable constraints:

- It restarts from the **original** messages, not from half-finished history.

Turns containing tool calls are stored in the previous provider's native format and cannot be handed to another. The turn's work is lost; a correct answer is gained.

- Fallback uses **free services only**. Silently moving to a paid one would

spend the user's money without their say-so; if the chosen service was paid, it still gets the first attempt.

And it moves on **only** for exhausted quota. A `401` or a malformed request would repeat identically everywhere: trying all providers and then reporting a random last error would hide the real cause.

2.8 Core Mode, and saying what is happening

The mode. One switch that changes four things at once, because none of them alone would move the needle: it switches to the most capable configuration among those with a key, raises agentic rounds from 10 to 16 (a long tool chain ran out halfway), requests maximum reasoning depth

from models that expose it, and adds a system-prompt instruction to work in verified steps rather than answer fast.

The model switch is **announced in the chat**, and if the new one is paid it says so: switching silently to a metered service would spend the user's money without telling them. For the same reason it is a switch and not the default — it costs more quota, and on a paid service it costs money.

Saying what is happening. Several seconds can pass between sending and the first word of the answer: the model reasons, calls tools, waits on the network. Before, the user saw a mute pause and concluded it had frozen.

The header now carries an atomic core whose orbitals change speed and colour with state — slow and cyan at rest, fast while reasoning, amber while a tool runs — alongside a line saying the same thing in words, for anyone not watching the animation. The states hook into callbacks the agentic loop already emitted: no mechanism was added, only made visible.

Three layout defects fixed in the same pass, all found by measuring rather than looking:

- The input field shared its row with four buttons and stayed 156 px wide: the

placeholder wrapped and was **clipped**. The text now has its own row with the controls below it. The placeholder length was chosen by testing at 360 px width, not by eye.

- The title "Spectra — il copilota AI di BioSpecInfo" did not fit and truncated

mid-word. Only the name remains; the rest became the status line, which now has a purpose.

- The core's orbitals, when rotating, escaped their box and overlapped the name.

2.9 Using what the provider actually offers

Three things the Anthropic API provides that were not being used. Checked against the official reference rather than recalled — two of the three carry a constraint that would not have been guessed.

Prompt caching. The fixed cost of every request is about 8,150 tokens: 2,000 of system prompt plus 6,128 of tool definitions, identical every turn. Marked for caching they are paid once and then read back at a fraction of the price.

The constraint is that caching works by *prefix*: one differing byte at the start invalidates all of it. But Spectra's prompt also carries user memory and question context, which change every message. Concatenated into one string the cache **would never have hit**. The prompt now travels in two blocks — stable and marked, volatile and not — and the marker also goes on the last tool, because composition order is `tools` → `system` → `messages`.

The Python sandbox, and why it excludes web search. `code_execution` gives the model an environment with sympy, numpy, scipy and matplotlib: numerical integration, symbolic algebra, matrix diagonalisation — things the 32 solvers do not cover because they cannot all be anticipated.

But the new-generation web search **already carries an execution environment** (it filters results by writing code), and declaring `code_execution` as well would create a second one, confusing the model. They are alternatives, not complements. The choice follows Core Mode:

normally web search and page fetching, in Core Mode the sandbox — because when maximum power is requested the problem is computational, not documentary. A test verifies the two never appear together.

Page fetching. `web_fetch` joins search: if the user pastes an article link, Spectra reads it instead of merely searching for its title.

Sandbox results arrive as `bash_code_execution_tool_result` and `text_editor_code_execution_tool_result` — not a generic `code_execution_tool_result` — and must be kept in the assistant turn like the web-search blocks, or a server-paused turn cannot be resumed.

---

3. The 35 tools

The list is the one **measured** in `window.BSI_AI_TOOLS` on the running application, not transcribed from memory: this document declared 32 when there were already 35, and three — `analizza_molecola`, `disegna_molecola`, `mostra_spettri` — did not appear at all.

Area	Tools
General computation	`calcola`, `risolvi_equazione`, `analisi_dati`
General chemistry	`bilancia_equazione`, `stechiometria`, `massa_molecolare`, `converti_unita`, `costante_fisica`
Physical chemistry	`termodinamica`, `equilibrio_acido_base`, `cinetica`, `gas_e_soluzioni`, `elettrochimica`
Structure & spectra	`spettroscopia`, `quantistica_e_spettroscopia`, `cristallografia`, `analizza_molecola`, `disegna_molecola`, `mostra_spettri`
Life sciences	`biochimica`, `farmacocinetica`, `valuta_druglikeness`
Physics	`astrofisica`, `nucleare`, `statistica_inferenziale`
External databases	`cerca_pubchem` (NIH), `cerca_letteratura` (PubMed)
Internal data	`cerca_nel_database` (9 datasets), `cerca_molecola`
App control	`naviga_sezione` (96 sections), `apri_strumento` (12 labs), `stato_app`
Memory	`ricorda`, `ricordi`
Animations	`apri_animazione` (6 reaction mechanisms)

> **Web search is not among these thirty-five**, and the distinction matters. The thirty-five are executed **by the application**, in the user's browser. On providers that offer them, the agent adds two tools to the schema that are executed **by the provider** — `web_search` and `web_fetch`, with a per-turn usage cap. Counting them with the others would make them look always available, whereas they depend on the provider chosen.

3.1 Internal datasets exposed

297 synthesis reactions · 118 elements · 67 amino acids · 263 drugs · 63 pathologies · 46 retrosynthetic strategies · 36 drug interactions · 29 metabolic pathways · 29 redox potentials.

Drugs and strategies are the two figures that `tools/verifica-affermazioni.js` measures against the running application on every battery run: here they had been left at 143 and 39 while the app counted 178 and 46.

These are the datasets curated by the author for the application: the agent treats them as a primary source and is instructed to **report discrepancies** between the database and its own memory rather than silently smoothing them over.

---

4. Notable engineering decisions

This section documents the non-obvious choices and the reasoning behind them.

4.1 Expression engine without `eval()`

Problem. Serving arbitrary physical-chemistry computations requires an expression evaluator. `eval()` would solve it in a few lines.

Why it was rejected. The agent reads untrusted external content (PubChem results, PubMed abstracts, web pages). An injection in that content could induce the model to emit a hostile expression; with `eval()` that string would execute in the page context, **where the user's API key lives**.

Solution. Recursive-descent parser with a *closed* set of 27 functions and constants. It has no access to global scope: it can only do arithmetic. Validated against 10 escape attempts (`window.localStorage`, `constructor`, `this`, `fetch(...)`, calls to non-whitelisted functions) — all rejected, including when executed in a real browser.

4.2 Preserving reasoning blocks

On models with adaptive reasoning, `thinking` blocks are part of the assistant turn and must be **echoed back unchanged, cryptographic signature included**, on the next tool-use round. Dropping them causes the request to be rejected.

The implementation accumulates `thinking_delta` and `signature_delta` from the stream and reinserts them **first** in the reconstructed turn. Verified in integration tests: signature preserved, ordering correct.

4.3 Resuming paused turns (`pause_turn`)

Web search is a server-side tool. When its internal loop exhausts its iterations, the response ends with `stop_reason: "pause_turn"`.

The turn must be resumed by **sending the assistant turn back as-is, without adding any user message**: the server recognises the trailing `server_tool_use` block and continues from there. Appending a "continue" message would break the mechanism.

4.4 Per-model configuration, not global

The four Claude tiers do not accept the same parameters:

Model	`effort`	Web search	Notes
Fable 5.1	`xhigh`	yes (8)	refusal fallbacks; reasoning always on
Opus 5	`high`	yes (6)	default
Sonnet 5	`high`	yes (5)	
Haiku 4.5	**unsupported**	**unsupported**	sending them returns HTTP 400

On all recent models `temperature`, `top_p` and `budget_tokens` were removed from the API and their presence causes an error: none of the four sends them. Every parameter is therefore conditioned on the provider rather than set globally.

4.5 Multimodal input and material reading

The agent accepts images, PDFs and text files, several at once, treated as consecutive pages of a single document. Three constraints shaped the implementation:

- **Images are downscaled to 2000 px** and re-encoded as JPEG before sending.

A phone photo drops from roughly 1 MB to 440 KB: without this step a single image would approach the request ceiling and multiply token cost. 2000 px is the minimum for reading small handwriting — at 1600 px fine script became illegible.

- **The PDF page ceiling depends on the model**, it is not a constant: 600

pages for models with a 1M-token window, 100 for 200K ones. Page counting is done by reading `/Type /Page` objects from the file bytes, without libraries.

- **History stores a 160 px thumbnail**, not the image that was sent. Storing

the large one (440 KB each) filled `localStorage` within a dozen photos, and once the quota is exceeded every subsequent write fails silently.

Six quick actions turn the material into a study artefact (summary, concept map, outline, flashcards, exam questions, transcription) and two translate it. For transcriptions the agent is instructed to write `[illegible]` rather than guess: an invented chemical term is worse than a declared gap.

4.6 In-house graph renderer, no CDN

Concept maps are produced by the model as Mermaid diagrams and drawn by a renderer written into the application (~190 lines): longest-path layering from the roots with cycle breaking, barycentre reordering over three passes to reduce edge crossings, SVG output with Bézier curves.

Not loading Mermaid from a CDN follows from the app's offline-first nature: a map must be drawable without a network. Unsupported syntaxes are recognised and left as a readable code block instead of being drawn badly.

4.7 Voice command with wake word

User-activated continuous listening: the phrase "Hey Spectra" followed by a command sends it automatically. Two non-obvious details:

- the browser's speech recognition **stops on its own** after a few seconds of

silence and must be restarted, otherwise listening dies at the first pause;

- **all transcription alternatives** are examined, not just the first, and the

most common phonetic variants are accepted: Italian recognition renders "Spectra" in many different ways.

Denying microphone permission stops the loop instead of retrying forever. The word "spettroscopia", extremely frequent in this domain, was verified as non-triggering.

4.8 Guard against non-finite results

A systematic audit revealed that, given legitimate degenerate inputs (zero wavelength, zero volume, zero half-life, a transition between identical levels), eight solvers returned `ok: true` with `Infinity` or `NaN` among their fields.

This is the most insidious failure in this context: an exception is noticed, a formally valid but meaningless value is reported to the user as correct. The fix is a single check at the point where all results pass through: if a numeric field is not finite, the result becomes an explicit error naming the fields and stating the probable cause. It also covers tools added in the future.

4.9 Graceful degradation

- **Models without reliable function calling** (some free tiers): on the first

tool-related error the request is retried once in text-only mode instead of failing.

- **Idle timeout (45 s)**, reset on every byte received: a long answer is never

truncated, but a dead connection is reported immediately with an explicit message.

- `localStorage` **quota**: history is capped (30 conversations, 100 messages)

with progressive pruning. Without this cap, exceeding the quota made every subsequent write fail *silently*, **including saving the API key**.

- **Classifier refusals** (`stop_reason: "refusal"`, HTTP 200): detected and

surfaced, instead of appearing as an empty answer.

---

5. Verification

Verification was carried out at three levels, with recorded outcomes.

5.1 Numeric correctness — against reference values

Area	Case	Expected	Obtained
Balancing	$KMnO_4$ + HCl → …	2:16:2:2:8:5	✓ exact
Molecular mass	$K_4[Fe(CN)_6]$ (nested)	368.35	368.345
Molecular mass	$CuSO_4 \cdot 5H_2O$ (hydrate)	249.68	249.677
Equilibrium	acetic acid 0.1 M, K_a $1.8 \cdot 10^{-5}$	pH 2.874	2.875
Thermodynamics	ΔG (N_2O_4/NO_2) at 298 K	+4.79 kJ/mol	✓
Kinetics	E_a from k(300 K), k(320 K)	55.3 kJ/mol	55.33

Area	Case	Expected	Obtained
Real gases	van der Waals CO_2	−10 % deviation	−10.13 %
Spectroscopy	bromine M+2	97.3 %	✓
Biochemistry	glycylglycine	132.12 Da	✓
Pharmacokinetics	t½ from V_d 50 L, Cl 5 L/h	6.93 h	✓
Astrophysics	solar emission peak (Wien)	501.5 nm	501.52
Astrophysics	Earth escape velocity	11.19 km/s	11.186
Nuclear	^{56}Fe binding energy	8.79 MeV/nucleon	8.790
Statistics	critical t, 95 %, df = 4	2.7764	✓
Crystallography	copper density (fcc)	8.96 g/cm³	8.935

P-values are not tabulated: they are computed with the regularised incomplete beta function (Lentz continued fraction) and the incomplete gamma function.

5.2 Robustness — cases that must fail

Verified as **rejected**: unbalanceable equations, formulas with unbalanced parentheses or non-existent elements, species absent from the equation, invalid database types, and the 10 injection attempts against the expression engine.

5.3 Integration — real browser

Automated tests with headless Chromium across all 14 application pages: no JavaScript errors, no missing resources. All 96 sections of `index.html` and the 18 tabs of the Astrochemistry module were traversed one by one. Request bodies were inspected for the four Claude tiers, and a full cycle with web search, turn suspension and resumption was simulated.

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6. Quantitative summary

Metric	Value
Component size	6,254 lines
Tools	32
Model configurations	10 (3 free)
Scientific areas covered	13
Records in exposed internal datasets	over 800
Max agentic loop rounds	10
History sent to the model	40 turns
Runtime dependencies	none

---

7. Stated limitations

Documented for technical honesty; these are deliberate choices, not omissions.

- **The API key lives in the browser** — unless the proxy is deployed. In

the default configuration the key never leaves the device, but it is accessible to anyone with access to that browser, and every user must obtain their own. The Worker in section 2.5 removes this limitation by moving keys to a server, at the cost of one component to maintain outside the page: a distribution choice, not a requirement.

- **Network tools depend on service availability.** PubChem and PubMed are

queried directly from the browser; when unreachable, the tool returns an explicit error and the agent is instructed to say so rather than substituting an estimate.

- **Solvers apply simplified models** where the literature offers more refined

ones (e.g. Woodward–Fieser for UV, or ATP yields using average P/O ratios). Each tool states the method used in its own result.

- **Verification is functional and numeric**, not formal: there is no proof of

solver correctness, but a set of reference cases drawn from the teaching literature.

Companion document to the BioSpecInfo technical dossier. See also: `01-Software-Architecture-Document.md`, `02-Verification-Validation-Report.md`, `03-Security-Privacy-Compliance.md`.

Scientific Accuracy and Data Provenance — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	`bsi-v219`
Purpose	Document how the scientific data shown by the application are generated, by what method they are verified, and what the declared limits are.

> **Why this document exists.** A chemistry teaching application can be impeccable from a software point of view and wrong from a scientific one, and the user has no way of noticing: a plausible molecular weight, a convincing-looking spectrum and a wrong structure all present themselves in the same way. This document describes the checks that make that error **visible and automatic**, and declares what stays beyond their reach.

1. Principle: every scientific datum must be open to contradiction

The criterion adopted throughout the application is that a scientific datum is not considered verified because it *looks* right, but because **a second, independent datum exists that would confirm or refute it**. Where that second datum does not exist, the limit is declared instead of being filled with an estimate presented as a measurement.

Three concrete applications of this principle:

Domain	First statement	Second, independent statement	Check
Drug database	structure (SMILES)	molecular weight from the literature	the weight recomputed from the structure must agree
IR / NMR spectra	functional groups recognised	molecular structure via RDKit	groups are sought on the graph, not on the text of the SMILES
Physical constants	tabulated value	symbol and unit of measure	the lookup refuses ambiguous matches instead of picking one

2. Pharmacological database — 263 entries

2.1 The verification method

Every entry declares **two** properties of the same molecule: the structure, in SMILES notation, and the molecular weight. They are independent statements — the first describes atomic connectivity, the second is a number taken from the literature — and therefore comparable:

```
weight computed by RDKit from the structure   ↔   weight declared in the table
```

A discrepancy above **0.6 u** (a tolerance that covers rounding and small differences between atomic-mass tables, but not a real error, which is worth tens of units) indicates that at least one of the two statements is wrong.

The check is automated (`tools/verifica-farmaci.js`, runnable by anyone holding the repository) and requires no human judgement: it does not ask anyone to trust a source, it compares two numbers.

2.2 What it found

Applied to the 143 pre-existing entries, the check identified **89 errors**. The distribution is significant: the **molecular weights were correct** — taken from the literature — while the **structures were wrong**.

Drug	Structure declared
Ketamine	an unrelated pyrrolidinone
Morphine	non-matching structure (in two separate entries)
Lorazepam	missing the 3-position hydroxyl that defines its identity
Atorvastatin	missing the anilide group
Testosterone	missing the Δ^4 double bond of ring A

Plus **7 duplicated drugs** in different categories, with data that sometimes diverged between the two copies.

2.3 The discrepancy as a diagnostic instrument

The numerical value of the difference is not merely an alarm: it indicates **which** fragment is missing or superfluous.

Δ (u)	Interpretation	Real case
28.05	C_2H_4	ketamine, fentanyl, morphine
16.00	one oxygen atom	tetracycline (one hydroxyl too many)
14.02	a CH_2 group	doxycycline
2.02	a missing double bond	testosterone (Δ^4)
0.98	cytosine ⟷ uracil	**sofosbuvir**

The last case deserves a note: a discrepancy of 0.98 u corresponds exactly to replacing an −NH group with an oxygen atom, that is, to the difference between cytosine and uracil. Sofosbuvir is a **uridine** analogue, and the proposed structure contained the wrong base. The number showed where to look.

2.4 Current state and declared limits

Drugs in the database	**178** (it was 143)
With a verified structure	**157** (it was 153)
Defects	**0**
Declared and accepted deviations	**21** (it was 25)

Verification is **conformant**: no entry presents a structure that contradicts its own molecular weight. The 21 deviations are entries **without a structure**, each recorded with its own reason in `../evidence/deviazioni-note.json` and reported in the verification report.

Defect versus deviation. The check distinguishes two situations that must not be confused:

- a structure that **contradicts** its own weight is a defect, and fails

verification;

- an **absent** structure may be an acceptable deviation — but only if

explicitly recorded, with its reason. An entry without a structure and **not** recorded fails verification like any other defect.

Recording a deviation is therefore a deliberate decision, traced in git and visible in the report, not a way of silencing a check.

And it must also be **revoked** when it is no longer needed. From version `bsi-v188` the bench also fails in the opposite case: an entry listed in the registry that **does** have a verified structure is a permission left switched on for nothing, and tomorrow it would silently cover a wrong structure put in its place.

2.4-bis Four entries that left the registry

Of the six entries without a structure, four were closed by taking the structure from **ChEMBL** — a source independent of this project — and verifying it with the same structure ⟷ molecular weight comparison that applies to all the others.

Entry	ChEMBL	Formula	Weight declared	Weight computed
Digoxin	`CHEMBL1751`	$C_{41}H_{64}O_{14}$	780.94	780.95
Vincristine	`CHEMBL90555`	$C_{46}H_{56}N_4O_{10}$	824.96	824.97
Tacrolimus topical (Protopic)	`CHEMBL269732`	$C_{44}H_{69}NO_{12}$	804.02	804.03
Tacrolimus systemic (Prograf)	`CHEMBL269732`	$C_{44}H_{69}NO_{12}$	804.02	804.03

The reason recorded for digoxin said "every structure tried departs by 14-30 u from the literature weight": the declared weight was right, it was the structures tried that were wrong. That the comparison is real and not complacent can be seen by adding a single carbon to the correct structure — the bench reports **Δ 14.04** and fails.

The two that remain are not single molecules, and no SMILES notation represents them:

- **Ivermectin** is a mixture of homologues: at least 80 % B1a ($C_{48}H_{74}O_{14}$) and no

more than 20 % B1b ($C_{47}H_{72}O_{14}$). The declared weight, 875.10, is that of the B1a component alone. Showing its structure would pass off the majority homologue as the whole drug.

- **Artemether/Lumefantrine (Coartem)** is a combination of two distinct active

ingredients.

In neither case is a structure missing: there is no single one to show.

Group	Entries	Nature
D-01 — not single molecules	2	ivermectin (mixture of B1a/B1b homologues) and artemether/lumefantrine (combination of two active ingredients). It was six: the other four were closed with the structure from ChEMBL, see §2.4-bis.
D-02 — not representable in SMILES	19	monoclonal antibodies, proteins and peptides (trastuzumab, pembrolizumab, insulin, semaglutide, ciclosporin A…). The absence is correct, not a gap.

> The choice is deliberate: a wrong datum replaced by another wrong datum is no progress, and a declared absence is preferable to an unverified presence.

---

3. Spectral prediction

3.1 Structural, not textual, recognition

Functional groups are identified with **SMARTS patterns applied to the molecular graph** through RDKit (`bsi-spettri.js`), not with regular expressions applied to the SMILES string.

The difference is not stylistic. The earlier textual method had two structural defects:

1. **Exclusivity.** The checks were organised in `else if` chains, so **only one

functional group per molecule** was recognised.

1. **Blindness to structure.** A regex reads characters, not bonds.

The emblematic case is **acetylsalicylic acid** (`CC(=O)Oc1ccccc1C(=O)O`), which contains both an ester and a carboxylic acid. Textual recognition identified only the ester; the resulting IR spectrum lacked the broad O–H band at ~3000 cm^{-1} and the acid C=O at ~1690 cm^{-1} — the two bands that identify the molecule — and the ^1H NMR spectrum lacked the carboxylic proton signal at 11.6 ppm.

Graph-based recognition is **additive by construction**: a molecule may present every group it possesses.

3.2 Representation conventions

Axis conventions are not uniform across techniques, and are verified automatically (`test_assi.js`, `test_assi_canvas.js`) by extracting the tick labels from the drawing produced:

Technique	Abscissa	Direction	Ordinate
IR	wavenumber (cm^{-1})	decreasing, 4000 → 400	**% transmittance**, bands pointing down
^1H NMR	δ (ppm)	decreasing, 14 → 0	intensity, TMS reference at 0
UV-Vis	λ (nm)	increasing, 200 → 400	absorbance
MS	m/z	increasing	relative abundance

3.3 Determinism

No spectrum contains random components. Earlier versions added pseudo-random noise to the trace on every redraw, with the consequence that the same molecule produced a different spectrum each time and two spectra were not comparable. A computed profile must be **reproducible**: where a realistic baseline is needed, a deterministic function of the wavenumber is used.

Likewise, in the NMR predictor the peak heights were generated at random. In a ^1H spectrum the height **is** the quantitative information — the integration — and inventing it is worse than omitting it. Currently:

- where the number of protons is known, the height is the real integration and

the label reports it (e.g. `7.26 (4H)`);

- where the prediction is a **range** (e.g. "6.5–8.5 ppm"), the band of the range

is drawn with a mark on the typical value and a uniform height: the chart shows *where* the signal falls without asserting how intense it is.

3.4 Chemical effects modelled

Effect	Implementation
Carbonyl conjugation	a ~25 cm^{-1} lowering applied **to the individual carbonyl**, not globally
Phenolic ester	raising to ~1760 cm^{-1}
C–H intensity	proportional to the number of CH_2/CH_3 groups present
Aromatic ring substitution	"out of plane" bands 900–690 cm^{-1} distinguished for ortho/meta/para/mono
Aldehyde Fermi doublet	bands at 2820 and 2720 cm^{-1}
Amide bands I and II	1655 and 1550 cm^{-1}

Verification: acetylsalicylic acid produces an aryl ester at 1760 cm^{-1} and a conjugated acid at 1690 cm^{-1} — the two values that identify it — while acetophenone (conjugated ketone, 1690) and ethyl acetate (unconjugated ester,

1. confirm that the effect is applied selectively and not indiscriminately.

3.5 Declared limits

> The spectra produced are **predictions based on functional-group tables**, not measured spectra nor quantum-mechanical simulations. They indicate where the bands of the groups present fall, with tabulated positions and intensities. **They do not replace an experimental database** (SDBS, NIST) nor an instrumental measurement, and the chart says so.

In particular the following are not modelled: second-order spin-spin couplings, solvent and concentration effects, combination bands and overtones, solid-state polymorphism.

3-bis. Cheminformatics and QSAR models

3-bis.1 Where the methods come from

None of the methods in the **Cheminformatics** section was invented for the occasion. They are the ones in use, with the references that define them.

Method	Reference	How it is implemented here
Tanimoto coefficient	Rogers & Tanimoto, 1960	Bits in common over bits in union, with the 0/0 = 1 convention declared in the code
Morgan / ECFP fingerprints	Rogers & Hahn, *J. Chem. Inf. Model.* 50 (2010) 742	From RDKit MinimalLib, radius 2, 2048 bits
MACCS keys	Durant *et al.*, *J. Chem. Inf. Comput. Sci.* 42 (2002) 1273	From RDKit MinimalLib, 166 keys
Rule of five	Lipinski *et al.*, *Adv. Drug Deliv. Rev.* 23 (1997) 3	Over the RDKit descriptors, with the violation count
Veber filter	Veber *et al.*, *J. Med. Chem.* 45 (2002) 2615	Rotatable bonds ≤ 10, TPSA ≤ 140 Å²
QED	Bickerton *et al.*, *Nat. Chem.* 4 (2012) 90	**Recomputed**: MinimalLib exposes 43 descriptors but not QED. The desirability-function parameters are those of the paper
Butina clustering	Butina, *J. Chem. Inf. Comput. Sci.* 39 (1999) 747	Threshold on the Tanimoto distance, leaders chosen by neighbour count
Bemis–Murcko scaffolds	Bemis & Murcko, *J. Med. Chem.* 39 (1996) 2887	Ring systems plus the bonds that connect them
PAINS	Baell & Holloway, *J. Med. Chem.* 53 (2010) 2719	A subset of the SMARTS patterns, with the reason for the flag
Brenk alerts	Brenk *et al.*, *ChemMedChem* 3 (2008) 435	Likewise
SALI	Guha & Van Drie, *J. Chem. Inf. Model.* 48 (2008) 646	Δactivity / (1 − Tanimoto), over the pairs above the similarity threshold

3-bis.2 The four things that make a QSAR honest

A QSAR model is extremely easy to make look good. The four precautions below are what separate a number from a measurement, and all four are verified by the `test_cheminfo` bench.

Scaffold split. With a random split, close analogues from the same series end up on both sides: the model recovers what it has already seen. The scaffold split groups the molecules by Bemis–Murcko scaffold and assigns **whole scaffolds** to one side only. The R^2 that comes out is lower — and it is the one that survives the new molecule.

Null model by scrambling. The same model is retrained eight times on shuffled labels. If the true score does not exceed the best of the twins, the model has learnt nothing transferable, and the panel says so in those words. It is the check most often missing: without it, an R^2 of 0.4 over thirty molecules is indistinguishable from chance.

Applicability domain. A prediction on a molecule far from everything the model has seen is an extrapolation, not a prediction. The distance to the nearest training neighbour is shown next to every predicted value.

Scaffold-grouped cross-validation. A single split, over a small set, is a noisy number: on 28 molecules at 25 % the test set holds seven, and moving one shifts the R^2 by tenths. The folds group by scaffold — no scaffold sits in two folds — and the result is given as **mean ± deviation**, not as a single figure.

> **How much this matters, measured on the 28-inhibitor example.** The single split gives R^2 **0.861**. Five-fold validation gives **0.605 ± 0.452**, with the individual folds at 0.861 · 0.538 · 0.795 · **−0.150** · 0.980. The single split had drawn the lucky fold. Together the two figures say what neither says alone: the model works, but it depends appreciably on which molecules it gets for training — and on a set of twenty-eight molecules that is exactly what one should expect.

3-bis.3 Declared limits

Limit	Consequence
The example sets are **small** (28 and 40 molecules)	They exist to show the method, not to produce a usable model. On sets that size, the null model is the only serious defence — which is why it is there
The model is **linear in the kernel** (kernel ridge, Tanimoto kernel) or logistic	No neural network, no gradient boosting: on a few hundred molecules these give no demonstrable advantage, and the cost would be a model one cannot inspect
PAINS are not a verdict	A PAINS pattern signals that the compound has often been a false positive in fluorescence assays, not that it is inactive. The panel writes this next to every flag
The **PCA is on the descriptors**, not on the fingerprints	On binary fingerprints PCA is not very informative; loadings on descriptors can be read, and are shown
There is no **large-scale substructure search**	The workbench is sized for the sets one pastes in by hand, not for a library of millions of compounds

3-bis.4 Two surfaces, two roles — and a number that was false

The application has **two** places where cheminformatics happens, and they serve different purposes. The distinction is written here because an assessor who opens the wrong page first would form the wrong impression.

	`rdkit_lab.html` — "RDKit Lab"	Cheminformatics section
Who it is for	Someone studying: one molecule at a time, to see what RDKit can say about it	Someone working: a set of molecules, to derive a defensible model
Input	One SMILES	A pasted set, with the measured activity
QSAR	**Empirical** estimates from the descriptors, declared as such in the panel	A model **trained** on the supplied data, with scaffold split, cross-validation and null model
Output	What is on screen	CSV and method report

> **A number that was false, and how the bench caught it.** The lab's "Pharma Pro" panel compared the molecule against eight reference drugs and displayed a similarity percentage. That percentage did not come from a fingerprint: it came from **eight bits of thresholded descriptors** — "has aromatic rings", "HBA > 4", "weight between 200 and 500" — over which a Tanimoto was computed. Measured: **caffeine against metformin gave 0.75**, while on Morgan fingerprints it is **0.024**. Two molecules with no fragments in common, shown at 75 %, on a page called "RDKit Lab" where RDKit was already loaded.
>
> Alongside that, three of the eight reference fingerprints were **written by hand** and had the "aromatic" bit wrong (caffeine, which has two aromatic rings, morphine and amoxicillin which have one), and the **SMILES for omeprazole was not omeprazole**: `COc1ccc2[nH]c(=S)cc2c1OC` is a structure RDKit rejects.
>
> All corrected: the comparison uses the Morgan fingerprint with the shared engine's Tanimoto, the fingerprints are computed rather than typed in, and omeprazole is omeprazole (monoisotopic mass 345.11, three aromatic rings — both verified). The two functions of the fake similarity were **removed**, not bypassed: leaving them around would have put them back in use at the first edit, and the bench checks that they no longer exist.

One implementation of Tanimoto. The lab had its own, which agreed with the engine's on ordinary cases but answered 0 where the engine answers 1 (two empty fingerprints). The page now loads `bsi-cheminfo.js` and calls that: `test_cheminfo` compares the two routes over all 28 reference pairs and demands a deviation of **exactly zero**.

3-ter. Alternative text for figures is not invented

The biochemistry guide carries **98 figures**. On replacing it, the accessibility bench rejected it: none had an `alt` attribute.

The easy way out was `alt=""`, and the bench would have accepted it — the comment in its code says, correctly, that for a **decorative** image this is the right thing: it tells the screen reader to skip it. But these are biochemistry figures, that is, information. Marking them decorative would

have passed the check by telling screen-reader users to ignore 98 diagrams: a hollow green, of the same family as the "contrast 0" that did not cover the new panels (§3-bis) and the 100 % coverage that summed the wrong ranges.

The honest way was available: every image sits inside a `<figure>` with its own `<figcaption>`, and **the captions were written by the document's author**.

Figures	62, holding 98 images
Single-image figures	26 — the alt is the caption, minus the "Figura N." label
Two-panel figures	36
Pair captions split between the two panels	**33 out of 36**
Pairs sharing one caption	3 — they do not split unambiguously

All 36 pair captions distinguish the two panels with "a sinistra" (left) and "a destra" (right): where the split is unambiguous, each image receives its own half of the sentence. Where it is not, the two images share the whole caption — verbose to listen to, but **it cannot be wrong**, and a wrong description of a figure is worse than a verbose one.

> **Why not attempt a more aggressive split.** The remaining 3 could have been squeezed out with looser regular expressions. A split that goes wrong produces "this image shows X" next to an image that shows Y, and the screen reader reads it with the same confidence as the correct ones. The fallback is noisy; the error is invisible.

Measured after the change: **98 out of 98 with alternative text, none empty**, contrast **0** over 1,897 text elements.

3-quater. From prototype to working tool

The functions described in §3-bis were enough to say something defensible about a set of molecules. They were not enough to **work** with it: on day one, a cheminformatics group asks to search a substructure across the whole set, to see the SAR table, and to know what happens to activity when a chlorine is replaced by a methyl.

3-quater.1 What the library had, and what it does not

Before designing anything, MinimalLib 2025.03.4 was **probed at runtime** rather than assumed. Probing first changed the design: three of the capabilities below had gone unused because nobody knew they were there.

Present	What it is for
`SubstructLibrary`	Substructure search with a pattern fingerprint: it discards in bulk the molecules that cannot match, instead of attempting isomorphism on each
`get_rxn` · `Reaction.run_reactants`	Chemical transformations. This is what makes it possible to **cut** a bond, and from there come matched pairs and the SAR table
`generate_aligned_coords`	Aligning molecules to a common core

Present	What it is for
`get_molblock` · `get_v3Kmolblock`	**SDF** export, the format in which sets are exchanged between groups and programs

Absent	Declared consequence
`cleanup`, `neutralize`	Standardisation stops at the largest fragment and the canonical SMILES: **charges are not neutralised**
`canonical_tautomer`	Two tautomers written differently remain two distinct entries
`FragmentOnBonds`, `RWMol`	No direct molecule editing: cutting goes through a reaction
Maximum common substructure (MCS)	A SAR table's core must be **supplied**, not deduced. The panel offers eight ready cores and a free field

3-quater.2 The cut, and why two rules are needed

A reaction SMARTS breaks a bond and marks both ends with a dummy atom. Verified at runtime: `CCOc1ccccc1` gives three cuts under the fine rule (CH_3–CH_2, CH_2–O, O–aryl) and two under the coarse one; benzene and methane give none.

Rule	Cuts	When it is needed
fine (default)	Any acyclic single bond, **including terminal substituents** Cl, CH_3, OH, F	SAR table and fine transformations (Cl → CH_3)
coarse (BRICS-style)	Excludes terminal atoms	Large sets, where fine cuts are too many; produces larger variable parts ("chlorophenyl → phenyl")

> **The fine rule was born of a defect.** The first draft used only the coarse one, and the SAR table of the test series **lost the chlorine on the ring**: it found the substituent on the nitrogen and missed exactly the column that matters, because chlorine is terminal. The fine rule is a superset of the coarse one: every cut the coarse rule finds, the fine one finds too.

3-quater.3 Matched molecular pairs

Two molecules form a pair when, cutting one bond in each, they are left with the **same context** and two different variable parts. The activity difference is attributed to that substitution and to nothing else — which is why this reading beats a correlation over the whole set, which averages over molecules differing in ten ways.

Verified on a series **built with known effects**: if the analysis does not recover them, it is not usable on real data.

Transformation	Expected	Found	Concordant pairs
chlorophenyl → phenyl	−1.0	**−1.0**	2 of 2
methylphenyl → phenyl	−0.5	**−0.5**	2 of 2
methyl → chlorine	+0.5	**+0.5**	2 of 2
ethylamide → propylamide	+0.2	**+0.2**	3 of 3

On the 28 example inhibitors, taken from ChEMBL: **449 pairs, 98 transformations** seen at least twice, 199 seen once, in 133 ms. At the top, **CF_3 → SO_2NH_2 with a median Δ of −2.65** over two concordant pairs.

Two declared choices:

- the **direction is normalised** on the alphabetical order of the two parts.

Without it the same effect would appear twice, with opposite medians, and neither would carry the right pair count;

- transformations seen **fewer than twice** are excluded from the table and

counted separately. A transformation seen once with Δ = +3 is not a discovery: it is an anecdote, and putting it at the top would have decisions made on it.

> **The "concordant" column is the one that decides whether to trust it.** A median of +1.0 over three pairs one of which is −2.0 is not the same as +1.0 over three pairs all positive, and the median alone does not say so.

3-quater.4 The SAR table, and how the position is found

The hard part is not finding the substituents: it is knowing which **position** each occupies. With no MCS and no molecule editing, the solution exploits a property of the cut: **the fragment containing the core carries the dummy atom exactly at the attachment point**. Matching the core onto that fragment reveals which of its atoms the dummy hangs off, and therefore which position the substituent occupies.

This also yields a necessary and free filter: if the dummy is **not** attached to a core atom, the cut happened inside a substituent and that piece is not the whole substituent. It is discarded.

On the 28 inhibitors, with a benzamide core: **20 molecules with the core, 2 positions, 8 excluded**. The excluded ones do not appear in the table: showing them with empty cells would make them look like part of the series with substituents that were not found.

> **A naming defect, and how it showed itself.** `scaffoldMurcko()` returned its result in a field called `smiles`, but that value **is not a SMILES**: it is a canonical fingerprint of the scaffold, built so that two molecules with the same scaffold produce the same string. It serves to **group** — folds, scaffold split — not to query. The SAR panel took it for a SMILES and proposed `6,6,6,6,7,…|0-13:1,…` as the core; decomposition answered "core not interpretable".
>
> The field is now called `chiave` (key) and declares `eUnoSmiles: false`; `smiles` remains as an alias. The bench pins that equal scaffolds give the same key, that different scaffolds give different keys, and that the key **does not read back as a SMILES**. A field's wrong name is a defect like any other.

3-quater.5 The rigour an assessor demands

Five things a modelling group puts in every report, and whose absence is the first objection in a meeting.

	Method	Why
Model comparison	Trivial reference (predict the mean), kNN over Tanimoto, kernel ridge — all on the **same folds**, grouped by scaffold	Comparing a model evaluated on one split with another evaluated on a different split is not a comparison. And the trivial reference sits at the top of the list, because it is the number every other must beat
Nested validation	Hyperparameters (k, lambda) are chosen on folds **internal to the training set**	Trying ten of them on the test set and reporting the best inflates the score: that number is not a prediction, it is the maximum of ten attempts
Conformal intervals	Split-conformal, quantile $\lceil (n+1)(1-\alpha) \rceil$ of the absolute residuals on a calibration portion split **by scaffold**	"7.2" and "7.2 ± 0.3" are two different pieces of information, and only the second says whether synthesis is worth it. Calibrating on analogues of the training set would give intervals too narrow precisely on new molecules
Enrichment	EF at 1 %, 5 %, 10 % with the **maximum achievable** alongside; BEDROC (Truchon & Bayly, *J. Chem. Inf. Model.* 47 (2007) 488)	In a screen one does not look at the whole list. A ROC-AUC of 0.80 can hide a top-of-list with not a single active, because AUC also rewards the ordering in the tail, which nobody will buy
Learning curve	Score against training-set size, with the test set **whole** at every point	It answers the only question that matters when the model is mediocre: more chemistry or more data?

The comparison's verdict is proven in both directions, like the null model. On a real signal — activity proportional to halogen count, over ten distinct cores — kernel ridge reaches R^2 0.943 with a margin of 0.943 against a dispersion of 0.035, and the verdict is *beats it*. On the **same molecules with random labels** the best stops at −0.265, margin −0.034, and the verdict is *does not beat it*. A guard that never fires and one that always fires are the same defect.

Enrichment is verified on the three cases where the right value is computable by hand:

Ordering (2 actives of 10)	AUC	EF@20 %	BEDROC
Perfect: both actives on top	1	**5** — and it is the maximum achievable	**1**
Worst: both actives last	0	0	**0**, not a negative
Actives at ranks 2 and 5	**0.75** = 12/16	**2.5** = 1/0.4	0.119

The worst case is the one that counts: without it one would not know whether BEDROC reaches zero or goes below, and a negative BEDROC is the sign that the formula's additive term is missing.

3-quater.6 Limits of this block

Limit	Consequence
Pairs grow with the **square** of the molecule count	Measured: 6 molecules 45 ms, 40 molecules 227 ms with 1,255 pairs. Beyond 300 molecules the panel refuses to start and says so, instead of freezing the page
The cut is **single**	A pair differing by two substitutions at once is not found. This is the limit of the single-cut scheme, shared with the standard tools
The same transformation appears in **several forms**	Cuts at different positions give different contexts for the same chemical change (`*CC>>*CCC`, `*NCC>>*NCCC`, `*C(=O)NCC>>*C(=O)NCCC`). This is expected redundancy in the single-cut scheme, not an error

Limit	Consequence
Charges are **not** neutralised and tautomers not canonicalised	MinimalLib does not allow it (§3-quater.1). A set mixing neutral and ionic forms of the same molecule counts them as distinct entries
The SAR table's core must be **supplied**	No MCS available. The panel proposes nothing automatically, because a wrong proposal is worse than no proposal

3.6 Per-atom NMR prediction: the source and the measured deviation

The $^{1}H/^{13}C$ predictor that assigns every signal to an ATOM (`bsi-nmr.js`) contains no numbers. The estimation tables live in **`bsi-pretsch.js`**, transcribed line by line from:

> E. Pretsch, P. Bühlmann, M. Badertscher, *Structure Determination of Organic Compounds — Tables of Spectral Data*, 4th ed., Springer. §4.1 pp. 82-84 · §4.5 pp. 100-102 · §5.1 p. 170 · §5.2 pp. 178-179 · §5.3 p. 182 · §5.5 pp. 188-189

They sit in a separate file so they can be **checked against the printed page** without reading the code that uses them, and so a wrong line is fixed in one place only.

scheme	formula	rows
monosubstituted benzenes, ^{13}C	$\delta = 128.5 + \Sigma Zi$	91
monosubstituted benzenes, ^{1}H	$\delta = 7.34 + \Sigma Zi$	66
substituted ethylenes, ^{1}H	$\delta = 5.25 + Zgem + Zcis + Ztrans$	42
substituted alkanes, ^{1}H	$\delta = base(CH_3/CH_2/CH) + \Sigma Z\alpha + \Sigma Z\beta$	31
terminal alkynes, ^{1}H	direct value per substituent	30
aliphatics, ^{13}C	$\delta = -2.3 + \Sigma Zi + \Sigma Sj$	24
steric corrections Sj	by degree of the observed C and of the α atom	4×4

The deviation is measured, on molecules that did not choose the parameters. The test molecules are split into two sets: *tuning*, used to choose values and rules, and *validation*, never used for that. The figure declared in the application's panel is the validation one.

	measured mean deviation
^{13}C, tuning (9 molecules)	0.71 ppm
^{13}C, validation (22 molecules)	**0.92 ppm**
^{13}C, worst case	4.9 ppm (cyclohexanone)
^{1}H (14 molecules)	0.06 ppm
^{1}H, aromatics only	0.03 ppm

The bench (`test_nmr`, 56 checks) also requires validation to stay **worse** than tuning: were they to become equal it would mean a molecule had been moved from one set to the other, and the figure would no longer say anything.

What is still wrong has not been corrected after the fact. Biphenyl misses its ipso carbon by 4.6 ppm, because the table's phenyl increment ($Z_1 = 8.1$) does not describe it. That number **has not been adjusted**: changing a transcribed value because a validation molecule does not fit is

exactly how the 0.92 ppm declared above would become a lie.

Declared limits. This is not a quantum calculation: it does not predict solvent effects, does not distinguish conformers (the source's conformational corrections K are transcribed but evaluate to 0, because conformation cannot be derived from a SMILES) and does no two-dimensional NMR. Coupling constants come from a table of **typical** values by geometric relationship, not calculated. **Substituted heteroaromatics** have, in this source, no positional increment table as benzenes do: they remain the module's least reliable prediction, and the elucidation tool declares it by lowering the confidence when it meets one.

3.7 Two-dimensional NMR: COSY, HSQC, HMBC

`bsi-nmr2d.js` holds no shifts: it asks the per-atom predictor for them and builds the correlations by **walking the graph**. This is why it could only be done once the Pretsch tables arrived: it needs the shift of *every* proton and *every* carbon, plus the index of the atom carrying them.

map	topological rule
HSQC	proton and carbon **one** bond apart (^1J); inverted sign on CH_2
COSY	two protons **three** bonds apart (H–C–C–H), excluding equivalents
HMBC	proton and carbons **two or three** bonds away (^2J, ^3J)

Exchangeable hydrogens (O–H, N–H) give no off-diagonal spots: they exchange with the solvent too fast for the coupling to be seen. They do appear on the COSY diagonal, because they are protons like any other.

Verified by counting paths (`test_nmr2d`, 40 checks), in both directions: hexafluorobenzene gives no HSQC spots, benzene gives no off-diagonal COSY spots, methane gives none in HMBC.

Declared limits. This is not a simulation of the experiment: no computed intensities, no artefacts, no residual coupling, no mixing-time dependence. In a real HMBC some two-bond correlations are not seen; here they all are. The shifts inherit the predictor's measured uncertainty (0.9 ppm on ^{13}C, 0.06 on ^1H).

3.8 3D geometry: built from the graph, not downloaded

`bsi-geom3d.js` builds coordinates by **distance geometry**: a bounds matrix from bonds, valence angles, ring diagonals and van der Waals contacts; initial positions random but **reproducible**, because the generator is seeded from the SMILES; iterative correction of the violations. Aromatic systems are projected onto their plane and linear triples are straightened by rotating the branches as rigid bodies, because those two things a distance alone cannot impose.

The indices are the NMR predictor's, and hydrogens are appended at the end: this is the condition without which the peak↔atom↔3D link would light up the wrong atom. The PubChem-fed 3D viewer could not be used for this — an SDF's numbering bears no relation to RDKit's.

Measured against known values (`test_geom3d`, 29 checks): C–C 1.54 Å, C=C 1.34, C≡C 1.20, aromatic 1.39, C–H 1.09; tetrahedral angle 109.5°, aromatic 120.0°, alkyne 180.0°. **In both directions**: benzene planar (0.00 Å), cyclohexane **not** planar (0.34). On real molecules the residual stays under 0.03 Å up to about thirty atoms and reaches 0.31 Å on a 76-atom molecule.

Declared limits. It is not a force field and does not minimise an energy: it minimises geometric violations. It does not choose the most stable conformer and does not handle stereochemistry — an R centre and its enantiomer come out the same.

3.9 From document to working: what is deduced and what is read

`bsi-documento.js` opens the file; `bsi-quesito.js` recognises the data in the text; `bsi-elucida.js` works through them. They are three separate steps **on purpose**, because they fail in different ways and conflating them hides the error: if recognition reads "1715" as an IR band when it was a mass, the working that follows is impeccable and the conclusion is wrong. This is why the panel shows **the table of data read above the working**, each entry with where it came from.

Recognition is by section, not by number. An IR band at 1738 and a mass at 150 are both numbers: telling them apart from the value alone is impossible. The technique's label is found (IR, MS, ^{1}H NMR, ^{13}C NMR) and what follows is read, up to the end of the paragraph or the next label.

What it does not do. There is no language model: there are regular expressions and a spectroscopic rule engine. If the text writes the data in an unforeseen form, the data are **not** read — and the module says so rather than working through half of them. The working goes as far as the rules go: compatible functional groups, fragments, counts. **The final structure is not proposed**, because proposing it would mean guessing; a structure can be written and the program **compares** it with the data, saying which signals fit and which do not.

There is no optical character recognition. A scanned page contains pixels, not letters: the text cannot be read, and the module declares it rather than returning an empty string as if the document were empty. The pages stay visible, and that is where pixel trace extraction takes over.

3.10 The two places where the predictor gives way, measured

Declaring "mean deviation 1.0 ppm" and stopping there hides what one actually needs to know: *where* it is wrong. There are two places, and they are known by measurement.

a) Substituted heteroaromatics — the table is missing, not the method

For benzenes the source gives positional substituent increments; for furan, thiophene, pyrrole and pyridine it **does not**. A substituted heteroaromatic therefore gets the values of the *un*substituted parent, and its β carbons all come out equal when in reality they are not.

Measured on the 23/04/2024 problem (structure `O=CC(=Cc1ccco1)CSC`):

observed	predicted	deviation	which carbon
192.04	192.0	0.04	CHO
150.77	150.0	0.77	furan α-C, substituted
146.05	142.7	**3.35**	furan α-C
135.55	135.0	0.55	=C
135.05	135.0	0.05	=C
118.55	109.6	**8.95**	furan β-C
112.97	109.6	**3.37**	furan β-C
26.67	34.0	**7.33**	CH_2 between C=C and S

observed	predicted	deviation	which carbon
15.72	15.6	0.12	S–CH_3

The two β carbons come out **identical** (109.6) because without positional increments nothing distinguishes them. Transferring the benzene increments to the heteroaromatic ring **was tried**: on the *ipso* carbon it works (151.6 against 150.77 measured), but on the β carbons it makes things **worse** — C3 would go from 3.4 to 5.7 ppm of deviation. It was not done: a generalisation that gets worse exactly where the problem is, is not a generalisation.

b) An sp³ carbon with TWO α substituents

The increments are derived from **mono**-substituted compounds. Adding two of them on the same carbon counts twice an effect that in fact saturates, and the prediction comes out high. Measured twice, same magnitude and same direction:

molecule	carbon	observed	predicted	deviation
benzyl acetate	OCH_2 (ester O + ring)	66.3	73.8	**7.5**
the problem above	CH_2 (C=C + S)	26.7	34.0	**7.3**

It is the module's declared worst case, and benzyl acetate entered the validation set precisely to keep it measured.

The comparison tool says so. When a proposed structure contains such a carbon, confidence drops from "high" to "medium" and the reason appears: *a signal that does not fit on that carbon may be a limit of the predictor, not evidence against the structure*. Without that warning, a low score on a **correct** structure would read as a refutation — which is exactly what happened to benzyl acetate, which scores 42 out of 100 while being the answer.

3.11 What could be consulted, and what could not

A table of increments for substituted heteroaromatics would close the weak point described above. **From the machine this project is built on, no primary source is reachable**, and that was verified, not assumed:

route	outcome
`curl` to any domain	the network policy allows **only** package registries (npm, PyPI, crates, Go) — the proxy says so itself
NIST WebBook, SDBS (AIST), PubChem, SpectraBase	**403**, blocked
CSIRO, RSC, ACS, Springer, Wikipedia, LibreTexts	**403**, blocked
the Hans Reich tables (`organicchemistrydata.org`, `chem.wisc.edu`)	**403**, blocked
NMRShiftDB2 (`nmrshiftdb.nmr.uni-koeln.de`)	**403**, blocked
PubMed (through its service)	no results: it indexes biomedicine, not pure chemistry

route	outcome
npm and PyPI	no package redistributes a reference NMR database: `nmr-predictor` **downloads** it from GitHub, which is blocked; `nmr-processing` does not contain it

What text search could establish is *where* the data are, and that is a useful result even without having them in hand:

> M. T. W. Hearn, "Carbon-13 chemical shifts in some substituted furans and thiophens", *Australian Journal of Chemistry* **29**(1), 107–113 (1976). DOI 10.1071/CH9760107

The abstract states that "the substituent effects in these heterocycles **resemble** those reported for substituted benzenes". *Resemble* is not *equal*: transferring the benzene increments was tried (§3.10 a) and on the β carbons it makes things worse. The real table is still needed.

Why it was not written from memory. It would have been easy: those numbers are well known. But a value nobody can trace to a source is exactly what this document exists to prevent, and putting one in a table next to ninety-one rows transcribed from a printed page would make those less credible too. **A declared hole is worth more than a badly patched one.**

One validation row with weaker provenance

For honesty this too must be said: benzyl acetate, which entered the validation set because it shows the two-α-substituent weak point, carries values **not verified against a primary source** from this environment. The phenomenon it shows is independently confirmed by the 23/04/2024 exam problem — a datum with real provenance — but the **declared worst case (7.5 ppm) rests on that row**, and the bench says so in plain words.

---

4. Physical constants and tabulated data

The lookup of physical constants proceeds by decreasing specificity (exact match → symbol → significant words → substring of at least 4 characters) and **collects every candidate**: if more than one remains, the request is refused with the list of possibilities rather than arbitrarily returning the first. A wrong physical value returned with confidence is worse than a request for clarification.

The exam molecules (`MOLECOLE_ESAME`) are tabulated with verified values; the recognition of groups from SMILES uses a `consuma` field, so that each group removes the portion of structure it has explained and no overlapping counts are produced.

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5. Automation of the verification

All the checks described are runnable and repeatable. At the time of writing the battery includes, among others:

Bench	Subject	Checks
`audit_farmaci`	structure ⟷ weight, duplicates, formula rebuilt from the graph against ChEMBL, deviations recorded with their reason	263 entries
`test_spettri`	group recognition on reference molecules	36
`test_assi` / `test_assi_canvas`	axis conventions (SVG and canvas)	19
`test_costanti`	physical-constant lookup and refusal of ambiguities	45
`audit_dati`	consistency of the tabulated data	29
`verifica_guida`	correspondence between what the documentation promises and what the code does	71

The last bench deserves a mention: it verifies that the statements contained in the user documentation correspond to the real behaviour of the code. A promise the documentation does not keep is a defect on a par with a computational error, and is treated as such.

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6. Transparency statement

This document describes checks that are **actually implemented and runnable**, with the results actually obtained and the limits of the predictors. At version `bsi-v188` the residual errors on the drug database are **zero**: the 21 deviations that remain are entries without a structure, each with its own recorded reason, not errors passed over in silence.

Where a datum is not verifiable with the instruments available, that is declared rather than presented as verified.

Document updated to version `bsi-v219`.

Software Bill of Materials (SBOM)

BioSpecInfo · document generated by `tools/genera-sbom.js`.

Application version	`bsi-v219`
Commit	`f39b42c40e242c612ebe7b49de358472e86de9dc`
Generated (UTC)	`2026-10-16T09:20:55.159Z`
Machine-readable format	`../evidence/sbom.cdx.json` — CycloneDX 1.5

> The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) _{RDKit / Greg Landrum}	`RDKit_minimal.js`	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) _{RDKit / Greg Landrum}	`RDKit_minimal.wasm`	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js _{mrdoob e collaboratori}	`three.min.js`	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass _{mrdoob e collaboratori}	`three_bloom.js`	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader _{mrdoob e collaboratori}	`gltf_loader.js`	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js _{University of Pittsburgh / David Koes}	`3Dmol-min.js`	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).

Component	Version/File	Licence	Size	Role
SmilesDrawer _{Daniel Probst}	`smiles-drawer.min.js`	MIT	246 kB	2D structure drawing from SMILES.
PDF.js (build legacy) _{Mozilla Foundation}	`vendor/pdfjs/pdf.min.mjs`	Apache-2.0	511 kB	Opening PDF documents in the spectrum reader: every page drawn and its text layer. It is the legacy build, because the default one uses very recent language features and breaks on browsers that do not have them yet.
PDF.js — worker _{Mozilla Foundation}	`vendor/pdfjs/pdf.worker.min.mjs`	Apache-2.0	1290 kB	The PDF decoding work, off the main thread: a complex page does not freeze the interface.
sql.js (JS glue) _{sql.js / SQLite}	`lib/sql-wasm.js`	MIT	45 kB	In-browser SQLite database for the data exercises.
SQLite (WebAssembly) _{SQLite Consortium}	`lib/sql-wasm.wasm`	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) _{CERN Open Data Portal}	`lib/dimuon.js`	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.
**Fists (2023)	First Person Animations (adattato)** _{1Matzh (Sketchfab)}	`models/braccia-atomic.glb`	CC-BY-4.0	6530 kB
Robotic Hand (modello 3D) _{SeanNicolas (Sketchfab)}	`models/mano-robotica-atomic.glb`	CC-BY-4.0	838 kB	The robotic left hand, over which the element power lights up.
Small Lab (adattato) _{Biohazard Rooms (Sketchfab)}	`models/mappe/small.glb`	CC-BY-4.0	1506 kB	ATOMIC PROTOCOL — laboratory (wings 1, 6).

Component	Version/File	Licence	Size	Role
Sci-fi lab (adattato) _{Agata Wilczek (Sketchfab)}	`models/mappe/scifi1.glb`	CC-BY-4.0	6683 kB	ATOMIC PROTOCOL — laboratory (wings 2, 7).
Flynn's Secret Lab [Tron Legacy] (adattato) _{jvouillon (Sketchfab)}	`models/mappe/flynn.glb`	CC-BY-4.0	2319 kB	ATOMIC PROTOCOL — laboratory (wings 3, 8).
Location underground laboratory (adattato) _{bazz4r (Sketchfab)}	`models/mappe/underground.glb`	CC-BY-4.0	10512 kB	ATOMIC PROTOCOL — laboratory (wings 4, 9).
Overgrown Laboratory (adattato) _{MichalTroye (Sketchfab)}	`models/mappe/overgrown.glb`	CC-BY-4.0	6227 kB	ATOMIC PROTOCOL — laboratory (wings 5, 10).
Quantum_Laboratory_X_Alpha (adattato) _{-X-ScornGames (Sketchfab)}	`models/mappe/reattore.glb`	CC-BY-4.0	3269 kB	ATOMIC PROTOCOL — apparatus (the reactor).
Sci Fi lab (adattato) _{IrisProcess (Sketchfab)}	`models/mappe/capsula.glb`	CC-BY-4.0	6548 kB	ATOMIC PROTOCOL — apparatus (the capsule).
hands animation (modello 3D, adattato) _{hugo colauto (Sketchfab)}	`models/mani-atomic.glb`	LicenseRef-Sketchfab-Standard	2059 kB	The first-person hands, with the skeleton and the animation the game derives its poses from.
Modern Scientist in a Lab Coat (ritaglio) _{restore50 (Sketchfab)}	`models/manica-atomic.glb`	CC-BY-4.0	275 kB	The first-person lab-coat sleeve, cropped from the scan.
Capsule Gun (modello 3D) _{sc8di (Sketchfab)}	`models/pistola-atomic.glb`	CC-BY-4.0	1366 kB	The capsule gun held in first person.

Digests

File	SHA-256
`RDKit_minimal.js`	`58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9`
`RDKit_minimal.wasm`	`e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8`
`three.min.js`	`9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2`
`three_bloom.js`	`42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090`
`gltf_loader.js`	`5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6`

File	SHA-256
3Dmol-min.js	06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff
smiles-drawer.min.js	955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2
vendor/pdfjs/pdf.min.mjs	bccc24ea711db8e44503629519904a5292d73b9daaa214bbe7cdcc282b0f4259
vendor/pdfjs/pdf.worker.min.mjs	145d2dd3ab0c86151011dba95acfa2d5336e2accd59388ea43dbee0efddaaec6
lib/sql-wasm.js	77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f
lib/sql-wasm.wasm	438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed
lib/dimuon.js	61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d
models/braccia-atomic.glb	3e72b0067366ba4906e8cb3d9077e3b800656705bfcdef5b0bb41339c0b765ff
models/mano-robotica-atomic.glb	3cadfde4777f5f609fb9fea7990f35ddc8f82d19fad4c60b8fc18a563ba7e5c6
models/mappe/small.glb	932e21a12c772ef3c36290fac8974a30b37b62e2000f7f1f26cb812a0572a7ab
models/mappe/scifi1.glb	f3e4434820cda4a2ebec5804749cd0a6278aa0cda6a19b74ec14623affb3dc5e
models/mappe/flynn.glb	d9c32659b5a75db09e88e97aeef645f46d30caee24522fc3a18b5048fa56911a
models/mappe/underground.glb	8cae8085cae16d6b496fe93a00eef7d687349492e3d74c7af1aaab407085fa75
models/mappe/overgrown.glb	c521f06e1da35eda66d585a0e6ab5b87a4da504d4b4fb09048518bf11683f403
models/mappe/reattore.glb	58f200ac13b072d40ee45bdd6c725ddd06591502e53825d60699862fc79a3e01
models/mappe/capsula.glb	be1ae03a8aa0c8cd160d517620985f83a515a2f05fc3d062aab4303496befb02
models/mani-atomic.glb	d27b4ea1689e677e9406944abaecd95d524ca7693b2e4b62aec6c906f46ece56
models/manica-atomic.glb	0034162e3fc772d3afb11c2df1a66222ae681ad7f404bcec59977d6557f14b98
models/pistola-atomic.glb	4e109a4ab7e2794b139a5c9414151b376b7a237baab5d9f183f98d8e9ac65f7f

2. Code written for this project

File	Size	Role
index.html	4671 kB	Main application: interface, teaching sections, chemical data.
bsi-ai-hub.js	422 kB	«Spectra» AI agent: agentic loop, tools, provider management.
bsi-spettri.js	35 kB	IR/NMR spectral prediction engine over the molecular graph.
bsi-cheminfo.js	125 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
bsi-pretsch.js	37 kB	The NMR estimation tables transcribed in full from Pretsch–Bühlmann–Badertscher, 4th ed.: ^{13}C and ^{1}H benzenes, ethylenes, alkanes, alkynes, ^{13}C aliphatics, steric corrections, ^{1}J(C,H). Numbers only, no code.
bsi-nmr.js	57 kB	Per-atom ^{1}H and ^{13}C NMR prediction: applies the bsi-pretsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.
bsi-geom3d.js	46 kB	3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link.

File	Size	Role
`bsi-nmr2d.js`	21 kB	Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions.
`bsi-moldraw.js`	44 kB	Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports.
`bsi-documento.js`	23 kB	Opening of any document — PDF, images, text, Word, OpenDocument, PowerPoint, Excel — with an explicit list of what could NOT be read.
`bsi-quesito.js`	19 kB	Recognition of spectroscopic data in a problem's text, with each datum's provenance, and step-by-step working through bsi-elucida.js.
`bsi-spettrolettore.js`	104 kB	Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses.
`bsi-mol3d.js`	42 kB	3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.
`bsi-lingue.js`	163 kB	Internationalisation engine: fourteen languages, writing direction included.
`bsi-molingue.js`	54 kB	Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.
`bsi-pannelli-lingua.js`	66 kB	Panels for languages and chemical languages.
`sw.js`	6 kB	Service Worker: offline operation and network strategy.
`rdkit_lab.html`	260 kB	Cheminformatics laboratory.
`astro.html`	2660 kB	Astrochemistry module.
`chimorga.html`	194 kB	Organic chemistry module.
`proxy/spectra-proxy.js`	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
`index.html`	`3d93251f7fb81768e8872911364e14e5861566ba8c9b03cd513258df784cbe4b`
`bsi-ai-hub.js`	`985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476`
`bsi-spettri.js`	`be757f15a70f884e3374d401596e2dfd18bcc852cafbb4584920e2b2d24eb3c9`
`bsi-cheminfo.js`	`59a3111b3b9436add5376fde1be6f90ded3dc08c095cebdc04da50c7f8268bbc`
`bsi-pretsch.js`	`27f727240363579fba3dba50ab3b567fbcdb26a8a9a06c0e43968b195c479db3`
`bsi-nmr.js`	`881917dbb141dfcbab9a25b9c518eaa4a8c1c52dc1020cbac90985a1cf859b72`
`bsi-geom3d.js`	`a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054`
`bsi-nmr2d.js`	`a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78`
`bsi-moldraw.js`	`45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e`
`bsi-documento.js`	`0e9ad6043a396652549b30879764cc401f8b7bd25689b5bb5f37cf26f7d7d8d9`

File	SHA-256
`bsi-quesito.js`	`6f686abcfdc3959d8c865a3e6ee7129224de52992ab1ba3da07372a986769cdf`
`bsi-spettrolettore.js`	`14daf41305b0a7271c6ff5f7c9078a4e5f0e0149f32460a9608ed1996f810256`
`bsi-mol3d.js`	`93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893`
`bsi-lingue.js`	`5a28570f930be99f9e07c46ed354dab3392805820209adb1eb38e28f2d20f410`
`bsi-molingue.js`	`e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb`
`bsi-pannelli-lingua.js`	`4af94d70ef104cf21be18ff7fbc7a2248c56e376e2ae48e8cac83878f4554f93`
`sw.js`	`469c7fa90d6a603b3b8f50e83d8bfa82033808d2c0776adf1e2303c1c5a550b8`
`rdkit_lab.html`	`066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad`
`astro.html`	`3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d`
`chimorga.html`	`047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d`
`proxy/spectra-proxy.js`	`808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962`

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
`playwright-core`	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

> **The application's licence is distinct from those of its components.** The code written for BioSpecInfo is **proprietary** — *All rights reserved* (see `LICENSE`): readable and open to assessment, but copying, reuse and commercial use require the Author's written permission. The permissive licences listed above cover **only** the bundled third-party libraries, and do not extend to the application.

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Traceability Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	`bsi-v219`
Purpose	Connect every declared requirement to the implementation that realises it and to the bench that verifies it.

> **How to read this matrix.** Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.
>
> The outcomes of the benches cited here are reported in full, with their complete output, in `../evidence/RAPPORTO-VERIFICA.md`, which is generated automatically.

1. Scientific requirements

ID	Requirement	Implementation	Verification bench
SCI-01	A drug structure must come from an external source and match that source, not merely weigh correctly	`FARM_DATA` in `index.html` — **263 entries**, 36 declaring a ChEMBL identifier and formula	`audit_farmaci` — the formula is **rebuilt by counting the graph's atoms** and compared with ChEMBL's: 36 of 36. Weight alone was not enough: three pre-existing entries (lenalidomide, palbociclib, aripiprazole) were **isomers** — same weight, different molecule — and were replaced. Two ratchets: verified provenance ≥ 36, readable structures ≥ 240
SCI-02	Functional groups must be recognised on the structure, not on the text of the SMILES	`bsi-spettri.js` §1, SMARTS patterns via RDKit	`test_spettri` — 8 reference molecules, additive recognition
SCI-03	A molecule with several functional groups must show them all	`bandeDaGruppi()`, additive construction	`test_spettri` §1 — acetylsalicylic acid: ester **and** acid

ID	Requirement	Implementation	Verification bench
SCI-04	Spectra must respect the conventions of their technique	`svgIR()`, `makeNMRsvg()`, `drawUVSpectrum()`, `drawMSSpectrum()`	`test_assi` (SVG), `test_assi_canvas` (canvas)
SCI-05	A computed spectrum must be reproducible	every `Math.random()` removed from the traces	`test_assi` §3, `test_spettri` §8 — byte-for-byte comparison
SCI-06	The height of ¹H peaks must represent the integration, or not be asserted at all	`makeNMRsvg()`; `_nmrSvg()` draws bands of uniform height	`test_assi` §4
SCI-07	Conjugation must act on the individual carbonyl	SMARTS `acidoConiug`, `estereConiug`, `chetoneConiug`, `aldeideConiug`	`test_spettri` §1 — aspirin 1760/1690; acetophenone 1690
SCI-08	An ambiguous physical constant must not be resolved arbitrarily	`costante_fisica` in `bsi-ai-hub.js` — collects every candidate	`test_costanti` — 45 checks, including the refusals for ambiguity
SCI-09	The limits of the predictors must be declared to the user	chart headers; `docs/06` §3.5	`verifica_guida` — consistency between promises and code
SCI-10	Molecular symmetry must determine the selection rules correctly	`ssimm` section in `index.html`	`test_simmetria` — 26 checks: normal modes, centre of inversion, mutual exclusion
SCI-12	A chromatogram must follow from the parameters, not be drawn by hand	`scroma` section: σ = t_R/√N, R_s from distance and widths	`audit_stabilita` (opening and drawing), inspection of the values against the definitions
SCI-13	A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads	`staratura` section: ordinary least squares, F-test line⟷parabola, deleted residual	Verified on the three example sets included: linear (F = 0.3), non-linear (F = 186.6 > 7.71), with an outlier (1 detected)
SCI-14	A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set	`bsi-cheminfo.js` §9 — `divisionePerScaffold()` alongside `divisioneCasuale()`	`test_cheminfo` — the scaffold split leaves no scaffold shared between training and test

ID	Requirement	Implementation	Verification bench
SCI-15	A model score must be compared against chance, not presented on its own	`bsi-cheminfo.js` §9 — retraining on scrambled labels, eight repetitions	`test_cheminfo` — verified **in both directions**: on a learnable signal R^2 0.394 beats all eight twins (margin 0.287); on random labels R^2 −0.199 does **not** beat them
SCI-16	Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split	`bsi-cheminfo.js` §1 — deduplication on the canonical SMILES after the largest fragment	`test_cheminfo` — `OC(=O)C` and `CC(=O)O` collapse into one entry; salts lose the counterion
SCI-17	Similarity between molecules must be computed, not estimated	`bsi-cheminfo.js` §3 — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints	`test_cheminfo` — compared against hand-computable values, including the 0/0 = 1 convention
SCI-18	A validation on a single split, over a small set, is a noisy number: it must come with its dispersion	`bsi-cheminfo.js` §9-bis — `validazioneIncrociata()` over **scaffold-grouped** folds	`test_cheminfo` — the folds cover every molecule once, **no scaffold sits in two folds**, and the same seed gives the same folds
SCI-19	A result that changes on every run is verifiable by nobody	seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report	`test_cheminfo` — two calls with the same seed give identical folds; the method report declares the seed
SCI-20	Results must be able to leave the tool, and the method must leave with them	`bsi-cheminfo.js` §9-ter — `esportaCsv()` and `rapportoMetodo()`, both formed in memory in the browser	`test_cheminfo` — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits
SCI-21	A number presented as structural similarity must come from a fingerprint, and one implementation must compute it	`rdkit_lab.html` loads `bsi-cheminfo.js` and uses its `tanimoto()` and `fingerprint()`; the eight-bit similarity has been removed	`test_cheminfo` — page and engine give the same Tanimoto over all **28 pairs**, deviation **exactly 0**; the fake-similarity functions no longer exist; every reference structure is valid and carries the **monoisotopic mass** of the drug it names

ID	Requirement	Implementation	Verification bench
SCI-22	Conversions between molecular languages must be checked against an external source or against hand calculation, and coordinates written to a file must be real and declared	`bsi-molingue.js` — **seventeen outputs**, from SMILES to CML	`test_lingue` — 7 **literature InChI keys**, full SMILES → molfile → SMILES round-trip, 26 hand-written names, **7 required refusals**; mass composition against hand calculation (alanine C 40.44%, aspirin C 60.00%); XYZ and PDB with non-null coordinates **and** declared 2D
SCI-23	Angles, lengths and dihedrals are measured on the coordinates, and on flat coordinates they are not shown	dot product for angles, cross product for dihedrals	`test_mol3d` — tetrahedron **109.4712°**, water 104.47°, CO_2 180°, BF_3 120°, ammonia 106.13°; dihedrals 0°, 60°, 90°, 180°; ethanol C–C 1.509 Å; and the refusal tested in both directions
SCI-24	A spectrum read from a file must be decoded according to the format, and the peaks found must be the ones that are there — no more, no fewer	`bsi-spettrolettore.js` — JCAMP-DX with full ASDF (PMAI, DIF, DUP) and the integrity check the format prescribes; peaks by **prominence**, noise from MAD × 1.4826	`test_spettrolettore` — **61 checks**. On three Gaussians built at 1715, 2950, 3400 cm^{-1} under noise it finds three: 1716, 2952, 3402. **Both ways**: on pure noise it finds **zero**. ASDF verified symbol by symbol (`abcdefghi` = −1…−9; `n` = −5, not −4)
SCI-25	A band must be declared *compatible with*, never assigned to a single possibility	`BANDE_IR` — 24 ranges; `assegnaIR()` returns **all** compatible assignments	`test_spettrolettore` — at 1715 cm^{-1} both a ketone C=O and a carboxylic acid C=O appear; the section states that it does not deduce the structure
SCI-26	A published model must be checked against **its own** declared error, and a learned model against chance	`bsi-cheminfo.js` §10 — ESOL (Delaney 2004) with its uncertainty, four drug-likeness filters reported separately, random forest with out-of-bag error	`test_spettrolettore` §4-5 — ESOL against four experimental values, mean deviation **0.68**, inside the declared ~1 log unit; forest out-of-bag R^2 **0.90** on a non-linear relationship and **−0.16** on pure noise; deterministic for a given seed and different for another

ID	Requirement	Implementation	Verification bench
SCI-27	A structure without a source must not get in, not even one that is known	collected from ChEMBL, `tools/dati/farmaci_v187.json` carries each entry's identifier	**Two refusals**: ivermectin (`structure_type NONE` , a mixture of homologues) and semaglutide (a protein, `SEQ`). And two entries kept under the **record's** name rather than the query's: «morphine» → apomorphine, «levothyroxine» → liothyronine
SCI-28	A predicted spectrum must say WHICH atoms produce each signal, and the deviation must be measured on molecules that did not choose the parameters	`bsi-pretsch.js` — the estimation tables transcribed in full from the source (Pretsch 4th ed.: 91 ^{13}C and 66 ^{1}H rows for benzenes, 42 for ethylenes, 31 for alkanes, 24 for ^{13}C aliphatics, 4×4 steric corrections); `bsi-nmr.js` — the reasoning that applies them: ring position obtained by walking the cycle, additive scheme with a cyclic reference compound, chemical equivalence by shell environment code	`test_nmr` — **56 checks**. Two separate sets: tuning **0.71 ppm** over 9 molecules, **validation 0.92 ppm** over 22 never used for tuning, worst case **4.9 ppm**, ^{1}H **0.06 ppm** (0.03 on the aromatics alone). The bench requires validation to stay **worse** than tuning: were they equal, someone would have moved a molecule between the two sets
SCI-29	Chemically equivalent atoms must give ONE signal, not one per atom	environment code built as concentric shells by distance (the HOSE-code principle), with aromatic bonds written as such and not in Kekulé form	`test_nmr` — eight molecules whose signal count is known: benzene **1** (not six), toluene 5, p-xylene 3, naphthalene 3, biphenyl 4, aspirin 9. All correct
SCI-30	From an image one can recover the SHAPE of a spectrum, not its calibration	`bsi-digitalizza.js` — `bsi-spettrolettore.js` delegates `daImmagine()` to it, a single implementation. The axis frame is found first, the trace is extracted at **sub-pixel** resolution (darkness-weighted centroid), the run is chosen by continuity **and** by darkness, both axes are calibrated linearly or logarithmically, %T is converted to absorbance with $A = -\log_{10} T$	`test_digitalizza` — **45 checks** over seven constructed figures: mean deviation **2.2 cm⁻¹**, worst **7.3** (one column is worth ~4 cm⁻¹). **Both ways**: from a blank sheet and from the app icon no curve comes out, and the crude fallback on the same figure with frame and labels finds **one** peak instead of three. `test_spettrolettore` for the routing

ID	Requirement	Implementation	Verification bench
SCI-31	A two-dimensional NMR map must say WHICH atoms produce each correlation, and must not connect what the topology does not connect	`bsi-nmr2d.js` — one-bond HSQC with inverted sign on CH_2, three-bond COSY between non-equivalent protons, two- and three-bond HMBC; exchangeables excluded from off-diagonal spots	`test_nmr2d` — **40 checks**, in both directions: hexafluorobenzene gives no HSQC spots, benzene **no** off-diagonal COSY spots (its six protons are equivalent), methane **no** HMBC spots despite having four protons
SCI-32	A program-built 3D geometry must carry THE SAME indices as the predictor, otherwise the peak↔atom link lights up the wrong atom	`bsi-geom3d.js` — distance geometry over the `bsi-nmr.js` graph, hydrogens appended at the end, generator seeded from the SMILES	`test_geom3d` — **29 checks**: every heavy atom has the same index and element across five molecules; bonds and angles within tolerance of literature values; **in both directions** benzene comes out planar and cyclohexane does not
SCI-33	A document must be opened in full, and whatever could NOT be read must be declared — a reader that returns little without saying so makes the document look nearly empty	`bsi-documento.js` — PDF page by page with its text layer (PDF.js), images, text, Office archives opened with `DecompressionStream` and no libraries	`test_documento` — **47 checks**: PDF, Word and text genuinely opened and their content verified; **in both directions** an image declares there is no optical character recognition and produces no working, an unknown format is refused rather than guessed, a binary file named `.txt` is recognised as binary
SCI-34	Spectroscopic data recognised in a text must carry their PROVENANCE, and the working must declare what it is missing	`bsi-quesito.js` — recognition by section (IR, MS, ^{1}H, ^{13}C) with the boundaries between sections, then `bsi-elucida.js` for the step-by-step working	`test_documento` — formula, bands, masses with intensities, ^{1}H signals with integral and multiplicity, ^{13}C signals; **in both directions**: $CDCl_3$ does not become the compound, a band outside 400-4000 is not a band, without an MS label there are no masses, a new paragraph closes the section, a text with no spectra produces no data

ID	Requirement	Implementation	Verification bench
SCI-35	A figure with a frame, labels and a grid must not be digitised as if it were only the curve: the axes and the numbers are dark pixels just like the trace	`bsi-digitalizza.js` — the frame is sought among the CONTINUOUS lines in the margins, and a border is told from a gridline by its **irregular spacing** relative to the family **or** by its thicker, darker stroke; the crop is moved past the line thickness; vertical gridline columns are marked as not measured and reconstructed from their neighbours	`test_digitalizza` §1-2 — on a figure with a frame **and** a grid (11 vertical families, 7 horizontal) it finds all four sides and the peaks stay within 2.2 cm^{-1}; **both ways**, on a grid-only figure it does **not** invent a frame and says so, and a **flat baseline** is not mistaken for an axis
SCI-36	A digitised curve must be able to leave the application, and the exported file must be valid	`bsi-digitalizza.js` — CSV output with the units in the header, and JCAMP-DX 4.24 in (XY..XY) form with FIRSTX/LASTX/NPOINTS and the provenance written inside the file	`test_digitalizza` §4 — the **full round trip**: the exported JCAMP is read back by `bsi-spettrolettore.js`'s `leggi()` and returns with the same number of points, the same end points and the same units
UI-14	One function must not have two entry points. The spectra lived in **nine sections** across two menus; Spectra in three places; and the ✨ panel showed **two "Data Science", two "File Manager", two "Articles", two "Updates"**, because the panel has a hardwired list that opened the very sections the menu module was inserting — and nobody compared the two lists	`bsi-centrospettri.js` — the panels are **moved** (`appendChild` moves, it does not copy) into a single tabbed section; the old buttons stay hidden because they are the target of `goSection()`, and a tab **clicks the button** instead of showing the panel, so lazy initialisation goes through the one road that already existed. `bsi-menu.js` — the "Utility" category is in the **main bar** with all its entries, and the rule is: if something has a **reachable** button in the bar it is not in the ✨ menu, if it has none it is only in the menu. "Reachable" is not measured with `offsetParent` — the bar shows one group at a time, and a perfectly reachable button has a null `offsetParent`	`test_articoli` §5-6 — the **ten** tabs are inside the centre, and each identifier exists in the document **exactly once** (the proof that it was moved and not copied); `goSection()` on each opens the centre on the right tab and shows **only** one. `test_menu` — Utility visible in the bar with all **fifteen** entries plus the two language ones, **no** menu entry repeating a bar section, **no** duplicate label in the panel, across **two viewports**: large screen and phone. **Both ways**: no repeated id, no duplicate button, Spectra and the reader out of the bar but with their button still in the document because `goSection()` uses it

ID	Requirement	Implementation	Verification bench
UI-15	A data table must not exist twice: two copies diverge, and the reader gets a different answer depending on where they came in	The J coupling constants existed in two independent tables with **different** numbers (aromatic ortho ^{3}J 6–9 Hz against 7–10 Hz). The one inside the Spectra centre remains, which sits in the section with its own reference list; `_apriCostantiJ()` is the single road and leads there	`test_articoli` §6 — asking for the J constants opens the centre on the reference-tables tab, on the right sub-tab, and the table has its rows
ING-02	A hand-written list of articles is stale the next day, and a prize year written from memory is an error nobody can see	`bsi-articoli.js` reads **live** from Crossref and from the Nobel Foundation; `bsi-articoli-nobel.js` holds **explanations only**, matched to the topic of the official motivation and not to the year, so no attribution comes from the application. The addresses are not written: they are **pure functions** of the identifier (DOI, year)	`test_articoli` §1-4 — the links tested as functions (stable for the same datum, different for a different one); the explanations matched against nine test motivations and **none** invented when the topic is not recognised; and the proof that counts: a response is **injected** with a name that exists nowhere, and the screen must show **that** name. **Three distinct failures tested**: failed request, empty answer, unrecognised shape — and they must be stated **differently**
UI-16	Changing language must cost a gesture, not a navigation	`bsi-sellingua.js` — a **round** button top right, at the title's height, with the flag in use; it opens into a disc with the fourteen flags on **two shells** (six inside, like a benzene ring) and, at the centre, the language name and **how much it is translated, measured** by `coperturaPerLingua()`. The actual switch stays in `bsi-lingue.js`: here there is only the way to call it, and the "Language" section remains because it INFORMS while this one ACTS	`test_atomic` §1 across **two viewports** — round, top, right, and "at the title's height" **measured** as the overlap of the two boxes (with `align-self:center` it lined up on a large screen and not on a phone, because there the flex aligns to the top and the logo block is two lines tall); an accessible name that says what it does rather than "flag of Italy"; the disc **inside the screen**; and **both ways**: the language really changes and going back to Italian the text is identical

ID	Requirement	Implementation	Verification bench
SCI-37	A didactic game's questions cannot be a second source of chemical truth, and cannot have two right answers	`bsi-gioco-domande.js` — it **contains no questions**: nineteen generators drawing from `ELEMENTS`, `SPEC_DATA`, `RXN`, `RETRO_STRATEGIES`, `AA_DATA` and the `BSIElucida` / `BSILettoreSpettri` engines, and every question carries its **provenance**. Range distractors are **disjoint** from the answer's, because different groups' ranges overlap (^{4}J meta 1–3 Hz against ^{4}J allylic 0–3 Hz)	`test_atomic` §2 — the nineteen generators **each tried** (one was dead and "available" called it running, because it looked at the source instead of trying it); the range reader on six known cases (it read "1–3 Hz" as [−3, 1], and the consequence was not a skewed number: it was that the four range generators stopped coming out and the "zero ambiguous" check passed vacuously); 600 sound questions; **zero ambiguous**, with the guard that those generators do run

ID	Requirement	Implementation	Verification bench
UI-17	A game must be playable from start to finish — exploration, puzzles, synthesis, guardians — and must not consume when nobody is watching	`bsi-atomic.js` + `bsi-atomic-livelli.js` — **Atomic Protocol**: **a hundred levels** in ten zones, generated from a seed (the level number) on a grid where two neighbouring rooms share the wall and the door, so doors match by construction; **ten guardians** (one every ten levels, in arenas at least 14 m wide) whose phases ask for a compound, an element or an answer; twelve recipes with equations **balanced atom by atom**; generated puzzles; **thirty audio logs** (three per zone, seven voices) heard while playing, with the device's voice, subtitles and an archive in the Codex, piecing together the mystery of the command phrase; **saving** at every level; the loop stops when the section is not visible; moving between levels **frees the previous level's geometries and textures**	`test_atomic` §3-5b, §6, §8 — the thirty logs all placed in the levels; the hundred levels generated and walked (twin doors, no unreachable or overlapping room, bench-terminal-exit in each, the elements needed to beat every guardian present before it, ten distinct guardians, wide arenas), the equations counted and **a wrong equation rejected**; level 1 played up to the elevator and the save; the guardians **both ways**; PROTOCOL 12 at level 100 ends the game; GPU memory that **does not grow** across ten levels with four guardians; the loop stopping when the section closes; the game's **floating button** above Spectra, aligned, with an icon and accessible name, opening the game, and the floating buttons not covering the scene in game and coming back on exit
UI-18	A view, difficulty and graphics choice must stay chosen	`bsi-atomic.js` — **first person by default** (third as an option), four difficulties (Easy, Normal, Hard, Experiment — where the reaction must be balanced before synthesising it), **high, medium or PS1 graphics**, sound and language: all in the settings menu, also reachable from pause, and all saved to local storage, with every read guarded because private browsing may refuse to write. The starting graphics is high (PS2 level), «medium» on a phone with few cores; PS1 remains a retro option	`test_atomic` §6-7 — the chosen view, difficulty and graphics really end up in storage and are read back; there are four difficulties and **«Experiment» is not «Easy» renamed**; changing graphics mid-game **does not lose the game**

ID	Requirement	Implementation	Verification bench
UI-19	A declared graphics quality — the PlayStation 2 level — must be built, not evoked, and without horror	`bsi-atomic-figure.js` + `bsi-atomic-grafica.js` — the protagonist with **thousands of polygons**: a dense sphere head shaped by hand (skull, brows, cheekbones, nose with wings, lips, chin) with **modelled eyes** (eyeball, iris, eyelid) and a face painted with soft shading and no outlines, a hair cap with strands, **five-fingered hands** in nitrile gloves, a lab coat with modelled lapels, collar, pockets, buttons and badge, shoes from an extruded profile; high-resolution textures with **normal maps**; **glow** from additive sprites on lamps, screens and cores; the ten guardians are **the institute's machines** (pump with hydraulic arms, extruder, laser bench, sequencing robot, electrolysis, cryostat, automated pharmacy, hot cell, clarifier, reactor), no creatures; residues are solvent spills. The **environments are art deco**, in the spirit of BioShock: checkered marble with brass inlays, walnut wainscot, green and gold wallpaper, marble pilasters, fan sconces, brass-ringed ceiling lights, coffered ceilings, Institute and Albedo Consortium posters, chalk writing on the walls, drips falling into puddles, amber light. High graphics is the default; PS1 remains a retro option. **The game is played in first person**, with an art deco HUD (segmented health and reagent in brass frames): the arms (from models supplied by the user, in `models/` and prepared by `tools/prepara-modelli.js`: **real skinned hands** in a nitrile glove whose poses — the grip, the finger on the trigger, the open palm, the idle gesture — are frames of their own animation blended finger by finger; the **lab-coat sleeve** cropped from a scan, with the light-blue shirt cuff; the **capsule gun** in the right hand, fingers closed on the grip, the capsule in the compound's colour and the spray from the muzzle; from v217 the arms are the **«Fists» rig with its eight animations** — walk, run, Equip on a switch, the left jab to throw and grab, Inspect at rest — with the gun attached to the right fist bone, gripped (index on top, muzzle forward), and on the left the **robotic hand**, its fingers articulated joint by joint (slightly bent at rest, opening and closing to grab, snapping open on the throw): with the power the palm turns up, a **sphere in the element's colour** with a mane of filaments lights above it and the sleeve crumbles into grains; until the models arrive the procedural arms stay; each finger follows its pose on a **damped spring**, with the micro-motions of a living hand, the trigger pulled sharply, the recoil in the wrist, the hand opening before grabbing, fingers flaring on the throw, the power rippling the fingers and, at rest, the gun raised to look at it). **The laboratories** (from v219): the maps supplied by the user — Small Lab, Sci-fi lab, Flynn's secret lab, the underground laboratory, the overgrown laboratory, one per wing in turn, with	`test_atomic` §6 — measured in the live scene: high is the starting graphics; ACES, sRGB, shadows, reflections, physical materials; **more than five glows** in scene; the protagonist with **two modelled eyes, five fingers**, a head above 3000 triangles and a figure above 15000; **no box** in the protagonist and the ten guardians; first person by default with the arms in their own scene, five fingers, the tool showing the compound held, spray droplets, a gaze that turns but not beyond a head; sodium sparking and chlorine as vapour on the left hand, the throw; brass and marble in the environments; the segmented health bar; a log with subtitles ending up in the Codex; the chemist moving forward facing where it goes and knees never backwards; the other way, PS1 at 480×360 with vertex lighting and 15-bit colour

ID	Requirement	Implementation	Verification bench
		the reactor and the capsule as apparatus — are really walked: in every level a door in the bench room leads to the wing's laboratory, with its walk grid computed from the geometry (floors, stairs, walls and furniture), the synthesis bench on the map's real counter and the door to go back. Over everything, the **final «serious PS2» pass** — film colour, soft glows, vignette and grain — as in Silent Hill 2, RE4, MGS3 live in their own scene, drawn after the world, so they never clip into walls; they sway with the stride, lag on turns, recoil when spraying, the left hand reaches to take and, **holding an element, keeps it alive on the palm like a plasmid** (sparks for metals, flame for fuels, vapour for gases, with a light illuminating the glove) and throws it; the gaze moves with the mouse (pointer lock) or by dragging a finger	
ING-01	One function must not have two implementations: the React/FastAPI stack must run THE SAME files as the application	`stack/api/worker/motore.mjs` — a Node process loads the repository modules with RDKit WebAssembly; no Python engine	`test_stack` — **15 checks**: compares the SHA-256 fingerprints declared by `GET /salute` with those of the files on disk, verifies the numbers match the browser's, and that no `.ts` / `.tsx` file contains a shift table

2. AI agent requirements

ID	Requirement	Implementation	Verification bench
AI-01	A model retired by the provider must not block the assistant	`modelloSuggeritoDaErrore()`, `bsi_modelli_ko` blacklist	`test_404` — 36 checks
AI-02	A provider unreachable from the browser must be remembered and demoted	`bsi_prov_ko`, 24 h TTL, `providerUtilizzabili()`	`test_ko` (35), `browser_ko` (37)
AI-03	A temporary overload must produce a retry, not an error	`erroreTemporaneo()`, `TENTATIVI_TEMPORANEO`	`test_503` — 40 checks
AI-04	An imposed wait longer than 10 s must switch provider if a ready one exists	`ATTESA_TROPPO_LUNGA_MS`	`test_attesalunga` — 14 checks
AI-05	The provider's token ceiling must be learned from the error, not hard-coded	`tettoDaErrore()`, `MARGINE_TETTO`	`test_tetto` — 32 checks
AI-06	Opaque reasoning signatures must be returned to the provider	capture and replay in `appendAgentTurn()`	`test_firma` — 31 checks
AI-07	Two overlapping submissions must not corrupt the history	`_turnoInCorso` in `send()`	`test_doppioinvio` — 7 checks

ID	Requirement	Implementation	Verification bench
AI-08	The user must be able to measure which providers they can actually reach	`provaTuttiIFornitori()`	`browser_prova` — 29 checks
AI-09	Paid providers must not be offered as free ones	separation by `PROVIDERS[id].free`	`browser_prova` §D2
AI-10	A documented route must exist to remove CORS and rate limits	`proxy/spectra-proxy.js` + in-app guide	`test_guidaproxy` (12), `browser_proxy` (15), `browser_proxyui` (17)

3. Stability requirements

ID	Requirement	Implementation	Verification bench
STA-01	A long session must accumulate neither DOM nodes nor timers	section lifecycle management	`audit_stabilita` §1 — 96 sections × 5 rounds
STA-02	Exhausting `localStorage` must not make the app unusable	guarded writes; persistent warning in the File Manager	`audit_quota` (10), `test_filemanager` (15)
STA-03	No rejected promise may remain unhandled	systematic `.catch()`	`audit_promesse` — 22 checks, healthy network and dead network
STA-04	Corrupted saved data must not prevent startup	`loadJSON()` with fallback	`audit_stabilita` §4
STA-05	A slow network must not be treated as an absent one	`ATTESA_RETE_MS = 3500` in `sw.js`	`test_sw` §3 and §3-bis — measured 3,507 ms
STA-06	An update must not destroy work in progress	`BSI_SW_UPDATED` message, decision taken page-side	`test_sw` §4
STA-07	Browsing molecules must not exhaust the WebGL contexts	reuse of the 3D viewer at all 5 creation points	`test_visore3d` — from 7 constructions down to 1
STA-08	An unavailable IndexedDB store must be declared, not fatal	`dbStore()` over a promise; persistent warning	`test_filemanager` §2

4. Interface and accessibility requirements

ID	Requirement	Implementation	Verification bench
UI-01	Every page must open without JavaScript errors	—	`audit_stabilita` §5 — 14 pages
UI-02	Charts must be sharp on high-density screens	`bsiNitido()` on the canvas contexts	`audit_grafici` — 40 canvases

ID	Requirement	Implementation	Verification bench
UI-03	The app must work at 390 px width, and content that does not fit must stay **reachable**	`auto-fit` grids, no fixed column, and wide tables inside a container with `overflow-x:auto`	`audit_mobile` — **two distinct measurements** across all 96 sections at 390 px: the document's `scrollWidth` (the horizontal bar the user sees) **and** elements wider than their section **with no scrollable ancestor**, that is content clipped and unreachable — a written column that cannot be read and gives no sign of existing. **Both ways**: a wide table inside a scrolling container is **not** reported, because it can be read by scrolling. `audit_stabilita` for errors in a phone viewport
UI-04	The user must be able to know which version they are running and force the update	"Updates" entry in the ✨ panel	`test_aggiorna` — 9 checks
UI-05	Data deletion must be selective and reversible in its choices	`bsiCancellaDati()` by group	`browser_reset` — 24 checks
UI-08	Surfaces that appear only after an action must also meet WCAG AA contrast	the six panels of the Cheminformatics section	`test_cheminfo` — the analysis is run, the panels opened in turn, **460 text elements measured, 0 defects**, with the WCAG formulas rewritten inside the bench
UI-09	Opening a section must not block the page	3Dmol's first `render()` moved out of the click; the 296 synthesis figures **and the 263 drug cards** drawn in 8 ms slices	`test_fluidita` — **96 sections traversed**, median switch 16 ms, worst 87 ms, **0 sections above 100 ms** (there were 3 with the new drugs, at 159 ms); plus the opposite proofs: the WebGL canvas appears anyway, all 296 figures exist shortly after, and printing does not come out mute. For the viewer the claim is **structural, not temporal**: immediately after the click handler returns, the WebGL context must not exist; shortly after it must — because a millisecond threshold was measuring SwiftShader rather than the code (14…192 ms on identical code)
UI-10	The interface must switch among many languages, declaring how much each is translated, and carrying the writing direction with it	`bsi-lingue.js` with **fourteen languages**, a searchable selector, `dir="rtl"` for Arabic	`test_lingue` — **170 skeleton elements out of 170 in each of the thirteen languages**, coverage computed by applying the dictionary and counting; writing direction tested in both directions; navigation categories and the search placeholder included
UI-11	The molecule must show how it holds together, and the viewer must choose the angle	`bsi-mol3d.js`: formation from dust with staggered arrival, and atom selection by click	`test_mol3d` — the canvas **pixels** at 130 ms and 2.3 s, the arrival order (skeleton before hydrogens) measured on each atom's progress, and the selection of 1, 2, 3 and 4 atoms

ID	Requirement	Implementation	Verification bench
UI-12	Whoever has a file must be able to open it: a picker listing extensions does not narrow the choice on a phone, it **hides** it	`bsi-spettrolettore.js` — a single `<input type="file">` **with no `accept`**, a single button, and `apriQualunque()` routing by type, by the extension the format declares and **by content**, announcing the decision in `#bsiSP-deciso`	`test_spettrolettore` §7 — one visible button, an **empty** `accept` (a non-empty `accept` here *is* the defect), and routing verified on PDF, `.docx`, `.txt`, `.jdx`, two `.csv` columns and an image; **both ways**, a binary disguised as `.txt` is not poured out as text but declared binary
UI-13	The version shown in the header must be the one actually running	`index.html` — the badge is filled from `window.BSI_APP_VERSION`, no longer a hand-written string	`test_spettrolettore` §8 — the badge text is compared with `BSI_APP_VERSION` measured in the running page. It was needed: for about eighty versions the header read `v13l` when the application was already past version 190, and no check looked at it

5. Security and privacy requirements

ID	Requirement	Implementation	Verification bench
SEC-01	No API key may be present in the repository	keys only in `localStorage` or in the Worker's secrets	`tools/verifica-sicurezza.js` — 388 tracked files, 8 credential shapes
SEC-02	No personal data may leave the device without an explicit action	local-first architecture, telemetry disabled	`audit_rete` — **direct verification**: a canary value seeded into 71 stores of the user's data, the application used across 96 sections on 6 pages, and the URL, headers and body of every request inspected. Plus `verifica-sicurezza` on the two exit mechanisms
SEC-03	Passwords must not appear in clear text in the source	SHA-256 in `file_manager.html`	`tools/verifica-sicurezza.js`
SEC-07	No credential may EVER have entered the repository, not even in a commit later fixed	no key has ever been committed; keys live in `localStorage` or in the Worker's secrets	`audit_storia` — **1,551 distinct versions of text files across 390 commits**, 9 patterns tested in both directions; fails on a shallow clone, because it would measure less surface
SEC-04	No conflict marker may reach publication	—	`verifica_guida` §12 — 58 text files

> **SEC-03 — declared limitation.** GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v188`. The old one remains in the history and no longer opens anything.

6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

ID	Requirement	Current coverage
UI-06	Full WCAG 2.1 AA conformance	the mechanical part is automated across **13 pages and 96 sections** (`tools/verifica-accessibilita.js`, 38,407 text elements: it was 19,751 before text over gradients entered the measurement, and 37,409 before the «Spectrum reader» section). **Of the 1,069 contrast defects that emerged, none remain: 0 measured** over those same 96 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are **zero**: see the box in `docs/09` §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09)
PERF-01	First-paint time on a low-end device	manual cross-device testing (`docs/02` §4)
SCI-11	Structures of 2 entries that are not single molecules (it was 6)	**not representable**: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see `docs/06` §2.4
PERF-02	Code coverage of the verification benches	**partially measured**: 49.89 % of statements over the widest path (`audit_copertura`). Whole-battery coverage and branch coverage remain unmeasured; see §8
UI-07	Behaviour on Firefox and WebKit	**not verified** — the battery runs on Chromium only; see §8

7. Overall coverage

Category	Requirements	Verified by a bench	Coverage
Scientific (SCI-01…37)	36	36	100 %
AI agent (AI-01…10)	10	10	100 %
Stability (STA-01…08)	8	8	100 %
Interface (UI-01…19)	17	17	100 %
Security (SEC-01…07)	5	5	100 %
Architecture (ING-01…02)	2	2	100 %
Automated total	**76**	**76**	**100 %**
Not automated (§6)	5	0	0 %
Declared total	**81**	**76**	**94 %**

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

> **Why two totals.** Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **54 requirements verified out of 59 declared** — and the five that are missing are listed by name in §6, not summarised into a percentage.
>
> This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

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8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

Gap	What it entails	Why it is so
Partial code coverage	Measured: **49.89 %** of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches	Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool (c8, istanbul), not to the problem. Value recorded; `audit_copertura` fails if it falls
Chromium only	Behaviour on Firefox and WebKit is verified by hand, not by a bench	The battery uses `playwright-core`, which downloads a single engine. In the verification environment the CDN for the other engines answers **403** to the network policy: they cannot be installed there
No visual regression	An unintended graphical change would not be caught	Spectra are deterministic and comparable byte for byte (SCI-05): the comparison exists on the traces, not on the whole page
2 entries without a structure	Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six	See `06-Scientific-Accuracy-Data-Provenance.md` §2.4
Performance on slow devices	First-paint time is not measured on low-end hardware	It requires physical devices; the test is manual

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Document updated to version `bsi-v219`.

Release Conformance Statement — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version	`bsi-v219`
Author and release owner	Samuele Pio Provenzano
Repository	`github.com/samupropio1-ship-it/BioSpecInfo-v11`
Distribution	GitHub Pages — `samupropio1-ship-it.github.io/BioSpecInfo-v11/`
Nature of the software	Client-side Progressive Web App, no backend

1. Subject of this statement

The undersigned declares that version `bsi-v219` of BioSpecInfo has been put through the verification procedure described in `02-Verification-Validation-Report.md` and that the outcomes are those reported, without selection, in `../evidence/RAPPORTO-VERIFICA.md` — a document generated automatically by `tools/genera-evidenza.js`, which transcribes the benches' output with no manual intervention.

This statement does **not** attest the absence of defects. It attests that:

1. an automated verification procedure exists and can be reproduced by third

parties;

1. the outcome reported is the one actually produced, failures included;
2. the known limitations are declared explicitly and not presented as features.

2. Intended use and limitations

2.1 Declared purpose

BioSpecInfo is a **teaching and study-support** tool for chemistry, biochemistry, pharmacology and astrochemistry. It is designed for university students and for exam preparation.

2.2 What the software is NOT intended for

These exclusions are an integral part of this statement.

Domain	Exclusion
Clinical use	It is not a medical device within the meaning of Regulation (EU) 2017/745. It does not support diagnosis, prescription, dosing or therapeutic decisions. The pharmacological data are for teaching purposes only, and the application says so at the head of the section.

Domain	Exclusion
Certified analytical use	The spectra are **predictions** from functional-group tables, not measurements nor quantum-mechanical simulations. They do not replace an experimental database (SDBS, NIST) nor an instrumental determination, and cannot be used for the formal identification of substances.
Regulated GxP environment	It is not validated for use in a GLP/GMP environment. The alignment with GAMP 5 described in `docs/03` is methodological, not a formal qualification.
Access control	The File Manager protects its contents with a client-side deterrent. On a static site there is no effective access control: whoever reads the source bypasses it. It must not be used for confidential material.

3. Summary of verification outcomes

The full outcomes, with each bench's textual output, the execution environment, the commit and the SHA-256 digests, are in `../evidence/RAPPORTO-VERIFICA.md`.

Family	What it demonstrates
Scientific data	The chemical data are verified against an independent source
AI agent	The assistant stays usable when the external provider fails
Stability	The app withstands long sessions, exhausted storage, degraded networks
Interface	The panels respond as documented
Documentation/code consistency	What the documentation promises exists in the code

4. Known deviations at the date of release

Listed in full. None has been removed from verification in order to make it pass.

#	Deviation	Impact	Treatment
D-01	**2 entries** (it was 6) that are not single molecules: Ivermectin, a mixture of B1a/B1b homologues, and Coartem, a combination of two active ingredients	For these two no 2D/3D representation and no spectral prediction is available	Entries left **without SMILES**: the pharmacological data remain, the structure is not shown. For Digoxin, Vincristine and Tacrolimus (topical and systemic) the structure was taken from **ChEMBL** and passes the comparison with the declared molecular weight: those four left the registry. The two that remain have no structure to show, not a missing one.

#	Deviation	Impact	Treatment
D-02	19 entries with no SMILES by their very nature (monoclonal antibodies, peptides)	None: for those molecules SMILES notation is not the appropriate representation	Not a substantive deviation; listed for completeness
D-03	~~SEC-02 verified only by proxy~~ — **remedied**: see S-10	—	`audit_rete` observes the traffic during real use, with a canary value seeded into the user's own data
D-04	~~The File Manager password in the git history~~ — **remedied**: see S-12	—	The password was **changed** at version `bsi-v188`. The old one is still readable in the history and no longer opens anything
D-06	Code coverage **measured but partial**: 49.89 % of statements, not branches	It is the coverage of the widest path a bench walks (96 sections plus the other pages), not of the whole battery; and an `if` entered from one side only counts as covered	`audit_copertura` measures it with Chromium's profiler, with no build and without rewriting the source. The value is **recorded**: if it falls, the battery fails
D-07	Verification on Chromium only	Firefox and WebKit are tested by hand	Declared in `docs/08` §8. In the verification environment the reason is checkable: the CDN from which `playwright-core` downloads the other engines answers **403** to the network policy, so Firefox and WebKit cannot be installed there. **Re-verified on 2026-10-08**: `npx playwright install firefox` fails with «Download failure», and `cdn.playwright.dev` still answers 403 to the tunnel. It is not a choice, it is a constraint of the environment, and it stands
D-08	~~Partial English translation~~ — **remedied**: see S-13	—	All **16 documents** are in English. `verifica-documenti` and `verifica-affermazioni` read `docs/en/` just like the Italian set: a disagreement between the two languages fails the battery

#	Deviation	Impact	Treatment
D-09	Full WCAG 2.1 AA conformance not verifiable entirely by automation	What stays outside is text inside SVGs (8,888 elements), text over a real background **image** (108) and everything requiring human judgement. Text over a **gradient** has entered the measurement: 591 elements, judged against the worst stop of the gradient	The skipped elements are **counted** and reported on every run
D-10	The `bsi-v181` tag points at the wrong commit	The release workflow, started without filling in the version field, used its own default — `bsi-v181` — and, being rerunnable by design, **moved that tag** from commit `cb0cf03` to the `bsi-v188` one, also attaching the v187 packages to it. The foreign assets were removed. **The tag cannot be moved back by an app**: GitHub refuses to let an app write a ref pointing at a commit whose `.github/workflows/*` differs from the current one, and at `cb0cf03` that file differs by 319 lines. Three routes were tried, all blocked by the same rule: `git push --force` («refusing to allow a GitHub App to create or update workflow»), `PATCH /git/refs` with `force` (403), and delete-and-recreate — where the **403 already lands on creating a throwaway tag** at that commit, so the real tag was left untouched. `GITHUB_TOKEN` cannot be granted the `workflows` permission. **Re-verified on 2026-10-08** with the session's own GitHub client, which uses a different credential: `PATCH /git/refs/tags/bsi-v181` answers **403 «Write access to this GitHub API path is not permitted through this proxy»**. It is a second block, independent of the first: even without the workflow-file rule, writing to git refs does not get through. The tag points at `64e8c316`; it should point at `cb0cf03e`	The three packages attached to the release are the correct ones; the «Source code (zip)» archive must be ignored, and the `bsi-v181` code is downloadable from commit `cb0cf03`. Declared in the CHANGELOG's `bsi-v181` entry and in the release body, with the command that restores the tag from personal credentials. The default was removed and the workflow now **refuses** a version that does not match `sw.js` and the package filenames — verified both ways

Contrast, from 1,069 defects to zero — how it surfaced and how it was closed

Up to version `bsi-v168` this document declared WCAG conformance "verified by sampling" and the bench reported **0 defects**. Those were two statements at once true and meaningless: the bench inspected only the **visible** elements, and `index.html` shows one section at a time. Out of 19,751 text elements it was looking at **41**.

Walking all 87 sections then present, **1,069** contrast defects surfaced. They had not appeared: they had always been there.

Today they are **zero**, measured with the same bench over every section — 88 of them now. Not by wholesale substitution — that road, attempted once, introduced 997 new defects — but by going back to the **common cause** every time and re-measuring after each change. The **40 unlabelled fields** have likewise fallen to **zero**.

The gravest causes were not wrong colours but **theme tokens whose value equalled the background**:

Structural defect	Measured effect
`--g800` is `#0d1522`, **identical to `--bg`** — and the rule `a { color: var(--g800) }` applied it to every link	Every link without a colour of its own was at contrast **1:1**: invisible
`--g700` is `#16263d`, nearly identical to `--white` (`#16273e`)	Text at **1.01:1**
Buttons that changed their **background** without changing their **text colour**	The selected state (or the deselected one, depending on the panel) became unreadable: **1.39:1**
`lightenColor(hex,pct){ return hex; }`	A function called "lighten" that returned the colour **unchanged**: an unfinished stub, in all likelihood written for this very problem
The quiz answer panel: light background, light text	The student **could not read the correct answer**

The structural remedy was an explicit token — `--testo-forte` — for prominent text on a dark surface, applied only to the uses as a *text colour*: `--g800` and `--g700` stay where they serve as background or border.

Cause	Defects	Remedy
A dark-theme safety net with higher specificity than the components: text meant for the **white** cards was forced to `#d4dce6`, contrast 1.38:1	159	Rewritten with `:where()` (zero specificity): it protects text that has no colour of its own and stops fighting text that has one
A per-category accent used as **text** on a dark header, with `#0d1522` (near-black) as fallback: 1.13:1	~400	`bsiAccentoLeggibile()` preserves the hue and raises the lightness just enough. Applied at the 14 points where the accent is chosen — not in the data, so it holds for future entries too
Two hues used only on dark backgrounds, just under threshold (3.75:1 and 3.04:1)	139	Replaced after **verifying** that they never appeared on a light background

Cause	Defects	Remedy
Table zebra striping white/near-black with always-light text: the white rows were unreadable	57	Two dark tones, consistent with the theme
Periodic-table cells just under threshold (4.46:1 and 3.88:1)	43	Text at `#eceff4`, verified across all nine categories
`"#var(--text)"` — one `#` too many makes the colour invalid: in SVG it falls back to black, on canvas the assignment is **ignored** and the previous colour persists	8 places	Replaced with the real value

The direction of the correction depends on the surface, not on the colour. The first attempt always lightened: it removed thirty defects on the dark header and added eighteen on the light formula box that uses the same accent. The function now darkens on light backgrounds and lightens on dark ones.

The final tail: 80 → 0. The last eighty were scattered over 33 distinct causes and looked like finishing work. They were not: three more structural causes lay underneath.

Final cause	Remedy
The "automatic contrast corrector" was producing defects instead of removing them. It used *perceived* brightness instead of WCAG luminance, so its thresholds corresponded to no real ratio; it **ignored alpha**, so `rgba(255,180,84,.08)` over a dark surface looked to it like a "light background" and it darkened the text to `#16273e` — **creating** a defect at 1.21:1 on top of a source that was already correct; and it looked only at elements carrying a background of **their own**, that is, never the commonest case	Rewritten: it walks the ancestors **compositing alpha** to obtain the effective background, measures the true WCAG ratio, and delegates the correction to `bsiAccentoLeggibile()`
`--g900` is `#0d1522` — **the same as `--bg`**, exactly like `--g800`: the legacy of a greyscale born for the *light* theme, where `--g900` was "the darkest text". With the theme inverted, that role no longer exists	The uses as a **text colour** (30 rules today) move to `--testo-forte`; the 16 uses as a **background** stay where they are
A colour chosen from data used as a **background** with the text hard-coded to `#fff`: it works while the hue is dark and stops as soon as someone adds `#e65100`	`bsiEtichettaLeggibile()` — the dual of the previous function: it picks the text by looking at the background and, if neither extreme suffices, **darkens the background while preserving the hue**, because a label must stay recognisable by colour

The steps as measured with the official bench, not estimated: **80 → 27 → 26 → 10 → 4 → 0**. Each step is a full re-measurement over the 87 sections after a group of changes; none of the figures above is attributed to a single cause, because the causes were not measured one by one.

The number remains **recorded as a baseline** in `evidence/accessibilita-riferimento.json`. Zero is not an achievement to file away: it is the value the bench defends. If a future change reintroduces even one defect, the battery **fails**.

> What stays outside automation is what automation cannot judge: text inside SVGs and text over gradient backgrounds (D-09). Those elements are **counted** and reported on every run, not passed over in silence.

4-bis. Deviations remedied in this version

They are listed because they concerned **the verification apparatus itself**, and a conformance statement that says nothing about the defects of its own evidence is not worth the paper it is written on.

#	Deviation remedied	Why it mattered
S-01	32 of the 39 benches were not in the repository: they lived in the working area of the session in which they had been written	It made **false** the sentence this whole document rests on — "anyone can re-run them". Whoever cloned the repository found 7 out of 39. They now live in `tools/banchi/`
S-02	A missing bench was counted but did not make the battery fail	A battery missing *all* its benches would have declared "CONFORMANT" with zero failures. A missing bench now makes the outcome **NON-CONFORMANT**
S-03	`package.json` and `package-lock.json` were excluded from the repository	`npm install`, the documented command for reproducing the evidence, installed nothing: the battery failed at the first bench with *Cannot find module*
S-04	The accessibility check inspected only the visible section	On `index.html` that meant one section out of 87. The "0 defects" held for the Dashboard, not for the application
S-05	`SCI-10` was defined twice in the matrix, and the coverage totals were wrong (36 declared out of 37 defined)	The first number an assessor reads was wrong. A bench now verifies it
S-06	50 broken links out of 127 in the delivery packages	Documentation handed to third parties with references that led nowhere
S-07	15 attachable PDFs stuck at documents 00-05 and at an earlier version	An obsolete attachment states false things with an air of authority
S-08	The numeric-claims bench looked at **the Italian documents only**	The English set drifted undisturbed: `docs/en/05` declared "84 sections" while the app has 87, and stayed stuck at version `bsi-v146` while the code was at 171. The English set is now compared against the measurement just like the Italian one, and a disagreement between the two languages fails the bench
S-13	The English translation stopped at 10 documents out of 16	A non-Italian-speaking assessor read less than two thirds of the documentation, and the missing part was the operational one: data model, interfaces, deploy, functional specifications, manual. All translated: the English set is now as complete as the Italian one, and the two consistency benches check it the same way

#	Deviation remedied	Why it mattered
S-12	The File Manager password had been left in clear text in the git history	Taking it out of the files does not take it out of the history: `git log -p` hands it to anyone, and as long as that password is in use the deterrent deters nobody. The only remedy is to **change it**, and that has been done: the source carries only the new SHA-256 digest, the old password remains in the history and no longer opens anything. Verified in the browser: the new one opens, the old one does not
S-10	SEC-02 — "no personal data leaves the device" — was verified by checking the two *mechanisms* of exit, not the exit	A check on mechanisms says "I cannot see how it would leave", which is not "it did not leave". `audit_rete` seeds an unrepeatable value into 71 stores of the user's data, then **uses** the application and inspects the URL, headers and body of every request. Proved in reverse with `BSI_PROVA_FUGA=1`, which causes a leak on purpose: the bench must fail, and it does
S-11	The deviations registry had no converse guard: an entry declared that **no longer deviates** simply stayed	An exemption nobody revokes is a permission left switched on, and tomorrow it would silently cover a wrong structure put in its place. The bench now fails for that too — verified by putting a remedied entry back
S-09	The "automatic contrast corrector" **created** the defects it was meant to remove	It ignored alpha: a `rgba(255,180,84,.08)` background over a dark surface looked light to it, and it darkened with `!important` a text that was already correct in the source. A remedy tool that worsens the thing it remedies is the hardest kind of defect to see, because it presents itself as the solution

5. Acceptance criteria applied

A version is published only if **all** the following criteria are satisfied.

- ☑ No non-network JavaScript error on any page
- ☑ `tools/verifica-farmaci.js` — no defect; every deviation recorded
- ☑ `verifica_guida` — no promise in the documentation without a counterpart in the code
- ☑ No conflict marker in the tracked files
- ☑ Service Worker cache version bumped and aligned with `BSI_APP_VERSION`
- ☑ Known deviations listed in §4 of this document
- ☑ Evidence report regenerated against the published commit
- ☑ **No missing bench** — a bench that is not there is not a bench that passed
- ☑ `tools/genera-pacchetti.js` closed with **0 broken links**
- ☑ `tools/genera-pdf.js` re-run: the attachments describe this version

6. Change management

Aspect	Procedure
Traceability	Every change goes through a pull request describing the defect, its cause and its verification
Versioning	The application version coincides with the Service Worker cache version (`bsi-vNNN`), incremented at every publication
Distribution	Automatic from `main` via GitHub Pages; the Service Worker signals the new version to open sessions without interrupting them

Aspect	Procedure
Regression	The full battery is re-run before every merge into `main`
Reversibility	Every version corresponds to a commit; rollback is a `git revert`

7. Intellectual property and licences

All third-party components are distributed under **permissive** licences (MIT, BSD-3-Clause, CC0, Public Domain), mutually compatible and free of strong copyleft constraints. The full bill of materials, with digests and residual obligations, is in `07-SBOM.md` and in CycloneDX format at `../evidence/sbom.cdx.json`.

The code written for this project is the author's own work.

8. Data handling

The application is **local-first**: there is no backend, no personal data is collected and no persistent identifier is present. Data generated by the user (notes, chats, API keys, uploaded files) stay in the device's `localStorage` and IndexedDB and can be deleted selectively from the interface itself.

Network calls are optional and go to documented public services (§4 of `07-SBOM.md`). If the user configures the proxy, the API keys reside on a Worker they own and not on the device.

Details in `03-Security-Privacy-Compliance.md`.

9. Signature

This document is drawn up by the author of the software, who takes responsibility for it. It does not constitute certification by a third-party body, and is not presented as such.

Anyone can verify what is declared here by re-running the procedure in §5 of `../evidence/RAPPORTO-VERIFICA.md` against the commit indicated, and comparing the SHA-256 digests of the files.

Samuele Pio Provenzano _Version `bsi-v219`._

Interface Reference — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v219`
Purpose	Document the project's HTTP interfaces and the external APIs the application consumes.

0. A necessary premise

BioSpecInfo exposes no API. It is an entirely client-side Progressive Web App, served as static files: there is no backend, no application server, no REST endpoints belonging to the product.

This document would therefore be empty, were it not for two interfaces that do exist and that an assessor is entitled to see documented:

	What it is	Documented in
Spectra proxy	An **optional** Cloudflare Worker, owned by the user, that forwards calls to the AI providers. It is the only HTTP interface written in this project.	§1, §2
External APIs consumed	Public third-party services the application queries, always optionally.	§3
Internal JavaScript APIs	The extension points of the code, for whoever has to work on it.	§4

> An OpenAPI specification of non-existent endpoints would be plausible and false documentation. Only what exists is described here.

1. Spectra proxy — overview

The proxy solves two problems that have no other solution from the browser:

1. **CORS.** Several AI providers do not send the headers that allow a web page

to call them directly. From the Worker the call originates from a server and the constraint does not apply.

1. **Custody of the keys.** With the proxy the API key lives among the Worker's

secrets and not on the user's device.

It is **optional**: without a proxy the application works using the keys the user enters locally.

Deploy

```
cd proxy
npx wrangler login
npx wrangler deploy
npx wrangler secret put GROQ_KEYS     # one or more keys, comma-separated
```

Configuration

Variable	Type	Default	Description
`ORIGINI`	secret/variable	*(empty = all)*	List of permitted origins, comma-separated. **Leaving it empty makes the proxy open to anyone:** not advisable.
`LIMITE_IP`	variable	`20`	Requests per single IP within the time window
`TETTO_GIORNO`	variable	`2000`	Total daily requests
`GROQ_KEYS`, `GEMINI_KEYS`, `ANTHROPIC_KEYS`, `OPENAI_KEYS`, `XAI_KEYS`, `NVIDIA_KEYS`, `ZAI_KEYS`, `DEEPSEEK_KEYS`	secrets	—	API keys. Several keys comma-separated: when the first one's quota is exhausted (429), the proxy moves to the next **within the same request**.

> The variable names are kept in Italian because that is what they are called in the Worker: translating them here would document a configuration that does not exist.

Endpoints

Method	Path	Description
`OPTIONS`	`/*`	CORS preflight
`GET`	`/stato`	Which providers have a key configured
`GET` `POST`	`/{provider}/{path...}`	Forwarding to the provider

Recognised providers: `anthropic` · `gemini` · `groq` · `xai` · `nvidia` · `zai` · `openai` · `deepseek`

Status codes

Code	Meaning
`200`	The provider's answer, forwarded as it is
`204`	CORS preflight accepted
`400`	Request body unreadable
`403`	Origin not permitted by the `ORIGINI` list
`404`	Unknown provider
`405`	Method other than GET, POST or OPTIONS
`429`	Per-IP limit or daily ceiling exceeded
`503`	No key configured for that provider

Security choices

- **No header is forwarded blindly.** Only `anthropic-version` and

`anthropic-beta` pass through; any credential sent by the client stays out. A proxy that forwards everything is an open relay.

- **The `key` parameter is not propagated.** If a client puts it in the query

string, the proxy discards it: the key is the server's.

- **The body is read once and kept**, because retrying with a different key

requires being able to send it again — and a stream is consumed.

---

2. OpenAPI 3.0 specification

> The `description` fields are translated here; the **error messages in the examples are left in Italian**, because that is what the proxy actually returns to a user of the Italian application. Translating them would document responses that do not exist.

```yaml
openapi: 3.0.3
info:
  title: Spectra Proxy
  version: "1.0.0"
  description: |
    Optional proxy (Cloudflare Worker) for calls to language-model providers.
    It removes CORS blocks and keeps the API keys on the server rather than on
    the user's device.

    It is not part of the BioSpecInfo application: it is published by the user
    on their own Cloudflare account, and the application uses it only if it
    knows its address.
  license:
    name: Proprietary — All rights reserved (Samuele Pio Provenzano)
servers:
  - url: https://{workername}.{account}.workers.dev
    description: The user's instance
    variables:
      workername: { default: spectra-proxy }
      account:    { default: example }

tags:
  - name: Status
    description: Querying the configuration without revealing secrets
  - name: Forwarding
    description: Forwarding requests to the model providers

paths:
  /stato:
    get:
      tags: [Status]
      summary: Configured providers
      description: |
        Lists the providers for which the proxy holds at least one key.
        It reveals no key. The application uses it to show only the models that
        actually work, instead of letting them fail one by one.
      responses:
        "200":
          description: List of active providers
          content:
            application/json:
              schema:
                type: object
                required: [fornitori]
                properties:
                  fornitori:
                    type: array
                    items:
                      type: string
                      enum: [anthropic, gemini, groq, xai, nvidia, zai, openai, deepseek]
              examples:
                twoKeys:
                  summary: Two providers configured
                  value: { fornitori: [groq, gemini] }
                none:
                  summary: Worker published but with no secrets
                  value: { fornitori: [] }
        "403":
          $ref: "#/components/responses/OrigineNonAmmessa"

  /{fornitore}/{percorso}:
    parameters:
```

```
    - name: fornitore
      in: path
      required: true
      description: Destination provider
      schema:
        type: string
        enum: [anthropic, gemini, groq, xai, nvidia, zai, openai, deepseek]
    - name: percorso
      in: path
      required: true
      description: |
        Path of the endpoint at the provider, reassembled as it is.
        Examples: `openai/v1/chat/completions` for Groq,
        `v1beta/models/gemini-3.6-flash:streamGenerateContent` for Gemini.
      schema: { type: string }
      example: openai/v1/chat/completions
  post:
    tags: [Forwarding]
    summary: Forward a request to the provider
    description: |
      The body is transmitted unchanged. The proxy adds the key in the way
      that provider requires (an `Authorization: Bearer` header, `x-api-key`,
      or a query parameter) and, if the call fails for exhausted quota,
      retries with the next key within the same request.
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            description: Body expected by the provider's API, transmitted unchanged
          examples:
            groq:
              summary: Streaming completion on Groq
              value:
                model: llama-3.3-70b-versatile
                stream: true
                messages:
                  - role: user
                    content: Which IR bands identify a carboxylic acid?
    responses:
      "200":
        description: |
          The provider's answer, forwarded without modification (typically
          `text/event-stream` when `stream: true`).
        content:
          application/json:
            schema: { type: object }
          text/event-stream:
            schema: { type: string }
      "400": { $ref: "#/components/responses/CorpoIlleggibile" }
      "403": { $ref: "#/components/responses/OrigineNonAmmessa" }
      "404": { $ref: "#/components/responses/FornitoreSconosciuto" }
      "429": { $ref: "#/components/responses/TroppeRichieste" }
      "503": { $ref: "#/components/responses/ChiaveAssente" }
  get:
    tags: [Forwarding]
    summary: Forward a read request to the provider
    description: |
      It serves, among other things, the resolution of the Gemini model, which
      queries `ListModels` with GET and could not pass without a key in the
      browser.
```

```
      responses:
        "200":
          description: The provider's answer
          content:
            application/json:
              schema: { type: object }
        "403": { $ref: "#/components/responses/OrigineNonAmmessa" }
        "404": { $ref: "#/components/responses/FornitoreSconosciuto" }
        "429": { $ref: "#/components/responses/TroppeRichieste" }
        "503": { $ref: "#/components/responses/ChiaveAssente" }

components:
  schemas:
    Errore:
      type: object
      required: [errore]
      properties:
        errore:
          type: string
          description: Message in Italian, meant to be shown to the user
  responses:
    OrigineNonAmmessa:
      description: The request's origin is not in the `ORIGINI` list
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Origine non ammessa da questo proxy." }
    FornitoreSconosciuto:
      description: The first path segment does not match a known provider
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Fornitore sconosciuto: pippo" }
    CorpoIlleggibile:
      description: The request body could not be read
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Corpo della richiesta illeggibile." }
    TroppeRichieste:
      description: The per-IP limit or the daily ceiling has been exceeded
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Troppe richieste da questo indirizzo: riprova fra un minuto." }
    ChiaveAssente:
      description: The proxy has no key configured for that provider
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example:
            errore: "Il proxy non ha una chiave per groq. Impostala con: wrangler secret put
GROQ_KEYS"

security: []
# CLIENT authentication is deliberately absent: the access control is the
# ORIGINI list plus the per-IP limits. Adding a client-side token would mean
# writing it into a static page, that is, making it public.
```

3. External APIs consumed by the application

All **optional**: with no network the application stays fully functional on local data, and says so to the user.

Service	Endpoints used	What it is for	If it does not answer

PubChem PUG-REST	`/compound/name/{name}/property/...`
	 `/compound/smiles/{smiles}/SDF?record_type=3d`
	 `/compound/cid/{cid}/PNG`
		IUPAC name, formula, CAS, GHS pictograms, 3D conformers, 2D images	The app uses the local tabulated data and says so

RCSB PDB	`https://files.rcsb.org/download/{pdb}.pdb`	Protein structures for the 3D viewer	Explicit message in the viewer
NASA / ESA	Public images and data	Astrochemistry module	The module shows the tabulated data
AI providers	At the user's choice, or via the proxy	Assistant answers	The assistant declares it and offers a reachable alternative

None of these calls transmits personal data. There are no persistent identifiers and no active telemetry (`BSI_TELEMETRY_URL` is empty).

> This is not asserted from the architecture alone: `audit_rete` seeds a canary value into the user's own data, uses the application across 6 pages and 87 sections, and inspects the URL, headers and body of every outgoing request.

---

4. Internal JavaScript APIs

Extension points for whoever has to work on the code. They are globals on `window` because the application is a set of classic scripts, not ES modules.

`window.BSISpettri` — spectral engine

Function	Signature	Description
`gruppi`	`(smiles, callback)`	Recognises functional groups via SMARTS/RDKit. Calls back with `null` if RDKit is unavailable: the caller decides whether to fall back or declare that it cannot compute.
`gruppiSync`	`(smiles) → object | null`	Synchronous version: answers only if the molecule is already cached
`bandeDaGruppi`	`(groups) → band[]`	Table of IR bands `{n, pk, w, i, f}`
`svgIR`	`(bands, options) → string`	IR spectrum in transmittance, SVG, deterministic

Function	Signature	Description
`legendaIR`	`(bands) → string`	Assignments in HTML
`quandoPronto`	`(callback)`	Notifies when the analysis of a molecule is available, so it can be redrawn
`precarica`	`()`	Warms up RDKit while nobody is waiting

`window.bsiProxy` — proxy state

Function	Description
`url()`	Configured address, empty string if absent
`stato()`	Promise: queries `/stato` and records which providers are covered
`copre(providerId)`	Whether that provider is covered by the proxy
`ricarica()`	Re-reads the configuration

Other notable points

Symbol	Description
`window.bsiLoadRDKit(cb)`	Loads RDKit once only and calls back with the instance
`window.irBandList(smiles)`	IR bands — replaced by the new engine, keeps the original signature
`window.makeIRsvg(bands, colour)`	IR drawing — likewise
`window.nmrPeakList(smiles)`	^{1}H peaks with integration
`window.bsiCancellaDati(groups)`	Selective deletion of the saved data
`window.bsiOpenUpdate()`	Opens the updates window
`window.bsiAccentoLeggibile(colour, background, threshold)`	Adjusts an accent so it reaches the WCAG threshold on that surface, preserving the hue
`window.bsiEtichettaLeggibile(background, threshold)`	The dual: picks the text colour from the background and, if neither extreme suffices, darkens the background while preserving the hue

---

Document updated to version `bsi-v219`.

Data Model — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v219`
Purpose	Document where the data live, with what structure, and what lifecycle they have.

0. A necessary premise

BioSpecInfo has no server database. There is no backend, so there are no centrally managed relational tables, no migrations, and no users stored anywhere.

There is, however, a real data model, and an assessor is entitled to see it documented. It is organised across three levels:

Level	Technology	What it holds	Where it lives
Preferences and progress	`localStorage`	Settings, API keys, AI history, study progress	The user's browser
Document archive	IndexedDB	Photos, documents, HTML sections, notes, activity log	The user's browser
Teaching datasets	SQLite via `sql.js`	Exercise tables, created in memory at each session	Memory, not persisted

All data stay on the device. They are not transmitted, there is no server-side user identifier, and the user can delete them selectively from the interface.

1. Entity diagram

```
erDiagram
    DISPOSITIVO ||--|| PREFERENZE : "possiede"
    DISPOSITIVO ||--|| ARCHIVIO : "possiede"
    DISPOSITIVO ||--o{ SESSIONE_SQL : "apre"

    PREFERENZE {
        string bsi_ai_provider "fornitore AI scelto"
        json   bsi_api_keys "chiavi per fornitore"
        string bsi_proxy_url "indirizzo del proxy"
        json   bsi_ai_threads "conversazioni salvate"
        json   bsi_ai_memory "memoria dell-agente"
        json   bsi_prov_ko "fornitori irraggiungibili"
        json   bsi_modelli_ko "modelli ritirati"
        json   bsi_tetti "tetti di token appresi"
        json   bsi_srs "ripetizione spaziata"
        json   bsi_quiz_progress "avanzamento quiz"
        string bsi_section "ultima sezione aperta"
    }

    ARCHIVIO ||--o{ PHOTOS : contiene
    ARCHIVIO ||--o{ DOCS : contiene
    ARCHIVIO ||--o{ SECTIONS : contiene
    ARCHIVIO ||--o{ NOTES : contiene
    ARCHIVIO ||--o{ ACTIVITY : registra

    PHOTOS {
        int    id PK "autoIncrement"
        string name
        string type "MIME"
        int    size "byte"
        string data "data URL"
        int    added "epoch ms"
    }
    DOCS {
        int    id PK "autoIncrement"
        string name
        string type
        int    size
        string data "data URL"
        int    date "epoch ms"
    }
    SECTIONS {
        int    id PK "autoIncrement"
        string name
        string icon
        string html "contenuto"
        int    size "caratteri"
        int    added "epoch ms"
    }
    NOTES {
        string key PK "sempre main"
        string text
        int    updated "epoch ms"
    }
    ACTIVITY {
        int    id PK "autoIncrement"
        string icon
        string msg
```

```
        string time "data locale"
    }
```

> The diagram keeps the entity and field names as they appear in the code, in Italian: renaming them in the translation would document a model that does not exist.

---

2. Preferences and state — `localStorage`

2.1 AI assistant

Key	Type	Content	Lifecycle
`bsi_ai_provider`	string	Identifier of the selected provider	Until the user changes it
`bsi_api_keys`	JSON	`{ providerId: key }`	Until explicitly deleted
`bsi_api_key`	string	Single key, legacy format	Migrated to `bsi_api_keys`
`bsi_proxy_url`	string	Proxy address	Until "Disconnect"
`bsi_ai_threads`	JSON	`{ threads: [...], activeId }` — max **30 chats**, **100 messages** each	Pruned automatically
`bsi_ai_memory`	JSON	Facts the agent remembers between sessions	Deletable
`bsi_ai_history`	JSON	Condensed history	Deletable
`bsi_nucleo`	string	"Core" mode active	—

2.2 What the assistant learns about providers

These three keys are the reason the assistant improves with use instead of repeating the same mistakes. They have an expiry because today's failure must not be a permanent sentence.

Key	Structure	Expiry	What it is for
`bsi_prov_ko`	`{ providerId: { t: epoch, n: count } }`	**24 hours**	Providers that do not answer from this device. Demoted in the selection, never removed.
`bsi_modelli_ko`	`{ digest: { modelli: [...], t } }`	**7 days**	Models retired by the provider, so as not to offer them again. Indexed on the digest of the key.
`bsi_tetti`	`{ providerId: tokenCeiling }`	Permanent	Token ceiling learned from the error message rather than hard-coded

2.3 Study and progress

Key	Content
`bsi_srs`	Spaced repetition: cards, intervals, next review
`bsi_sm2`	Parameters of the SM-2 algorithm
`bsi_quiz_progress`, `bsi_quiz_history`	Quiz progress and history
`bsi_studypath`	The study path chosen

Key	Content
`bsi_notes`, `bsi_note_<section>`	Personal notes, global and per section
`bsi_ds_prog`, `bsi_ds_proj`	Progress in the data-science module
`bsi_section`	Last section opened, to resume from there
`bsi_guide_v3`	Introductory guide already seen

2.4 Identifiers and licence

Key	Content	Privacy note
`bsi_device_id`	Random local identifier	**Not transmitted.** It only distinguishes installations within the same browser.
`bsi_user_email`	Email, if entered by the user	Entered voluntarily, not transmitted
`bsi_pro_license`, `bsi_trial_start`	State of Pro mode	Local
`bsi_rdkit_lib2`	Molecule library of the laboratory	Local

> That none of these ever leaves the device is verified directly by `audit_rete`, which seeds a canary value into these very keys and then inspects every outgoing request.

---

3. Document archive — IndexedDB

Database name	`bsi_filemanager_v1`
Schema version	`2`
Object stores	`photos`, `docs`, `sections`, `notes`, `activity`

3.1 Structure of the stores

Store	Key	Fields	Notes
`photos`	`id` (autoIncrement)	`name`, `type`, `size`, `data`, `added`	Images as data URLs. Per-file limit: **20 MB**
`docs`	`id` (autoIncrement)	`name`, `type`, `size`, `data`, `date`	Generic documents
`sections`	`id` (autoIncrement)	`name`, `icon`, `html`, `size`, `added`	HTML pages uploaded by the user
`notes`	`key`	`key` (`"main"`), `text`, `updated`	A single entry
`activity`	`id` (autoIncrement)	`icon`, `msg`, `time`	Log of the most recent operations

> The `files` store, present in earlier schema versions, is no longer created from version 2.

3.2 Opening and unavailability

Opening IndexedDB is **asynchronous**, and in some configurations (Safari in private browsing, site data blocked) it **never happens**. The code awaits a promise rather than reading a variable that might not yet be set, and distinguishes two cases:

- **reading** with the store absent → returns an empty set, the page shows

"nothing yet";

- **writing** with the store absent → the error reaches whoever pressed the

button, and a persistent warning appears at the top of the page saying nothing will be kept.

If another tab requests a version upgrade, the connection is closed (`onversionchange`) so as not to block both.

---

4. Teaching datasets — SQLite in memory

The data-exercise module creates an SQLite database **in memory** through `sql.js` (SQLite compiled to WebAssembly). The tables are generated at each session from the datasets bundled in the repository and are **not persisted**: closing the tab makes them vanish.

The choice is deliberate: the exercise must start from a known state every time, and no user data end up in a database they did not create.

---

5. Reference data — constants in the code

The scientific data are not in a database but in constant structures inside `index.html` , loaded with the application. They are read-only and versioned together with the code.

Structure	Content	Verification
`FARM_DATA`	**263 drugs**: name, category, SMILES, molecular weight, mechanism of action, indications, adverse effects, class; 36 entries also carry `chembl` and the record's `formula`	`audit_farmaci` — structure ⟷ weight, duplicates, and the formula rebuilt from the graph against ChEMBL's
`UV_DATA`	Chromophores, λmax, extinction coefficients, Woodward's rules	`audit_dati`
`MS_DATA`	Characteristic fragmentations by class	`audit_dati`
`MOLECOLE_ESAME`	Tabulated exam molecules	`test_spettri`
`PHYS_SIMBOLI`	Physical constants with symbol and unit	`test_costanti`
`SMARTS` , `SMARTS_H`	Patterns for functional-group recognition	`test_spettri`

---

6. Lifecycle and deletion

The user can delete data **by group**, from the interface, without wiping everything. The groups are defined in `bsiCancellaDati()` .

Group	What it removes
`chiavi`	API keys, chosen provider, proxy address, memory of unreachable providers
`chat`	Conversations and the assistant's memory
`studio`	Spaced repetition, quizzes, study path, notes
`tutto`	Every `bsi_*` key and the entire IndexedDB archive

Deleting the archive proceeds even if a single entry fails, and reports how many were not removed: a "delete everything" operation that stops halfway without saying so would be worse than the problem it solves.

---

7. Retention and space limits

Constraint	Value	Behaviour when exceeded
`localStorage`	5–10 MB depending on the browser	Writes fail. The app stays operational: chats are pruned (30 chats, 100 messages), and every write is guarded.
IndexedDB	Quota granted by the browser	Error reported to whoever pressed, with a persistent warning
Single image	20 MB	Refused with an explicit message; the upload of the others continues

Behaviour with space exhausted is verified by the `audit_quota` and `test_filemanager` benches: ten pages out of ten stay usable with `localStorage.setItem` forced to fail.

Document updated to version `bsi-v219`.

Deploy Guide — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v219`
Purpose	Operational procedure for publishing, updating and rolling back the application.

0. What gets published, in practice

BioSpecInfo is a set of **static files**. There is nothing to compile, no process to keep running, no database to provision.

Build	None. The repository files *are* the application.
Server runtime	None
Environment variables	None for the application. Only the optional proxy has any (§4).
Database	None server-side
Server requirement	Serve static files over HTTPS with the correct MIME types

The one non-obvious requirement: the server must return `application/wasm` for `.wasm` files, otherwise RDKit and SQLite do not start. GitHub Pages, Netlify, Vercel and Cloudflare Pages already do.

1. Environments

Environment	How it is obtained	What it is for
Development	`python3 -m http.server 8899` from the root	Daily work and verification benches
Preview	A branch published on Pages, or `npx serve`	Checking before merging
Production	GitHub Pages from the `main` branch	`samupropio1-ship-it.github.io/BioSpecInfo-v11/`

> **Why a server is needed even in development.** Opening `index.html` with a double click (`file://`) does not work: RDKit and SQLite are WebAssembly modules and require an HTTP context. It is the reason the verification benches also run against a local server.

2. Publishing to production

2.1 Procedure

```
# 1. from the working branch, the full battery
python3 -m http.server 8899 &
node tools/genera-evidenza.js          # must close with 0 failures

# 2. bump the version — with the tool, NEVER by hand
node tools/porta-versione.js bsi-v<NNN+1> --prova   # look first
node tools/porta-versione.js bsi-v<NNN+1>           # then write

# 3. regenerate SBOM, evidence, PDFs and packages — IN THIS ORDER
#    The packages bundle the PDFs and the evidence: built first, they carry
#    the typeset documents and the report of the PREVIOUS version.
#    genera-pacchetti now notices and stops, but better not to get there.
node tools/genera-sbom.js
node tools/genera-evidenza.js       # the battery; needs a server on :8899
node tools/genera-pdf.js            # 36 PDFs; every footer carries version and commit
node tools/genera-pacchetti.js      # the three packages; fails on broken links, stale PDFs or
stale evidence

# 4. merge into main
git add -A && git commit -m "…"
git push -u origin <branch>
#    then pull request and squash merge
```

GitHub Pages publishes automatically from `main`: the *pages build and deployment* workflow takes about a minute.

2.1-bis Publishing the version's release

The application is online as soon as `main` updates; the **release** serves a different purpose: giving an assessor a fixed point to download — a tag, the version notes and the delivery packages attached.

It is published from the repository, not from outside:

> **Actions → release → Run workflow**, field `versione`: `bsi-v<NNN>`.

The workflow creates the tag on the current commit, opens the release and attaches the three packages from `consegna/`. The full archive — `models/` included, about 124 MB — is generated by GitHub itself as *Source code (zip)*.

Why from the repository and not from the command line. The credentials a remote session uses on the repository are authorised to write to branches and **not to tags**: the `POST` to `git-receive-pack` for `refs/tags/…` answers `403 Forbidden`, while a new branch goes through. The Actions token, by contrast, writes to the repository that hosts it.

What the workflow **refuses** to do:

Condition	Outcome
`CHANGELOG.md` has no `## [bsi-v<NNN>]` entry with content	fails: a release without notes looks like a version and does not say what changed

Condition	Outcome
fewer than three packages in `consegna/`	fails: a missing attachment would make the release silently incomplete
the tag does not point at the commit the packages came from	fails: a release must have a single provenance

That last row exists because of a real defect. On re-running the workflow to refresh the packages, `gh release edit` updated notes and assets but **did not move the tag**: the *Source code (zip)* archive, which GitHub generates from the tag, stayed at the first publication's commit while the packages came from a later one. Two provenances, and nothing saying so. The tag is now moved, and a following step does not take that on trust: it asks GitHub where the tag points and compares.

The workflow is **re-runnable**: if the release already exists it updates notes, assets and tag instead of failing halfway. Moving an already-published tag is not free — whoever downloaded it already has different content — so it applies to a release just published, not an old one: for that, publish a new version.

2.1-ter Why the version bump is not done with a find-and-replace

It looks like the most harmless operation in the world: search `bsi-v<NNN>`, write `bsi-v<NNN+1>`. **For three releases in a row it was done that way, and every time it rewrote historical sentences** — sentences recording a fact that happened at a past version. What came out of it sat inside a signed conformance statement:

the document said	the truth
the password had been **changed at the current version**	it was changed at version `bsi-v188`
the tag had been **moved to the current version's commit**	the incident happened at version `bsi-v188`
the thirty-six entries were **added in the current version**	they were added in `bsi-v188`
"bump `CACHE` (e.g. `bsi-vN` → `bsi-vN`)"	an example equal to itself

No check could catch it: every line named a version that **exists**, in the document describing that version.

> This table uses placeholders rather than the real numbers because the two checks described below **would reject it** — which is the proof that they work.

`tools/porta-versione.js` touches **only** the lines that DECLARE the version — headers, footers, the README badge, package names, the two code lines — and **lists** everything else so it gets read. With `--prova` it writes nothing.

Two checks in `tools/verifica-documenti.js` now close the door: no past-tense sentence may name the current version, and no upgrade example may show the same version on both sides.

> **Name the thing, not the number.** "From the Pretsch tables" rather than "from version such-and-such": it reads better and does not age.

2.2 The two version lines

`CACHE` in `sw.js` and `BSI_APP_VERSION` in `index.html` **must match**.

- `CACHE` determines whether the Service Worker considers the cache stale:

without a bump, devices keep serving the old files and the publication reaches nobody.

- `BSI_APP_VERSION` is what the user sees in the "Updates" window and what the

update check compares with the `sw.js` on GitHub.

They drifted apart once in the past — the second had been stuck for months — and the result was that the update check was comparing the wrong number.

2.3 How the update reaches users

1. The browser detects a different `sw.js` and installs the new Service Worker.
2. `skipWaiting()` activates it at once; the old caches are deleted.
3. The Service Worker **notifies** the open tabs (`BSI_SW_UPDATED`) — it does not

reload them on its own authority.

1. The page decides: if no work is in progress it reloads itself; if the

assistant is answering or there is unsaved text, it shows "Update now / Later".

Whoever opens the app at that moment already receives the new version.

---

3. Publishing alternatives

The application does not depend on GitHub Pages. Any static file host will do, provided it is over HTTPS (the Service Worker requires it).

Platform	Command / configuration	Notes
GitHub Pages	Settings → Pages → branch `main`, folder `/`	In use. Free, custom domain possible
Netlify	`netlify deploy --prod --dir=.`	No build command
Vercel	`vercel --prod`	Static project, no framework
Cloudflare Pages	`npx wrangler pages deploy .`	Convenient if the proxy is already in use
VPS with nginx	Copy the files to `/var/www/biospecinfo`	See the configuration below

nginx configuration

```
server {
    listen 443 ssl http2;
    server_name biospecinfo.example.org;
    root /var/www/biospecinfo;

    # Without this MIME type, RDKit and SQLite do not start
    types { application/wasm wasm; }

    # sw.js must not be cached by the CDN: it is what announces new versions
    location = /sw.js {
        add_header Cache-Control "no-cache, no-store, must-revalidate";
    }

    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

A rule that holds everywhere: `sw.js` must not be cached

If a CDN keeps `sw.js`, the browser keeps seeing the old version and **no update ever arrives**, however much one publishes. This holds on every platform.

---

4. Optional proxy — a separate deployment

The proxy is a **distinct** component, owned by the user, published on their own Cloudflare account. It is not required in order to use the application.

```
cd proxy
npx wrangler login
npx wrangler deploy
# the address is printed: https://spectra-proxy.<account>.workers.dev

npx wrangler secret put GROQ_KEYS      # one or more keys, comma-separated
npx wrangler secret put GEMINI_KEYS    # optional
```

The address is then pasted into the assistant's "Proxy" box, inside the app. The same four steps are available **inside the application**, with the commands ready to copy.

Recommended configuration

```
# Without this variable the proxy accepts requests from any origin
npx wrangler secret put ORIGINI
# value: https://samupropio1-ship-it.github.io
```

Variable	Default	Why change it
`ORIGINI`	*(empty = all)*	**Always set it.** An open proxy is usable by anyone, at the expense of your quota.
`LIMITE_IP`	`20`	Requests per IP within the window
`TETTO_GIORNO`	`2000`	Overall daily ceiling

> **Keys must never go into `wrangler.toml`.** That file is versioned: a key written there ends up on GitHub, and the bots that scan it find it within hours. Use `wrangler secret put`, which keeps them out of the repository.

---

5. Domain and certificate

Aspect	Situation
HTTPS	Mandatory: without it the Service Worker does not register and the app does not work offline
Certificate	Provided and renewed automatically by GitHub Pages
Custom domain	Settings → Pages → Custom domain, plus a `CNAME` record pointing to `<user>.github.io`

---

6. Monitoring

There is no server to monitor: there are no processes that could fall over, nor resources that could run out.

What to watch	How
Outcome of the publication	GitHub → Actions → *pages build and deployment*
Availability	An HTTP check on `index.html` (UptimeRobot or similar)
Client-side errors	The browser console. Remote telemetry exists but is **disabled** (`BSI_TELEMETRY_URL` empty): enabling it requires a deliberate decision about data handling.
Proxy consumption	Cloudflare → Workers → Analytics

---

7. Rollback and emergency procedures

7.1 Going back to the previous version

```
git revert <commit>
git push origin main
```

Pages republishes in about a minute. Devices receive the restored version as a normal update — **provided `CACHE` changes**: a revert that also takes the version number backwards is fine, because what counts is that the value be *different*, not that it be greater.

7.2 "I published but I do not see the changes"

In order:

1. **Was the version bumped?** It is the cause in ninety per cent of cases.

Without it, the Service Worker serves the cache.

1. **Has Pages finished?** GitHub → Actions.
2. **Is `sw.js` cached by a CDN?** It must answer `no-cache`.
3. From the app: ✨ menu → **Updates** → *Check*, which forces a

`registration.update()`.

1. As a last resort: developer tools → Application → Service Workers →

Unregister, then reload.

7.3 "The assistant has stopped answering"

This is not a publishing problem: it concerns the AI provider. In the app, the 🔌 **Test** button measures in a few seconds which providers the device can actually reach, without needing any key.

7.4 Restoring the user's data

The data live on the device and **there are no copies on a server**: they cannot be restored remotely. This is a direct consequence of the local-first architecture, and it is declared to the user. Exports (notes, documents) remain the only form of backup.

---

8. Pre-publication checklist

- ☐ `node tools/genera-evidenza.js` closes with **0 failures**
- ☐ `CACHE` in `sw.js` and `BSI_APP_VERSION` in `index.html` bumped and **equal**
- ☐ `node tools/genera-sbom.js` re-run
- ☐ `node tools/genera-pdf.js` re-run — otherwise the attachments describe the previous version
- ☐ `node tools/genera-pacchetti.js` re-run and closed with **0 broken links**
- ☐ Evidence report regenerated against the commit being published
- ☐ Any new deviations noted in `09-Release-Conformance-Statement.md` §4
- ☐ No conflict markers (`verifica_guida` checks this across 58 files)
- ☐ No API key in the versioned files
- ☐ The accessibility debt **has not grown** — if it has fallen, regenerate the

baseline with `node tools/verifica-accessibilita.js --aggiorna-riferimento` and record it in the commit

- ☐ Code coverage **has not fallen** — `audit_copertura` fails if it has

Document updated to version `bsi-v219`.

Functional Specifications — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v219`
Purpose	Describe what the product does, for whom, under which rules and with which limits.

1. Product description

BioSpecInfo is an **interactive cheminformatics** platform for university study of chemistry, biochemistry, pharmacology and astrochemistry.

The whole scientific logic — interpretation of molecular structures, spectral prediction, calculations, 2D and 3D visualisation — runs **in the student's browser**. There is no server: the application installs as an app, works with no network, and transmits no data.

1.1 Objectives

Objective	How it is pursued
Total accessibility	No registration, no cost, no server. Whoever opens the link is already using it.
Independence from the network	Everything needed is bundled and cached: it works in a lecture hall, on a train, in a library with no signal.
Verifiable correctness	Scientific data are put through automatic checks; the limits of the models are declared rather than hidden.
Privacy by construction	There is no backend that could collect data: the absence of collection is not a promise, it is a consequence of the architecture.

1.2 What the product is not

Not a medical device	It does not support diagnosis, prescription or dosing (Reg. EU 2017/745)
Not a certified analytical instrument	The spectra are predictions, not measurements
Not a collaborative platform	No accounts, no sharing, no synchronisation between devices

2. Actors

Actor	Description	What they can do
Student	Primary user. Studying for a university exam.	Every function: consultation, calculation, quizzes, spaced repetition, AI assistant
Teacher	Uses the app as teaching support	The same functions; uploading their own material into the File Manager

Actor	Description	What they can do
Technical assessor	A company or committee examining the work	Reading the documentation and re-running the verification benches

There are no technical roles: there is no authentication, so there is no distinction of permissions. The File Manager has a local protection that is a **deterrent**, not an access control (see §6).

---

3. Functional areas

The application has **96 sections**, grouped by area of study.

3.1 Spectroscopy

Function	Description
Spectroscopy centre	A molecule is entered (SMILES, name, formula) and one obtains IR, ^{1}H/^{13}C NMR, MS, UV-Vis, Raman, 2D and 3D structure, functional groups
Visual IR spectra	Characteristic bands by class of compound
Mass spectrum	Fragmentation patterns and diagnostic rules
UV-Vis	Chromophores, λmax, Woodward's rules
Per-atom NMR prediction	Draw or type a structure and get ^{1}H and ^{13}C with WHICH atoms produce each signal; interrogable spectrum (zoom, intensity, click a peak), assignment table, SMILES/InChI history, CSV/JCAMP-DX/PNG exports
Two-dimensional NMR	COSY, edited HSQC and HMBC built by walking the graph; clicking a spot highlights the atoms producing it. It is not a simulation of the experiment, and the panel says so
Open any file, and the program decides	One entry point, with no extension filter: open what you have and the program **states what it decided the file is**. A document (PDF, photograph, Word, OpenDocument, PowerPoint, Excel, text) opens, **everything** it contains is shown (every page drawn, all the text), the recognised spectroscopic data appear in a table **with their provenance**, and the working proceeds step by step; a JCAMP-DX file or two columns of numbers become a measured spectrum; an image does both. Whatever could not be read is declared
Header language switcher	A round button top right, at the title's height, with the flag of the language in use; it opens into a disc with the fourteen flags on two shells and, at the centre, the language name and **how much it is translated, measured**. The actual switch stays in the languages module: this is only the way to call it
Game: "The Element Run"	A retro platformer where the power-ups are **periodic table elements**, each with an ability that comes from its chemistry (twenty elements). Twelve levels in four zones, a boss per zone and the final boss; levels generated from a seed — therefore random every run and **repeatable** — and verified passable against the jump's physics. Five difficulties with chemistry names, from "dilute solution" to "nuclear fusion". At the checkpoints a question **generated from the application's tables** (nineteen generators: elements, IR/^{1}H/^{13}C/J spectra, unsaturation, masses, reactions, retrosynthesis, amino acids, languages), with the provenance shown after the answer. On-screen and keyboard controls. **It works offline**: rectangle graphics and synthesised sounds, no file to download

Function	Description
Spectrum from a figure (digitisation)	From a photograph of a chart recorder, a crop from a paper or the instrument screen, the trace is extracted from the **pixels**: the axis frame is recognised and only its inside is digitised, the line position is the **darkness-weighted centroid** (sub-pixel), the grid is worked around by continuity and darkness, both axes are calibrated linearly or logarithmically, %T is converted to absorbance with $A = -\log_{10} T$. The chart, the peaks and the assignments appear by themselves and are redone at every change of the calibration; a diagnostics line says what was decided and why. CSV and JCAMP-DX outputs. Measured deviation: mean 2.2 cm^{-1}, worst 7.3 over seven constructed figures
3D model from the drawn structure	Coordinates built from the graph, carrying the predictor's own indices: a clicked peak lights up the atom in three dimensions too
NOE and 2D NMR (tables)	Reference values for spatial correlations
Reference tables	Tabulated values for quick consultation
Spectrum quiz	Recognising the functional group from a spectrum

3.2 General and physical chemistry

Interactive periodic table · chemical calculator · reaction balancing · degrees of unsaturation · distribution diagrams · VSEPR · molecular orbitals · advanced thermodynamics · electrochemistry (Nernst) · pKa and buffers · laboratory calculator · constants and formulae.

3.2-bis Separation — chromatography

Chromatography section: a chromatogram computed from the parameters (theoretical plates, retention factor, selectivity, dead time) with an overview and an expanded view of the peak pair; a **van Deemter** curve with the three terms separated and the minimum computed in closed form; a **Kováts retention index** calculator; a table of the quantities with their working ranges; GC/HPLC comparison; a guide to detectors; diagnostics for peak-shape defects.

> **Rule applied.** The chromatogram is not an illustration: the Gaussians have σ = t_R/√N and the resolution shown is the one that would be measured on those peaks. The section explicitly declares what it does **not** do — predict a molecule's retention time from its structure, which would require experimental parameters of the stationary phase that cannot be derived from the SMILES alone.

3.2-ter Calibration and uncertainty

Calibration section: ordinary least-squares regression on data pasted from a spreadsheet, with slope and intercept **accompanied by their standard deviation**; a plot of the line with a 95 % confidence band and a **residual plot**; the unknown's concentration with a full confidence interval (the three sources 1/M, 1/N and distance from the centroid are shown separately); **LOD and LOQ** per ICH Q2(R1); uncertainty propagation indicating which contribution dominates; significant-figure rules.

Two diagnostics that an r^2 does not provide:

Diagnostic	Method	Why r² is not enough
Non-linearity	F-test between line and parabola on the same data	An r² of 0.984 accompanies an obvious saturation curve: r² measures how much variance the line explains, not whether the line is the right model
Outlier	**Deleted** residual: the judging line is rebuilt without the point being judged	The naive criterion `

> **The second row is a trap met while writing the section**, not a textbook case: the 3 σ criterion was already written, and the example meant to demonstrate it was not being flagged. The check passed without seeing anything.

3.2-quater Cheminformatics and QSAR

Cheminformatics section: a workbench that goes from the raw set to a verdict on the model, entirely in the browser. SMILES are pasted in — with a name and an activity, where there are any — and six panels return what a cheminformatics group produces in a day.

Panel	What it does
Descriptors	Standardisation (largest fragment, canonical SMILES, deduplication), 43 RDKit descriptors, Lipinski's rule of five, Veber filter, QED recomputed — MinimalLib does not expose it
Similarity and groups	Morgan/ECFP4, RDKit, pattern and MACCS fingerprints; Tanimoto and Dice; similarity matrix; nearest neighbours; **Butina clustering** and **MaxMin** selection for diversity
Chemical space	PCA on the standardised descriptors, with explained variance per component and the **loadings** that say which descriptor moves which axis; **Bemis–Murcko** scaffolds with the molecule count per scaffold
QSAR model	Kernel ridge (Tanimoto kernel), kNN over Tanimoto or logistic regression; **scaffold split**; **grouped cross-validation** over 3, 5 or 10 folds; **model comparison on identical folds** with the trivial reference at the top; hyperparameters chosen by **nested validation**; **conformal prediction intervals**; **learning curve**; applicability domain
Search	Substructure search across the whole set with RDKit's `SubstructLibrary`, SMARTS or SMILES query, occurrence counts, eight ready queries
SAR table	R-group decomposition around a core: one row per molecule, one column per substitution position, with each substituent's median activity. Molecules without the core are **excluded**, not shown with empty cells
Matched pairs	Pairs differing by a **single** substitution, grouped by transformation with median, range and **number of concordant pairs**
Activity cliffs	Structurally close pairs with distant activities, ranked by **SALI**: these are the pairs every model gets wrong, and it is honest to show them
Structural alerts	PAINS and Brenk, with the matched fragment highlighted and the reason it is flagged

Three choices that separate a workbench from a demonstration:

Choice	Method	Why
Scaffold split	The test set contains **only scaffolds never seen** in training	A random split puts close analogues on both sides: the model recognises rather than predicts, and the R² that comes out does not survive the first new molecule
Null model by scrambling	The same model is retrained on **shuffled labels**, eight times; the verdict is the comparison	An R² of 0.4 can arise by chance when the molecules are few. If the random twins reach the same score, the model has learnt nothing — and the panel **says so**, instead of showing only the good number
Deduplication on the canonical form	`OC(=O)C` and `CC(=O)O` are the same molecule	The same molecule in training and in test is an information leak, and the written string is not enough to see it

> **The verdict is written out in words, not left to the reader.** The QSAR panel does not stop at the metrics: it states whether the model beat all its random twins, and by how much. Measured on the 28-inhibitor example: R² 0.861 against a maximum of 0.574 among the eight twins — margin 0.287. On the same molecules with random labels the model does **not** beat the twins, and the panel says so in the same way.

The results can be taken away

Two exports, both formed in memory by the browser: there is no network request, which is why the note beside the buttons can say that no data leaves the device without asking anyone to take it on trust.

Export	Content
CSV table	One row per molecule: name, canonical and original SMILES, activity, scaffold, nine descriptors, QED, Lipinski violations, Veber outcome, alerts found. With cross-validation active, **predicted** and **residual** are added — and only then, because cross-validation is the one scheme that gives every molecule a prediction from a model that had not seen it
SDF	The structures with **generated 2D coordinates**, the name on each block's first line and the data fields attached: it is the format in which sets are exchanged between groups and programs. Verified by round-trip
Method report (.md)	What is needed to **redo** the same analysis: RDKit version, fingerprint, split type, number of molecules and scaffolds, **the seed**, the metrics, the null-model verdict, the folds one by one, and the declared limits

> **Why the seed is in the report.** Without it, "random split" and "scrambled labels" mean nothing: two runs would give different numbers and nobody could say which was the right one. The generator is a seeded mulberry32, and the same seed gives the same folds.

A measurement no other bench could make

`verifica-accessibilita` walks the 96 sections and measures what is **visible**. The six panels of this section live inside a container that stays `display:none` until the analysis has been run: that bench never saw them, and its "contrast 0" was not speaking about them.

`test_cheminfo` runs the analysis for real, opens the six panels in turn and measures WCAG contrast on every piece of text that appears there — **460 elements, 0 defects**. The formulas are rewritten inside the bench rather than taken from the application: a bench that asked the code under test how good its own contrast was would be measuring nothing.

The engine lives in `bsi-cheminfo.js`, is exposed as `window.BSIChem` and is verified by the `test_cheminfo` bench (189 checks) against hand-computable values.

3.3 Organic chemistry

Retrosynthesis (98 exercises) · synthesis and mechanisms with electron arrows · animated mechanisms · structure editor · *Organic Chemistry 1+2* module with 25 modules.

3.4 Biochemistry

Amino acids · biomolecules · animated metabolic pathways · enzyme kinetics (Michaelis-Menten) · 3D macromolecules · interactive biological structures · biochemistry guide · exam biochemistry.

The **biochemistry guide** is a document of its own (`Biochimica_Guida_Definitiva.html`, 8.2 MB) loaded into a frame by the section: 12 chapters plus the index, 78 subsections, 98 embedded figures, and two chapters devoted to the exam — "Oral exam guide" and "Real exam questions and correct answers". It contains **no scripts and makes no network requests**: it cannot raise JavaScript errors, and the alternative text of the 98 figures comes from the captions the author wrote, not from invented descriptions (see `docs/06` §3-ter).

The document closes with a **self-test of 12 multiple-choice questions**, recovered from the previous guide — where it was a JavaScript quiz — and rewritten **without a line of code**: the four options are in plain sight, the solution sits in a native `<details>` that opens on a click. In print the `<details>` disappears and a dedicated copy of the solution appears, so the printed copy does not lose the explanations.

3.5 Pharmacology

Function	Description
3D Atlas	One chooses a body region, a disease and a drug: the model highlights the target organ
Database	**263 drugs** with structure, molecular weight, mechanism, indications, adverse effects, class. For **36** the entry carries the ChEMBL identifier and the record's formula, and the formula is rebuilt from the graph and compared
Interactions	Checking interactions between active ingredients
Pharmacokinetics	Calculator for half-life, clearance, volume of distribution
Clinical cases	Diagnosis-therapy paths for teaching purposes

3.6 Astrochemistry

Molecules of the interstellar medium · exoplanets and JWST data · nebulae · stellar spectra · nucleosynthesis · cometary chemistry · quizzes.

3.6-bis Spectrum reader

You load a file from your own instrument, or paste a text, and it is read.

Function	Description
Formats	**JCAMP-DX** (`.jdx` , `.dx`) with full **ASDF** compression — PMAI for digits, DIF for differences, DUP for repeats — or two columns of numbers separated by spaces, commas or tabs
Integrity check	The JCAMP format prescribes that, after a run of differences, the next value written in full must equal the computed one. That check is performed here
Peak finding	By **prominence**, with noise estimated from the median absolute deviation (MAD × 1.4826) rather than the standard deviation, which one tall peak inflates
IR assignment	For each band, **all** compatible assignments among 24 ranges are listed, not the first
Mass spectra	A peak list is recognised as such; differences between ions are matched against 17 **neutral losses**
NMR	Integration of areas by chemical-shift region
Drawing	IR with **decreasing** wavenumbers, the way it is read

What it does not do, stated in the section itself: it does not deduce the structure. A set of bands is compatible with many molecules, and a program that produced a name from three peaks would make a claim the data cannot support.

3.6-ter Languages and chemical languages

Function	Description
Interface language	**14 languages** with a selector and a search field: Italian, English, Spanish, French, German, Portuguese, Dutch, Polish, Romanian, Greek, Russian, Chinese, Japanese, Arabic. Each one's coverage is **measured**, not declared
Writing direction	Arabic carries `dir="rtl"` ; every other language `ltr` , including on the way back
Chemical languages	**25 outputs** from one SMILES: formula and mass composition, canonical/flat/explicit-hydrogen SMILES, CXSMILES, InChI and InChIKey, the key's skeleton block, Murcko scaffold, CIP stereo tags, SMARTS and CXSMARTS, molfile V2000 (also aromatic) and V3000, XYZ, PDB, CML, JSON, and the **six fingerprints** (Morgan, MACCS, RDKit, atom pairs, topological torsions, pattern)
IUPAC naming	Over a **declared class** (acyclic, neutral, C H O N F Cl Br I, one kind of principal group, linear substituents). Outside that class it **refuses** instead of guessing

3.7 Study tools

Function	Description
Spaced repetition	SM-2 algorithm on cards generated from the contents
Quizzes	By subject, with score and history
Study path	A guided sequence of topics
Notes	Global and per section
Pomodoro	Study timer
File Manager	Personal archive of photos, documents and HTML pages

3.8 The "Spectra" assistant

An AI agent with **35 tools** that can interrogate the application itself: compute descriptors, generate spectra, look up constants, balance reactions, open sections. It supports ten providers, and stays usable when one of them fails (§5.2).

---

4. Main flows

4.1 Analysing a molecule

```
1. Open the Spectroscopy centre
2. Enter the molecule:
     · SMILES                → used directly
     · name or formula       → resolution via PubChem (needs network)
     · drawing in the editor → SMILES generated
3. The application shows: 2D structure, functional groups, descriptors,
   predicted spectra (IR, NMR, MS, UV-Vis), drug-likeness
4. The 3D structure loads if there is a network (conformer from PubChem)
```

With no network: point 3 stays complete for SMILES entered directly; resolution by name and the 3D structure are unavailable, and the application says so instead of hanging.

4.2 Asking the assistant

```
1. Tap the "Spectra" button
2. On first use: choose a provider and enter an API key,
   or connect a proxy (and then no key is needed)
3. Type the question
4. The assistant answers, using its tools when needed:
   every tool used is shown, not hidden
```

If the provider does not answer: the assistant notes it, demotes it in the selection for 24 hours and moves to another configured provider. The 🔌 **Test** button measures in a few seconds which providers the device actually reaches, **without needing any key**.

4.3 Studying with spaced repetition

```
1. Open the study section
2. Generate a deck from the contents
3. For each card, rate how well it was remembered
4. The SM-2 algorithm computes when to show it again
5. On reopening, only the cards that are due appear
```

4.4 Archiving your own material

```
1. File Manager → local password
2. Upload photos, documents or HTML pages
3. The material stays ON THE DEVICE (IndexedDB), it is not transmitted
```

---

5. Business rules

5.1 Scientific data

Rule	Rationale
A molecular structure enters the database only if the recomputed weight agrees with the literature one (±0.6 u)	A plausible but wrong datum is indistinguishable from a correct one for the user
If a structure does not pass verification, the entry stays **without a structure**	A wrong datum replaced by another wrong one is no progress
Every absence must be **recorded with its reason**	An unjustified exception becomes a shortcut
Spectra are deterministic	Two spectra of the same molecule must be comparable
An ambiguous physical constant is not resolved at random	A wrong value given with confidence is worse than a request for clarification

5.2 AI assistant

Rule	Behaviour
A provider that does not answer is remembered for **24 hours**	Then it is retried: today's failure is not a sentence
A retired model is learned from the error message	Hand-coded lists grow stale
An imposed wait longer than **10 seconds** switches provider, if a ready one exists	Beyond that threshold the user believes the app has frozen
One turn at a time	Two overlapping submissions corrupt the history
Paid providers are not offered as free ones	Without active billing one hits the limit at the first question

5.3 User data

Rule	Behaviour
At most 30 conversations, 100 messages each	Beyond that, `localStorage` fills up and **every** subsequent save fails
Images up to 20 MB	Beyond that, explicit refusal; the others proceed
Selective deletion by group	Wiping everything to free space is an avoidable loss
A failed write **must** be declared	"✅ Saved" when it is not true is worse than an error

6. Current limitations

Declared, not hidden. The detail is in `09-Release-Conformance-Statement.md` §4.

#	Limitation	Practical consequence
1	**2 entries with no structure to show** (it was 6)	Ivermectin is a mixture of B1a/B1b homologues and Coartem a combination of two active ingredients: a single SMILES notation does not represent them, and for these two the 2D/3D and predicted spectra are missing. Digoxin, vincristine and tacrolimus (topical and systemic) now have their structure, taken from ChEMBL and verified against the molecular weight
2	**Predicted, not measured, spectra**	Useful for recognising functional groups; not for formal identification
3	**No synchronisation**	Data do not pass from one device to another, and there are no copies on a server
4	**File Manager: deterrent, not security**	On a static site, whoever reads the source bypasses any page-side control. The source carries only the SHA-256 digest of the password, never the password
5	**Functions that need a network**	IUPAC name, CAS, GHS, 3D conformers, AI assistant
6	**Accessibility automated, not complete**	Contrast, accessible names, labels, heading hierarchy across 13 pages and all 96 sections: **0 defects**, including text over gradients (judged against the worst stop). What stays outside is text inside SVGs and text over a real background image, counted on every run, and everything requiring human judgement
7	**Chromatography: no retention prediction**	The section computes resolution, efficiency and indices **given** k, α and N; it does not predict k from a structure, which would require experimental parameters of the stationary phase
8	**Partial code coverage**	Measured: **49.89 %** of statements over the widest path a bench walks. It is not whole-battery coverage, and it is statements, not branches; see `08-Traceability-Matrix.md` §8
9	**Verification on Chromium only**	Firefox and WebKit are tested by hand, not by a bench

7. Possible developments

Not commitments: directions consistent with the architecture.

Area	Proposal	Why
Data	The 2 remaining entries have no structure to show	It is not a gap to close: ivermectin is a mixture, Coartem a combination. At most one could show the **components**, declared as such
Spectra	Comparison with experimental databases (SDBS/NIST) inside the app	It would make the distance between prediction and measurement visible
Accessibility	Add `axe-core` alongside, for the rules the bench does not implement	ARIA roles, tab order and focus management stay outside the current measurement
Export	Full export of the user's data	The only possible remedy for the absence of backups
Assistant	A preconfigured proxy for those who do not want to publish one	It would remove the last barrier to AI access

Document updated to version `bsi-v219`.

A guide for the people who use it — BioSpecInfo

This guide is written for people who study, not for people who program. You do not need to know anything about computing.

What it is

BioSpecInfo is an application for studying chemistry, biochemistry, pharmacology and astrochemistry. It opens like a website, but you can **install** it on your phone or computer and it works **without internet** too.

Three things to know straight away:

- **No sign-up needed.** No account, no email, no password.
- **It costs nothing.**
- **Your data stay yours.** Notes, files and conversations never leave your

device: there is no server they could reach.

👉 **samupropio1-ship-it.github.io/BioSpecInfo-v11**

Installing it (optional, but worth it)

Installing gives you an icon on your screen, full-screen opening and offline operation.

Device	How to do it
Android / Chrome	⋮ menu → *Install app* (or *Add to Home screen*)
iPhone / iPad	Share button → *Add to Home Screen*
Computer	Install icon in the address bar, on the right

After installing, open the app once with the network on: it will download everything it needs. From then on it works with no signal too.

Finding your way around

Element	Where it is	What it is for
🔍 **Search**	At the top	Searches the whole app. Shortcut: `Ctrl+K`
Section menu	Navigation bar	The 96 sections, grouped by subject
✨	Bottom right	Tools, settings and updates
Spectra	Bottom right	The assistant that answers questions

The things you will do most often

Analysing a molecule

1. Open the **Spectroscopy centre**
2. Type the molecule into the box. You can use:

- the **name** — `aspirin`, `caffeine`
- the **SMILES formula** — `CC(=O)Oc1ccccc1C(=O)O`
- or **draw it** with the editor

1. Press Enter

You get the 2D and 3D structure, the functional groups, the molecular weight, the predicted spectra (IR, NMR, mass, UV-Vis) and the pharmacological properties.

> **If you have no network:** searching by *name* does not work, because it has to consult an external database. The SMILES formula, on the other hand, always works.

Reading an IR spectrum

The spectrum is drawn as on a real instrument: the scale at the bottom runs from 4000 to 400 cm^{-1} (**from left to right the numbers go down**) and the bands point **downwards**.

The coloured area below 1500 cm^{-1} is the **fingerprint region**: that is where the specific molecule is recognised, not just its class. Below the chart you will find the list of bands with their assignment.

> The spectra are **predicted** from the structure, not measured in a laboratory. They are there to help you recognise functional groups and to understand the link between structure and spectrum. For experimental data, the app sends you to the official databases.

Everything about spectra, in one place

What you do with a spectrum used to live in **nine sections** across two different menus. Now there is 🔬 **Spectra centre**, a single section with ten tabs:

tab	what it is for
Read your spectrum	a file, a photo, a PDF
Predict NMR	^{1}H and ^{13}C assigned per atom
2D NMR and NOE	COSY, HSQC, HMBC
Mass	fragmentation and neutral losses
IR	bands and functional groups
UV-Vis	chromophores and λmax
Reference tables	^{1}H, ^{13}C, IR, J constants
Everything on one molecule	editor, spectra, groups, comparison
Elucidate a structure	from the data to a proposal

tab	what it is for
Practise	recognise the group from the spectrum

The panels are the same ones as before: they were **moved**, not copied — there is no second copy of anything. The old links still work: if something in the application used to take you to "Mass spectrum", it now opens the Spectra centre on the **Mass** tab.

> **Two tables of J constants, with different numbers.** There were two: one gave the aromatic ortho ^3J as 6–9 Hz, the other as 7–10. Whoever looked that number up got a different answer depending on where they came in. **One** remains, the one inside the Spectra centre, which sits in the section with its own reference list.

Changing language: the round button, top right

At the title's height, at the far right, there is a **round button** with the flag of the language in use. It opens into a **disc**: the fourteen flags on two shells — six inside, like a benzene ring, and eight outside — and at the centre appears the name of the language you are moving to and **how much it is translated**. That percentage is not hand-written: it is measured by applying the dictionary and counting what stays in Italian.

Touch a flag and the interface changes at once. `Esc` closes the disc.

> The 🌍 **Language** section remains, and does something else: it explains the languages, shows each one's coverage and says that Arabic carries the writing direction with it. That one **informs**, this one **acts**.

🎮 The Element Run

A retro platformer, in its own section. The power-ups are not flowers: they are **periodic table elements**, and each one transforms you with an ability that comes from its chemistry.

element	what it gives you	why
H	you jump higher	the lightest there is
He	you fall slowly	noble gas lighter than air
C	four shots in a fan	four bonds
F	the hardest shot	the most electronegative
Al	you take one hit	it passivates with its own oxide
Fe	you smash through falling	heavy and ferromagnetic
Hg	you run	liquid metal
Au	oxidising enemies cannot touch you	it does not oxidise

…and twelve more. Name and atomic number are read from **this application's periodic table**: they are not rewritten in the game.

Twelve levels in four zones, a **boss** at the end of each zone and the **final boss** at the twelfth. Levels are generated at random every run, but never impassable: no gap wider than a jump covers, no step higher than a jump clears — and before every boss you are guaranteed a shooting

element, because a boss without one would be invincible.

Controls: `◂ ▸` move, `⤒` jumps (hold it to jump higher), `✦` uses the element's ability. Keyboard: arrows or `A/D`, space jumps, `K` acts. Jumping **on** an enemy squashes it.

The five difficulties have chemistry names, and they are not decorative — they say how reactive the system is:

difficulty	what changes
Dilute solution	5 lives, everything slow, basic questions
Saturated solution	4 lives, more enemies and more power-ups
Exothermic reaction	3 lives, and **a wrong answer costs**
Enzymatic catalysis	2 lives, everything faster, 25 s to answer
Nuclear fusion	**one life**, fierce enemies, 20 s to answer

At the checkpoints (the flags) a chemistry question arrives. The difficulty does not make the same question harder: it changes **which** questions come — from an element's symbol to the right retrosynthetic disconnection for a target, by way of IR bands, ^{1}H and ^{13}C shifts, J constants, degrees of unsaturation, masses, reactions, amino acids and chemical languages.

> **The questions are not written in the game.** They are **generated** from this application's tables, and after each answer I tell you **where** that question comes from. If the table is right, the question is right — and there is no second list of chemistry that could diverge from the real ones.

It works offline: there is not a single file to download — the graphics are drawn as rectangles and the sounds are synthesised.

Where things are

The sections live in the **navigation bar**, grouped by subject, with 🛠️ **Utility** among them: laboratory, GHS, concept map, molecule comparison, molecule search, atlas, resources, materials, notes, file manager, statistics, data science, pomodoro, lab calculator, **articles**, and the two language ones.

The ✨ menu holds only what is *not* a section: the editors, the overlay labs, and actions such as exporting a figure, printing, sharing, updating. The rule is simple and always holds: **if something is in the bar, it is not also in the menu**. It used not to be, and the menu showed two "Data Science", two "File Manager", two "Articles" and two "Updates".

Two exceptions, and they have a reason:

- **Spectra** is not in the bar: it has its own dedicated button, the pulsing

one at the bottom. Three roads to the copilot were too many.

- The **spectrum reader** is not in the bar: it is a tab of the Spectra centre.

> **Two File Managers, and they are two different things.** In the bar, 📁 **File Manager** is for uploading and organising your own notes (PDF, images, Word). In the ✨ menu, 🔒 **Protected File Manager** is a separate page with the password, for the gallery and the custom sections. They had the same name and looked like a duplicate: now they are named for what they do.

The articles and the Nobel Prizes

📰 **Articles** (from the ✨ menu) has two tabs.

Latest articles shows what came out in the last two months, by topic: spectroscopy, synthesis, catalysis, drugs, materials, biochemistry, computational. It is not a hand-written list — that would be stale tomorrow: it comes from **Crossref**, the DOI registry, at the moment you open the tab. Each title leads to its **DOI**, which is a permanent address, and beside it there is a link to **read it in Italian**.

Nobel Prize in Chemistry lists every prize. Years, laureates and motivations come from the **Nobel Foundation**, live: they are not written inside the application. The **explanation** in Italian is — and it is matched to the *topic* of the official motivation, not to the year, so no attribution comes from here. For each prize you get the official page, the Foundation's scientific and popular material, the Italian translated view and an Italian Wikipedia search — all addresses **computed** from the year, not written.

> **If a prize has no explanation, I say so.** I would rather write "for this prize I have no explanation" and leave you the official motivation and the Foundation's material, which is the source, than make something up.

> **It needs the network.** Without it you do not see a stale list passed off as current: you see what I asked for, from whom, and why it did not arrive — with the address of the request, which you can open by hand.

Reading a spectrum you measured yourself

The **Spectrum reader** section does the opposite of the one above: it does not predict a spectrum from a structure, it reads one you already have.

There are **two buttons**, and they are not duplicates.

`📂 Open a file` filters nothing: whatever you have in your folder, open it. `📷 Photo or image of a spectrum` instead opens the **phone gallery** and also lets you take the picture there and then — on a phone these are two different system pickers, not two filters of the same one, and without the second there is no way to hand over a photograph.

A line under the buttons then **tells you what it decided the file is**:

What you open	What happens
a **PDF**, a **Word** file, an OpenDocument, a spreadsheet	it opens, shows you all of it and **works through** what it contains
a **photograph** (`.png` , `.jpg` , …)	both things: the **trace** is extracted and the image is also shown as a page, because it may be a photo of a spectrum or of a sheet

What you open	What happens
a **JCAMP-DX** (`.jdx`, `.dx`), compressed form included	it is a spectrum: the format itself says so
a **text** file or a `.csv`	decided **from the content**: if the numbers form a curve it is a spectrum, otherwise it is **problem text** and gets worked through

If you prefer, **paste** the data or the problem text into the box and press **Read and work it out**: the same reasoning applies.

> There used to be two cards, and the spectrum one allowed only `.jdx` and `.csv`. On a phone that filter did not narrow, it **hid**: anyone arriving with the PDF of an exam saw an empty folder. Now there is one door and it filters nothing.

If you have no file at hand, the three buttons **IR example**, **MS example** and **NMR example** load a sample spectrum, and **example: an exam problem** loads a complete problem.

Inside the section there is 📖 **How to use it — full guide**: seven short chapters covering the two buttons step by step, the calibration, what the program does by itself, what it cannot do, how to recognise a wrong result from the symptom you see, and how to take the numbers away.

A spectrum you only have as a figure

The trace is extracted from the **pixels**, and the chart appears at once. Then it must be **calibrated**, because the axis numbers are not in the image:

field	what you write in it
x at left / x at right	the values at the two edges of the plot — for an IR typically 4000 and 400 **in that order**, because an IR is written with decreasing wavenumbers
y at bottom / y at top	the values of the vertical axis; for a figure in %T, 0 at the bottom and 100 at the top
y unit	`%T`, `T`, `A` or intensity. Picking `%T` derives the absorbance with $\mathbf{A = -\log_{10} T}$, which is the definition
x/y logarithmic	if an axis is logarithmic, say so: applying a linear map to it does not raise an error, it produces wrong numbers that look right
frame	normally I find it; "the whole figure" is for when the image is already cropped to the plot only

Change a field and the chart redraws by itself: there is nothing to confirm. Below it the **diagnostics** appear — how many columns, whether the frame was found and on which sides, background and threshold, how many columns were grid — because a number without its how cannot be challenged.

What I do by myself: I find the **axis frame** and digitise only what is inside it (the numbers written on the axes are dark pixels just like the trace); I **survive the grid**, picking in each column the run near where the trace was a pixel earlier and dark enough to be the trace; I read at **sub-pixel**, because the centre of a two-pixel line sits in between.

> **How much I get wrong.** Measured over seven constructed figures whose band centres are known: **mean 2.2 cm⁻¹** deviation, **worst 7.3**. One column of a figure 900 pixels wide over 3600 cm⁻¹ is worth about 4 cm⁻¹: below that there is nothing to gain, it is the limit of the figure.

> **If the calibration is wrong**, the peaks come out at wrong numbers while sitting in the right place on the figure: the shape is recovered from the pixels, the calibration is not. And a photo taken at an angle carries a distortion I do not straighten — frame the figure head-on.

With **download CSV** and **download JCAMP-DX** the numbers leave this place: any spectroscopy program opens the JCAMP, and this very reader reads it back identical. Inside it is written that the calibration was given by hand.

Below the chart you will find the peaks found with their signal-to-noise ratio, and for each band **all** compatible assignments. If you read two rows at 1715 cm⁻¹, that is not a defect: at that wavenumber a ketone and a carboxylic acid are both possible, and naming only one would be inventing.

> The reader **does not tell you which molecule it is**. A set of bands is compatible with many different molecules: the step from spectrum to structure is yours to take, with everything else you know about the sample.

Opening a problem and having it worked through

The same button, `📂 Open a file`. It takes the PDF of the exam, a photograph of the sheet, a Word file, plain text — and shows you **everything** inside: every page drawn (click to enlarge) and the full text.

Then, if the text contains spectroscopic data, it recognises them and puts them in a table **with where it took each one from**. That table sits above the working on purpose: look at it first. If it mistook a mass for an IR band, the working that follows will be impeccable and the answer wrong, and you catch it there.

The working lists the steps with the **evidence** for each and how certain it is: degrees of unsaturation are arithmetic, the carbon count from the M+1 peak is an estimate with a margin, an IR band is *compatible with* several things.

> **It does not give you the structure.** A set of spectroscopic data is compatible with more than one molecule, and a program producing a single one would offer a certainty the data do not contain. What you can do is the opposite: write a structure yourself and the program **compares** it, signal by signal, saying where it fails.

> **If the document is a scan**, the text cannot be read: a scanned page contains pixels, not letters, and there is no optical character recognition. The app tells you instead of pretending the sheet was empty. The pages stay visible, and if one is a photograph of a spectrum the trace can be extracted with **Open an image**.

Choosing the language

The **Languages** section holds **fourteen** languages, with a search field to find them quickly: Italian, English, Spanish, French, German, Portuguese, Dutch, Polish, Romanian, Greek, Russian, Chinese, Japanese and Arabic. Choosing Arabic also flips the page's **direction**, right to left.

Next to it is **Chemical languages**, which is a different thing: it converts a molecule among the twenty-five notations chemistry writes it in — SMILES, InChI, SMARTS, molfile, XYZ, PDB, CML, the fingerprints — and, for a declared class of simple molecules, writes its **IUPAC name**. Outside that class it says so, rather than attempting a wrong name.

Studying with cards

1. Open the study section
2. Generate a deck of cards
3. For each card, after answering, say how well you remembered it

The app works out **when** to show it to you again: the things you know well come back rarely, the uncertain ones come back soon. When you reopen, you only find the cards that are due.

Looking up a drug

Open **Pharmacology**. For each of the 263 drugs you will find the structure, molecular weight, mechanism of action, indications, adverse effects and class.

> ⚕ **For teaching purposes only.** This information does not replace the judgement of a doctor or a pharmacist, and is not there to decide a therapy.

Spectra, the assistant

Spectra answers chemistry questions and can use the app's own tools: calculate, generate spectra, look up constants, balance reactions.

Getting it working

Spectra relies on an external artificial-intelligence service, and you need a free **key**. Two routes:

The simplest — a key of your own

1. Go to console.groq.com and create a key

(free, it takes two minutes)

1. Open Spectra, paste the key into the 🔑 box
2. Done

The definitive one — a proxy

If services often fail to answer for you, or you would rather not keep the key on your phone, you can publish a *proxy*: a small free program that keeps the key safe and makes even the services that are blocked from the browser work.

In the assistant's **Proxy** box you will find **"How to set it up, in 4 steps"**, with the commands ready to copy.

When Spectra does not answer

Almost always it is not the app's fault but the external service's. Press 🔌 **Test** in Spectra's panel: in a few seconds it tells you **which services your device can actually reach** — and you need no key to find out.

If a service stops answering, Spectra remembers it for 24 hours and moves it to the bottom of the list, under *"Did not answer from this device"*. It does not delete it: tomorrow it might work again.

The File Manager

A personal archive for photos, documents and HTML pages, protected by a password.

> ⚠️ **Something you need to know.** This protection is a **deterrent**, not real security: whoever knows where to look can get around it. Do not put confidential documents in it.

The files stay on your device. **There is no copy anywhere else:** if you clear your browser data or change phone, they are gone. Use the export for the things you care about.

Frequently asked questions

Do I have to pay for anything? No. The app is free. If you use Spectra, the AI service's free key has daily limits, but nothing is paid.

Does it really work without internet? Yes, after the first opening. The only things left out are those that by their nature have to ask someone else: searching molecules by name, 3D structures downloaded on the spot, and Spectra.

Are my notes saved somewhere? Only on your device. There is no server: nobody can read them, but nobody can recover them for you either if you lose them.

How do I delete my data? ✨ menu → Settings → delete data. You can choose **what**: only the keys, only the conversations, only the study data, or everything.

I updated but I do not see anything new. ✨ menu → **Updates** → *Check*. There you can also see which version you are running.

Can I trust the spectra for a report? For recognising functional groups and understanding the structure-spectrum relationship, yes. For a figure to be cited as experimental, no: they are predictions. The app sends you to the official databases (NIST, SDBS) with direct links.

I found an error in a chemical datum. Report it: open an *issue* on the GitHub repository. The data go through automatic checks, but no check finds everything.

If something goes wrong

Problem	What to try
The app does not open	Reload it. If it persists, ✨ menu → Updates → Check
I do not see the new features	The version is in the ✨ menu → Updates
Spectra does not answer	The 🔌 **Test** button: it tells you which services you reach
"Out of space"	✨ menu → Settings → delete the old conversations
A section stays empty	Check the network: some sections load data from the internet
The charts are blurred	Reload the page; the charts are redrawn at your screen's resolution

Contacts

Author: Samuele Pio Provenzano **Repository and reports:** github.com/samupropio1-ship-it/BioSpecInfo-v11

Guide updated to version `bsi-v219`.

Test Documentation — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v219`
Purpose	Describe how the tests are organised, how to run them, what they cover and where they leave gaps.

1. Kinds of test present

Kind	Present	Notes
End-to-end (E2E)	✅	Playwright on headless Chromium against a local HTTP server. It is the prevailing form.
Data verification	✅	Scientific data compared against an independent source
Stability tests	✅	Long sessions, exhausted storage, degraded network
Documentation/code consistency	✅	Verifies that what the documentation promises really exists
Isolated unit tests	❌	See §6
Instrumented code coverage	✅	`audit_copertura`, with Chromium's profiler: no build, no source rewritten. See §6

Why E2E and not unit tests

The application is made of classic scripts operating directly on the DOM, with no modules and no dependency injection: there are no units that could be isolated without a deep rewrite. The defects that actually occurred — a blurred canvas, a button unreachable because its container is collapsed, an AI provider that stops answering — **would not have surfaced** from a unit test anyway: they live in the integration between code, DOM and browser.

The choice is therefore reasoned, not endured. Its limits remain, and are declared in §6.

2. How to run the tests

Requirements

```
npm install                      # playwright-core only
python3 -m http.server 8899 &    # RDKit WASM requires an HTTP context
```

> Without a server every test fails: opening the files over `file://` prevents the WebAssembly modules from loading.

The whole battery, with a report

```
node tools/genera-evidenza.js
```

It runs every bench, transcribes its full output and produces `docs/evidence/RAPPORTO-VERIFICA.md` with environment, commit and SHA-256 digests. It exits with code `0` only if no bench fails.

```
node tools/genera-evidenza.js --veloce   # skips the benches that need a browser
```

The `--veloce` option is useful for a quick check but **is not sufficient for an attestation**: the benches it skips are precisely the ones measuring real behaviour.

A single bench

```
node tools/verifica-farmaci.js           # a stand-alone tool
node tools/verifica-farmaci.js --json    # machine-readable output

cd tools/banchi && node test_spettri.js  # one E2E bench
```

The E2E benches live in **`tools/banchi/`**, versioned like the rest of the repository; the `BSI_BANCHI` variable allows another folder to be pointed at.

> **It has not always been so, and that is worth saying.** Up to version `bsi-v168` the default folder was the working area of the session in which the benches had been written: **32 of the 39 benches were not in the repository**. Whoever cloned it and ran the documented command found seven, and received no warning — because a missing bench was counted but did not make the battery fail, which closed with "CONFORMANT" and zero failures.
>
> Both things have been corrected: the benches are versioned, and **a missing bench makes the outcome NON-CONFORMANT**. A bench that is not there is not a bench that passed.

---

3. Composition of the battery

65 benches, grouped by what they demonstrate.

3.1 Scientific data

Bench	Verifies	Checks
`audit_farmaci`	Structure ⟷ weight, duplicates, and the **formula rebuilt from the graph** against ChEMBL's for every entry declaring a provenance	263 entries, 36 with a source
`test_spettri`	Functional-group recognition on reference molecules	36
`test_spettri_ui`	Grafting of the engine into the app's spectrum-drawing points	16
`test_assi`	Axis conventions in the SVG spectra	13
`test_assi_canvas`	Axis conventions in the canvas spectra	6

Bench	Verifies	Checks
`test_costanti`	Physical-constant lookup and refusal of ambiguities	45
`audit_dati`	Consistency of the tabulated data	29
`test_farm_ui`	That the Drug Atlas **draws** every drug it holds in memory, category by category, and that a **never-labelled** category still appears	11
`test_astro`	The astrochemistry data: molecular weight ⟷ formula for the 30 interstellar molecules against IUPAC atomic weights, no repeated celestial body, every quiz question with a valid answer among its options	13
`test_datasci`	The maths the Data Science section **displays**: R^2 on an exact relation, standardised weights in the true ratio, ROC on hand-computable values, behaviour on pure noise, and that **no section is left unreachable**	23
`test_simmetria`	Point groups, normal modes, IR/Raman selection rules	26
`test_cheminfo`	The cheminformatics engine: Tanimoto and Dice against hand-computable values, Morgan and MACCS fingerprints, Butina clustering, Bemis–Murcko scaffolds, PCA, kernel ridge, scaffold split, **null model by label scrambling**, grouped cross-validation, CSV and SDF exports, **contrast of the nine panels**, agreement between lab and engine, fragmentation, **matched pairs against known effects**, substructure search, SAR table, model comparison in both directions, enrichment (EF and BEDROC) on hand-computable cases, conformal intervals and their **refusal on classification**	189
`test_spettrolettore`	The spectrum reader and the models that follow from it: ASDF symbol by symbol, explicit and DIF/DUP-compressed JCAMP, prominence peaks on **built** spectra **and on pure noise**, curve versus peak list, neutral losses on toluene, ESOL against four experimental values, four drug-likeness filters reported separately, random forest both ways, **a single entry point** (one button, empty `accept`, routing checked on PDF/ `.docx` / `.txt` / `.jdx` / `.csv` /a disguised binary/an image) and the version badge compared with `BSI_APP_VERSION`. **And the relationship between the two IR tables**: `SPEC_DATA.ir` is for studying, `BANDE_IR` for assigning, they are not a duplicate — but the assignment umbrella must **cover** the reference for each of the 23 groups present in both, and not be arbitrarily wider (margin derived from the data, not invented)	66
`test_nmr`	Per-atom NMR prediction: two separate molecule sets (tuning and **validation**, the latter never used to choose parameters), the signal count on molecules whose count is known, refusals on an unreadable SMILES and on an empty string, the **history** with duplicate detection by InChI key, and the **peak ↔ structure ↔ 3D** sync verified on the indices, not on how many atoms light up	64
`test_nmr2d`	Correlation maps counted on the graph: HSQC with inverted sign on CH_2, COSY between non-equivalent protons, HMBC two and three bonds away. **In both directions**: hexafluorobenzene gives no HSQC spots, benzene **no** off-diagonal COSY spots, methane **no** HMBC spots despite having four protons	40

Bench	Verifies	Checks
`test_digitalizza`	A spectrum extracted from a **figure**, over seven constructed images whose Gaussian centres are known to the cm⁻¹: bare figure, with frame and labels, with a grey grid, with a black grid, with frame **and** grid, dashed, in %T. **Mean 2.2 cm⁻¹** deviation, **worst 7.3** — one column is worth ~4 cm⁻¹, which is the limit of the figure and not of the method. **In both directions**: from a blank sheet and from the app icon **no** curve comes out; the crude fallback on the same framed figure finds **one** peak instead of three; on a logarithmic axis the midpoint is the geometric mean and on a linear axis the arithmetic one. %T→A conversion on exact values (50 %T → A = 0.301) and a **full round trip** of the exported JCAMP, read back by the reader itself	45
`test_atomic`	The round language switcher and **Atomic Protocol**. The switcher: round, top right, **at the measured title height**, fourteen flags on two shells, and **both ways** the language actually changing and coming back identical. The questions: **28 generators each tried**, 600 sound questions and **none ambiguous** on ranges. The **hundred levels** generated and walked: twin doors, no unreachable or overlapping room, bench-terminal-exit in each, the elements needed to beat every guardian present before it, the same level from the same seed; the twelve equations counted atom by atom, ten distinct guardians in wide arenas and **a wrong equation rejected**. Level 1 played up to the elevator: elements, card, synthesis (and a pair that does not react, and a recipe not yet known, producing nothing), the residue dissolved by chlorine and not by salt, the terminal that **stays shut on a wrong answer** and the bulkhead that does not let you through while shut, the save and the resume with the inventory. The guardians **both ways**: PUMP 7 phase by phase, HELIX punishing the wrong answer, PROTOCOL 12 ending the game. The engine **measured in the live scene**: high graphics (PS2 level) as the default, with more than five glows in scene; **first person** by default with the arms in their own scene, five fingers, the tool showing the compound held, spray droplets and a gaze that turns but not beyond a head; the protagonist with **modelled eyes and five fingers**, a head above 3000 triangles and a figure above 15000; the painted face, coat tails, ankles, **ten machine guardians with no box at all**; PS1 at 480×360 as an option, with smooth vertex lighting and the 15-bit shader; the chemist **moving forward facing where it goes** and knees never bending backwards; memory not growing across ten levels; and modern graphics (ACES, sRGB, shadows, PBR) as an option, without losing the game. The thirty logs placed, one played with subtitles and archived in the Codex; the left hand with sodium sparks and chlorine vapour, and the throw; brass and marble in the environments; segmented health. The game's floating button above Spectra, opening the game, and the floating buttons not covering the scene in game; in first person A and D strafe. The **first-person models**: the rig arms with the skeleton and the eight animations, the robotic left hand, the gun in the right fist (grip inside the ring of fingers, index above the little finger, muzzle forward, attached to the hand bone), the nitrile glove, the procedural arms hidden; the walk weighing in while walking, Equip on a switch, the spray recoil, the robotic hand following the arm, with five articulated fingers closing to grab; with the power the palm up, the sphere in the element's colour and the sleeve crumbling; the final «serious PS2» pass; the jab and the inspect moving only the left arm. The **laboratories**: the door in all hundred levels, in the bench room; for each of the five maps the sign, entering the right map with the level hidden, thousands of triangles, the floor underfoot, walls that stop you, the synthesis bench opening and the way back to the door with the laboratory freed. Settings remembered, loop stopped with the section closed, thumb-sized controls at 390 px. **245 checks**, declared minimum 245	245

Bench	Verifies	Checks
`audit_statico`	The classes of bug this project has already had, searched across **all** files instead of where they showed up: `calc()` without spaces (the browser drops the rule), secrets in logs, API keys in the code, plaintext passwords, `target="_blank"` without `noopener`, duplicate ids in the markup, unbalanced scripts. The report **flags and does not block**, because those classes have false positives. Three measures **do** block, and have a written baseline: verbatim duplicated code blocks (**0**), redefined global functions (**11**, the file's declared "additive patch" style), of which with identical bodies (**0**). If they grow, the battery fails	3 gates
`test_articoli`	That no function has two entry points, and that the articles are not hand-written. Ten tabs inside a single Spectra centre, with each identifier present in the document **exactly once**; no repeated id, no duplicate button, no section reachable by two roads; Spectra out of the menu and with its dedicated button. Then the articles: the links tested as **pure functions** of the identifier, the Nobel explanations matched to the topic of the motivation and **none** invented when it is not recognised, and the decisive proof — an **injected** response with a name that exists nowhere, which must appear on screen. **Three distinct failures**: failed request, empty answer, unrecognised shape	58
`test_geom3d`	The 3D geometry built from the graph, measured against literature values: bonds, valence angles, alkyne linearity. **In both directions**: benzene comes out planar and cyclohexane does **not**. And the proof that makes the peak↔atom link legitimate: every heavy atom has **the same index and element** as the predictor's	29
`test_documento`	Opening any document and working through the problem inside: PDF, Word and text genuinely opened and the content verified; the data recognised **with their provenance**. **In both directions**: an image declares there is no optical character recognition and invents no working, an unknown format is refused rather than guessed, $CDCl_3$ does not become the compound's formula, without an MS label there are no masses	47

> **A bench that verifies a model must also verify it when the model is supposed to fail.** `test_cheminfo` exercises the null model in both directions: on a learnable signal the true R^2 (0.394) beats all eight label-scrambled twins, with a margin of 0.287; on the **same molecules with random labels** the R^2 (−0.199) does **not** beat them. A check that verifies only the first case would pass even if the guard were wired to "true".

> **A "contrast 0" that did not cover the new surface.** `verifica-accessibilita` measures what is visible, and the six panels of the Cheminformatics section stay hidden until the analysis has been run: it had never seen them. `test_cheminfo` now runs the analysis, opens the panels in turn and measures contrast over **460 text elements**, with the WCAG formulas rewritten inside the bench — asking the code under test how good its own contrast is, is not measuring. The bench also demands that more than 250 were measured: if the panels failed to open, the zero would be empty and the check would fail.

> **Three peaks found prove nothing on their own.** A detector that is too sensitive finds thirty where there are three; one that is too strict finds none. Neither case raises an error. `test_spettrolettore` starts from three Gaussians **built** at 1715, 2950 and 3400 cm^{-1} under noise and demands that the reader find three — 1716, 2952, 3402 — and then, on the **same noise without the Gaussians**, that it find **zero**. Without the second check the first would pass even with a detector that marked every wiggle. The same holds for the random forest: out-of-bag R^2 0.90 on the real relationship and **−0.16 on pure noise**.

> **The defect was in my counter, not in the data.** The first run of `verifica_farmaci_v187` returned **0 formulas out of 36**: every molecule came out as `C<n>`, all carbons and no hydrogens. RDKit's `get_json` format is **commonchem**, where an atom carries only the fields that **differ** from the defaults — `z` is 6 and `impHs` is 0 — so that an atom written `{}` is a carbon. I was reading `a.element`, which does not exist in that format. Thirty-six data points that look wrong all at once are almost always a wrong counter.

3.2 AI agent

Bench	Condition simulated
`test_404`	The provider has retired the model
`test_503`	Temporary overload
`test_ko` · `browser_ko`	Provider unreachable from the browser
`test_firma`	Opaque reasoning signatures to be returned
`test_tetto`	Token ceiling learned from the error
`test_attesa` · `test_attesalunga`	Timeouts and imposed waits
`test_doppioinvio`	Two overlapping submissions
`browser_prova`	Reachability measurement without a key
`test_nucleo` · `browser_prov`	Agentic loop and provider selection
`caccia_ai`	Malformed answers, truncated stream, 502 with HTML, Stop during the answer, panel closed halfway, hostile text

3.3 Stability

Bench	What it puts to the test
`audit_stabilita`	96 sections opened 5 times, full storage, corrupted data, bursts of clicks
`audit_promesse`	Rejected and unhandled promises, with a healthy network and a dead one
`audit_quota`	`localStorage.setItem` forced to fail on 10 pages
`test_sw`	Offline, degraded network, update while working
`test_filemanager`	Store unavailable, space exhausted, failed writes
`test_visore3d`	WebGL context consumption while browsing molecules

3.4 Interface

`browser_reset` · `browser_proxy` · `browser_proxyui` · `browser_rdkit` · `browser_lab` · `browser_frontiera` · `test_aggiorna` · `test_guidaproxy` · `test_fluidita` · `test_lingue` · `test_mol3d` · `test_spettrolettore`

`test_fluidita` — 15 checks. It measures how long the page stays *blocked* at every section switch, across all 91. It found three defects no other bench could see: the "3D PRO Viewer" froze the page for **797 ms** on the first click — the profiler attributes them to 3Dmol's first `render()`, which compiles the WebGL context's shaders; the **296 figures** in "Synthesis" (5,351 SVG nodes) were all drawn before anything was shown, while three are visible on opening; and **296 timers 40 ms apart** were scheduled when that section opened — eleven and a half seconds of wake-ups, each calling `getTotalLength()` on every stroke of every figure, including those off screen.

After the fixes: median switch **16 ms**, worst **87 ms**, **no section above 100 ms** (there were two, the worst at 1,166 ms).

At version `bsi-v188` the same bench caught two things. First: with 263 drugs instead of 233, «Pharmacology» blocked the page for **159 ms** and the count of sections over 100 ms went back to **three** — the battery had passed by a hair the time before, and a limit exceeded intermittently protects nothing. The cards are now drawn in 8 ms slices: **55 ms**, zero sections over 100.

Second concerns the bench itself. The viewer threshold was **80 ms absolute**; with the same published code, on different containers, the same click measured 14, 22, 98, 105 and 192 ms, because WebGL here is **SwiftShader** and the cost is shader compilation. Such a threshold measures the machine: it fails on identical code and would pass on worsened code. Even a **ratio** against a calibration taken in the same run does not hold — by the time the bench reaches the viewer WebGL is already warm (2.9 ms) while the viewer pays the cold cost: the ratio swung between 14× and 36×. The claim became **structural**, which is also the original defect: immediately after the click handler returns the WebGL context **must not exist**, shortly after it **must**. Measured 0 and 1; and the measurement is not vacuous, because on a second click — when the canvas is already there — it returns 1.

`test_lingue` — 64 checks. It watches two new features and two different ways of lying without noticing.

The language. A switch that "translates the application" is easy to write and hard to keep honest: someone changes an Italian label and its translation is no longer found. The defect is invisible — the label stays Italian in the middle of English — unless someone counts. The bench counts: **160 skeleton elements, 160 translated, none left**. It also demands the two opposite proofs: that switching to English **actually changes** the text (an empty dictionary would pass "everything translated" without moving a letter) and that switching back to Italian restores it **character for character**.

The molecular languages. Conversions are checked against external facts — seven literature InChI keys **written inside the bench** — and the full SMILES → molfile → SMILES round-trip is required. Formulas are counted by hand.

The IUPAC namer. 26 names written by hand in two languages, and seven molecules that **must be refused**: a namer that never refuses is a namer that invents.

Tested in both directions: restoring the `trim()` that breaks molfiles and removing one translation makes the bench fail three checks out of twenty.

The doors. Since v184 the bench also checks that the two features are **reachable**: a navigation button, a section that really draws, and — a check born from a real defect — that neither shows the "🚧 Section under construction" placeholder that the empty-section fallback stamps 800 ms after a click into any section left empty. It also checks that the ✨ menu entry **navigates** to the section rather than opening an overlay window, and that no duplicate id exists with both sections drawn.

The six languages. For each one two things are demanded together: full skeleton coverage (**162 of 162**) and that the text **really changes** — an empty dictionary would pass "everything translated" without moving a letter.

`test_mol3d` **— 46 checks.** It watches the molecule-formation animation and the bond-angle measurement, two additions with two different ways of looking right without being right.

The animation. "It is there" or "it is not" cannot be seen by inspecting the code: a loop that spins and redraws the same thing passes any check on the function's existence. The bench looks at the **pixels**: at 130 ms the canvas must be nearly empty — dust — and at 2.3 s there must be a molecule. Measured: **532 lit pixels → 2,334**. If the two numbers were equal there would be no animation, however the program is written. It also checks that the loop **stops** when stopped: a removed canvas with the loop still running is a drain paid in battery.

The angles. A number with a "°" next to it looks like a measurement. They are checked on hand-built geometries whose value is known **by construction**, not taken from a table: regular tetrahedron **109.4712°**, water 104.47°, CO_2 180°, BF_3 120°, ammonia 106.13°. Tolerance 0.05°.

The refusal. On a **flat** structure (z = 0) the angles button must be disabled and the reason written: on 2D coordinates a bond angle is not a bond angle, and showing it would be the worse of the two errors because it looks like data. Tested both ways: a structure with $z \neq 0$ must not be mistaken for flat, or no angle would ever be shown.

The first draft of this bench read the control bar **after** stopping the loop, which removes it: it found zero buttons and then declared "each has an accessible name" over an empty set. A check that passes because it has nothing to look at is exactly the defect this project chases; the check now demands two buttons **and** two names.

The selection. The viewer chooses the angle by clicking atoms: one shows all the angles at that atom, two the bond length, three the angle with the vertex at the second click, four the **dihedral**. The bench checks all four cases on ethanol — C–C 1.509 Å, C–C–O 111.9°, H–C–C–O 60.7° — and demands that two **unbonded** atoms be declared as such rather than passed off as a bond: a distance is not a bond.

The dihedrals. Checked on four hand-built conformations: eclipsed 0°, gauche 60°, orthogonal 90°, anti 180°. Here too the hand-written table corrected an error of mine, not of the program: with the coordinates I had chosen the dihedral is 120°, not 60°, because looking along the central bond the projected vector sits at (−0.5, 0.866).

The arrival order. "Skeleton first, hydrogens after" is a **measurable** property, not an intention: the bench reads each atom's progress halfway through the formation and requires every heavy atom to be ahead of every hydrogen. That check found a real defect: the elastic function that produces the bounce is not monotonic, and using it as the progress too meant an atom that

started earlier could look "less arrived" than one that started later. The two quantities are now separate — the progress is monotonic, the elastic factor only does the bounce — and the bounce, previously clipped away, is visible.

Half the checks are the *opposite proofs*, because a page that builds nothing is instantaneous: the WebGL canvas must appear anyway, all 296 figures must exist shortly after, no frame may stay empty, printing must not come out mute, and the search must filter even cards whose figure is not yet born. And the bench counts the sections it traversed: on ten instead of 89 the timings would be excellent and the measurement worthless.

3.5 Security and accessibility

Bench	Verifies
`verifica-sicurezza`	API keys in tracked files, clear-text passwords, secrets in `wrangler.toml`, telemetry, scripts from external domains
`audit_storia`	That no credential has EVER entered the repository: **1,551 distinct versions of text files** across **390 commits**, with 9 patterns. It refuses to pass on a shallow clone
`verifica-accessibilita`	WCAG AA contrast, accessible names, field labels, alternative text, heading hierarchy, `lang` attribute — across 13 pages
`audit_mobile`	That at **390 px** the page does not scroll horizontally, across all 96 sections — **and** that no section clips content into unreachability: an element wider than its section, with no scrolling ancestor, is text that is present and cannot be read. A wide table inside a scrolling container is not reported
`audit_rete`	That **no user data leaves the device**: a canary value seeded into 71 stores, the app used across 6 pages and 96 sections, the URL, headers and body of every request inspected
`audit_copertura`	How many bytes of JavaScript are **actually executed** while walking the application: 49.89 %, recorded and defended

`audit_storia` — 6 checks. `verifica-sicurezza` examines *tracked* files, i.e. the present state: it answers "there is no key in the repository today", not the question that matters for a public repository. A key added in one commit and removed in the next passes `verifica-sicurezza` forever and stays readable to anyone who clones, because git does not forget. If that happens, the remedy is not a fix-up commit: it is revoking the key.

This bench examines every *version* of every text file ever committed. Result: **zero credentials across 1,551 versions and 390 commits**.

Two defences against itself, because here a perfect number can come from less surface or from a blind pattern:

- **The clone must not be shallow.** A `--depth 1` holds a fraction of the

history: the bench would run, find nothing, and claim "no credential anywhere in history" having seen a tenth of it. If it finds `.git/shallow` it fails and names the command to run.

- **The patterns are tested before being trusted.** A wrong pattern finds

nothing and is indistinguishable from a clean repository: each of the 9 must match a crafted example and must *reject* a near-miss, and both are tested inside a tail of noise containing digits, as a real file would have.

The second defence was born from a defect in this very bench. The AWS pattern required a digit via `(?=.*\d)` — always satisfied in a large file, because the digit lies further ahead: it found 53 matches, all the same string `AKIAAAAAAAAAAAAAAAAA`, sixteen "A"s, the zero region of a base64 image. The near-miss did not catch it because it was too short to contain digits. An AWS identifier is now judged by the *variety* of its sixteen characters, and the near-misses carry their tail.

Tested in the other direction too: with a local commit containing a well-formed AWS key the bench fails naming the blob; once the commit is removed, it passes again.

> **The obstacle belonged to the tool, not to the problem.** For three versions code coverage was declared unmeasurable, with this reason: "instrumenting would require introducing a build, which the application does not have". That was true of `c8` and `istanbul`, which rewrite the source before running it. But **Chromium collects it by itself**, inside the engine, without touching a byte of the file served: `Profiler.startPreciseCoverage`, which Playwright exposes as `page.coverage.startJSCoverage()`. No build was needed: what was needed was noticing that the constraint belonged to the tools chosen.
>
> **The first result was 100 % on every file**, including 3.6 MB of `index.html`. That was the bench measuring nothing, in its most insidious form: a perfect number. V8 emits, for each function, an outer range with the number of entries, and inside it ranges with `count: 0` for what was not executed. Summing the positive ranges sums the whole file. One does the opposite: start from the total and subtract the union of the empty ranges.
>
> The real number is **49.89 %**, and it holds the same pact as the accessibility debt: recorded, and the bench fails if it falls. An absolute threshold ("80 %") would be an invented figure; "it must not get worse" is a promise that can be kept.

> **Why a canary and not a list of requests.** SEC-02 — "no personal data leaves the device" — was verified by checking the two *mechanisms* of exit: empty telemetry variable, no script from an external domain. That is real evidence but indirect: it says "I cannot see how it would leave", which is not "it did not leave". Looking instead at the list of hosts contacted would be just as weak, because it concludes from absence of evidence.
>
> The canary reverses the burden. An unrepeatable string is written into the user's own data — notes, chats, API keys, settings, progress — then the application is **used**, and every outgoing request is opened and read. If the string appears nowhere, it is because it really did not leave.
>
> ```
> BSI_PROVA_FUGA=1 node tools/banchi/audit_rete.js   # must FAIL
> ```
>
> The bench carries its own proof of working: with that variable it causes a leak on purpose, towards a **declared** host — contacting it is legitimate, sending it the user's data is not. If it does not fail then, the instrument is the thing that is broken.

Why a bench on horizontal overflow. UI-03 says "the app must work at 390 px", and the matrix gave it as verified by `audit_stabilita`, which in a phone viewport looks at **JavaScript errors**: a page can have not a single one and still run a hundred and sixty-three pixels off the screen. It was a requirement declared covered and not measured. The bench measures the document's `scrollWidth` — exactly what makes the scrollbar appear — and not any element wider than the screen, which inside a container built to scroll is correct and would fill the output with noise until nobody reads it any more.

What the accessibility check does not cover, and it must be said: text inside SVGs (the colour comes from `fill`, the background is a drawn shape) and text on gradient backgrounds (they do not have *a* colour). The skipped elements are counted and reported, not hidden.

One section at a time. The inspection skips elements that are not visible — and rightly so: a hidden element has no contrast to measure. But `index.html` alternates 96 sections and shows only one: out of **19,751** text elements the bench was looking at **41**, and printing "0 defects". It was not a false result, it was a result on a sample nobody had declared. The sections are now opened one by one, and **the number of sections traversed is printed**.

That count earned its keep on the very first run: the first attempt traversed **zero** — Playwright's click waits for the element to be visible, and the buttons sit inside collapsed navigation groups — and the bench said so ("87 sections present, none traversed") instead of printing another reassuring zero.

3.5-bis The accessibility debt, and the pact that it must not grow

Walking every section, **1,069** contrast defects surfaced. Today they are **zero** — the list of causes and remedies is in `docs/09` §4 — and unlabelled fields have fallen from 40 to **zero**.

The mechanism described below remains, and matters more than the number it guards. Demanding zero *from the outset* would have left only two roads, both bad: a verification red for ever, or one loosened until it turned green again. The third road is to **declare the number** and forbid it from growing, whatever it is. That is how 1,069 became zero: every step recorded, none of them reversible in silence.

File	Role
`docs/evidence/accessibilita-riferimento.json`	The recorded count

Situation	Bench outcome
The count **rises**	✗ FAILED — it is a regression
The count **falls**	✓ passes, and asks for the baseline to be updated
The count is **unchanged**	✓ passes silently: debt declared, not grown — today that value is **0**, and the bench defends it

```
node tools/verifica-accessibilita.js --aggiorna-riferimento
```

To be used **only after reducing** the defects, never to turn a regression green. Even at zero this is not full WCAG conformance: text inside SVGs and text over gradients stay outside, which automation cannot judge and which the bench **counts and reports** on every run (`docs/09` D-09). It is the honest measure of what can be measured, with the guarantee that it will not silently get worse.

3.6 Consistency

Bench	Verifies
`verifica_guida`	That the documentation's promises exist in the code; absence of conflict markers across 58 files
`verifica-documenti`	Internal links, version alignment, index consistency, justification of the deviations, **internal consistency of the traceability matrix** (no duplicated identifiers, coverage totals equal to the identifiers actually defined)
`genera-pacchetti`	Rewrites the references of the extracted documents and **verifies that no link is left broken** in the three delivery packages. It also verifies that **no PDF is older than the document it comes from** and that the bundled verification report declares **this** version: a typeset document or an evidence report from the previous version, inside a current package, is a silent lie
`genera-pdf`	Converts the documents to PDF and fails if a file falls below the plausibility threshold (empty conversion)
`verifica-affermazioni`	Compares every number declared in the documents — sections, drugs, diseases, tumours, strategies, modules — with the one **measured in the running application**, in Italian *and* in English
`test_stack`	That the React/FastAPI stack runs **the same files** as the application: it compares the SHA-256 fingerprints `GET /salute` declares with those of the files on disk, checks the numbers match the browser's, and that **no `.ts` / `.tsx` file contains a shift table** — because that is where the second engine would be born

> **Why numbers must be measured, not remembered.** The technical dossier opens with figures: "63 diseases (23 tumours)", "46 strategies", "25 modules". They are the first thing an assessor reads, and they are enough for everything else to be believed or doubted. Nobody was checking them: written once, they stayed put while the application changed. And indeed `chimorga.html` declared **"25 modules"** at the top and **"17 modules"** further down — two statements about the same object, in the same file, one of them false.
>
> **A lesson paid for while writing the bench.** I had already concluded that "63 diseases" was an exaggeration: in the source I had found a `DISEASES` array with 12 entries, and I was about to "correct" the document. Measuring in the DOM I saw that the menu really does list 63 — the array I had found was a different, smaller one. **A count on the source is not a measurement: it is a clue.** That is why the bench opens the page, opens the section, and counts.
>
> It also checks that the documents agree **with one another**: two documents saying different things about the same object are a defect even when one of them is right. The English translation counts as one of those documents — `docs/en/05` had been left at "84 sections" while the app had 87, and had been stuck at version `bsi-v146` for three releases — and no bench saw it, because none looked at the English set.

> **Why a check on the matrix.** The matrix declared `SCI-10` twice — once as a requirement verified at 100 %, once as an explicitly **unverified** one — and the summary table added up to 36 requirements while 37 were defined in the document. Both errors sat in the number an assessor reads first. The bench now counts the identifiers actually present and fails if the totals written do not match: the proof that the check really measures something is that, with the two errors reintroduced one at a time, it reports both.

4. Notable test cases

Some benches deserve a mention because they verify properties a functional test does not observe.

Case	Method	Measured result
Memory leaks	96 sections × 5 rounds, DOM nodes counted each round	+26,876 on the first round (construction), **+0** on the four that follow
Exhausted storage	`setItem` replaced with a function that always throws	10 pages out of 10 stay operational
Degraded ≠ absent network	Requests held for 20 s, interception at context level	Answer from cache in **3,507 ms** (threshold 3,500)
Spectrum reproducibility	Drawn twice, compared byte for byte	Identical
WebGL contexts	Viewer constructions over 10 consecutive molecules	From **7 to 1**
Corrupted history	Two `Enter` presses 500 ms apart	`user,assistant` instead of `user,user,assistant,assistant`
Text contrast	WCAG formula on every element with text of its own, 13 pages **and 96 sections**	19,751 elements examined (it was 41): **1,069** defects surfaced, **1,069 corrected**, **0 recorded** as a value that must not grow
Immediate cancellation	Stop pressed at 1.5 s, state sampled every second	Send button available from the **1st** second (it was the 10th)

5. How to add a bench

Minimal structure

```
/* A comment explaining WHAT this bench demonstrates and why it is needed. */
const { chromium } = require('playwright-core');
const CHROME = '/opt/pw-browsers/chromium-1194/chrome-linux/chrome';

(async () => {
  const b = await chromium.launch({ executablePath: CHROME, args: ['--no-sandbox'] });
  // serviceWorkers:'block' if the bench modifies files: without it the SW
  // might serve a cached copy and the old file would be measured
  const ctx = await b.newContext({ serviceWorkers: 'block' });
  const pg = await ctx.newPage();
  const err = [];
  pg.on('pageerror', e => err.push(e.message));

  let ok = 0, ko = 0;
  const att = (d, atteso, avuto) => {
    if (String(atteso) === String(avuto)) { ok++; console.log('  ✓ ' + d + '  → ' + avuto); }
    else { ko++; console.log('  ✗ ' + d + '\n      expected: ' + atteso + '\n      got:      ' +
avuto); }
  };

  await pg.goto('http://127.0.0.1:8899/index.html', { waitUntil: 'load', timeout: 60000 });
  await pg.waitForTimeout(2500);

  // … measurements …

  att('no JS error', 0, err.length);
  await b.close();
  console.log('\n' + (ko ? '✗ ' + ko + ' FAILED, ' : '') + ok + ' passed');
  process.exit(ko ? 1 : 0);   // the exit code is what counts
})();
```

The name is then added to the appropriate family in `tools/genera-evidenza.js`.

The rule that matters

> **A bench that measures nothing passes.** It is the most insidious way to obtain a false guarantee: no alarm, no failure, and no verification.

Therefore **every bench that simulates a condition must count how many times the simulation actually fired, and fail if that count is zero.**

Traps actually met while building the battery:

Trap	How it showed itself
`page.route` does not intercept Service Worker requests (`context.route` is needed)	The "degraded network" test measured a healthy network and concluded in 16 ms instead of 3,500
Probing exhausted `localStorage` with a one-byte write	After saturation one byte always finds room: the bench "passed"
Looking for `[data-tab]` where navigation uses `[data-s]`	Zero tabs found, therefore zero measurements — and zero failures

Trap	How it showed itself
The ticks of an axis on `<canvas>` are not text	They must be collected by intercepting `fillText`; and the bottom-most line is the axis *title*, not a tick
A probe installed on a library loaded **afterwards**	It is replaced along with the library: zero calls counted while the code was working
A `var` inside an IIFE is not `window.something`	Assigning it from outside creates a different variable

6. Coverage gaps

Declared. The coverage reported in `08-Traceability-Matrix.md` measures **how many requirements have a bench**, not how many lines of code are executed: they are two different things and must not be confused.

Gap	Current situation	Recommendation
Code coverage	Measured: **49.89 %** of statements over the widest path. What stays outside is whole-battery coverage and **branch** coverage: an `if` entered from one side only counts as covered	Extend collection to every bench, and move from statement coverage to branch coverage
Accessibility	Automated over 13 pages and all 96 sections: WCAG contrast, accessible names, labels, alternative text, heading hierarchy. Text inside SVGs and over gradients stay outside, and are **counted** on every run	Add `axe-core` alongside, for the rules this bench does not implement (ARIA roles, tab order, focus management)
Security	`verifica-sicurezza` runs 10 checks over 388 tracked files and covers SEC-01, SEC-03, SEC-05, SEC-06; SEC-04 is covered by `verifica_guida`. **SEC-02 remains indirect**: see `docs/09` D-03	Observe the network traffic during real use, the only direct verification of SEC-02
Browsers other than Chromium	No automatic test on Firefox or WebKit	Extend the main benches to `webkit`, where the differences on IndexedDB and Service Worker are greatest
Performance	Manual cross-device testing	Automatic measurement of first-paint time
Visual regression	Absent	Screenshot comparison for the charts, which are the heart of the product

7. Acceptance criteria

A version is not published if even one of these is unsatisfied.

- ☐ `node tools/genera-evidenza.js` closes with **0 failures**
- ☐ `tools/verifica-farmaci.js`: no defect, every deviation recorded with its reason
- ☐ `verifica_guida`: no promise in the documentation without a counterpart
- ☐ No non-network JavaScript error on any page
- ☐ Evidence report regenerated against the commit being published

Document updated to version `bsi-v219`.